

## CHAPTER 1

# Introduction

Each day people make many decisions; decisions which have both immediate and long-term consequences. Decisions must not be made in isolation; today's decision impacts on tomorrow's and tomorrow's on the next day's. By not accounting for the relationship between present and future decisions, and present and future outcomes, we may not achieve good overall performance. For example, in a long race, deciding to sprint at the beginning may deplete energy reserves quickly and result in a poor finish.

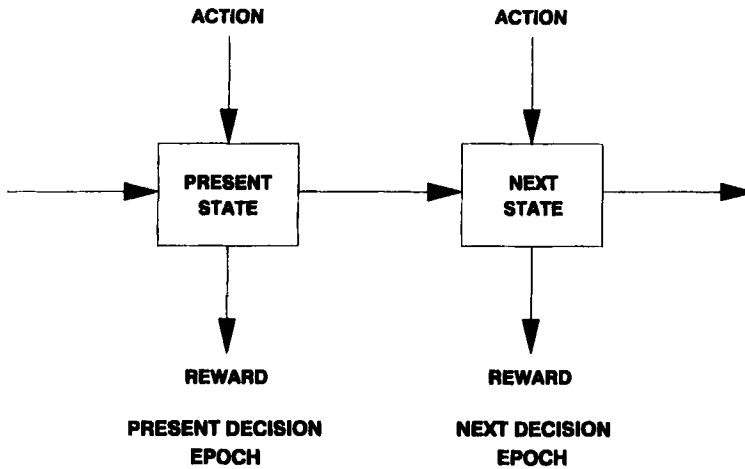
This book presents and studies a model for sequential decision making under uncertainty, which takes into account both the outcomes of current decisions and future decision making opportunities. While this model may appear quite simple, it encompasses a wide range of applications and has generated a rich mathematical theory.

### 1.1 THE SEQUENTIAL DECISION MODEL

We describe the sequential decision making model which we symbolically represent in Figure 1.1.1. At a specified point in time, a decision maker, agent, or controller observes the state of a system. Based on this state, the decision maker chooses an action. The action choice produces two results: the decision maker receives an immediate reward (or incurs an immediate cost), and the system evolves to a new state at a subsequent point in time according to a probability distribution determined by the action choice. At this subsequent point in time, the decision maker faces a similar problem, but now the system may be in a different state and there may be a different set of actions to choose from.

The key ingredients of this sequential decision model are the following.

1. A set of decision epochs.
2. A set of system states.
3. A set of available actions
4. A set of state and action dependent immediate rewards or costs.
5. A set of state and action dependent transition probabilities.



**Figure 1.1.1** Symbolic representation of a sequential decision problem.

With the exception of some models which we refer to in the Afterword, we assume that all of these elements are known to the decision maker at the time of each decision.

Using this terminology, we describe the probabilistic sequential decision model as follows. At each decision epoch (or time), the system state provides the decision maker with all necessary information for choosing an action from the set of available actions in that state. As a result of choosing an action in a state, two things happen: the decision maker receives a reward, and the system evolves to a possibly different state at the next decision epoch. Both the rewards and transition probabilities depend on the state and the choice of action. As this process evolves through time, the decision maker receives a sequence of rewards.

At each decision epoch, the decision maker chooses an action in the state occupied by the system at that time. A *policy* provides the decision maker with a prescription for choosing this action in any possible future state. A *decision rule* specifies the action to be chosen at a particular time. It may depend on the present state alone or together with all previous states and actions. A policy is a sequence of decision rules. Implementing a policy generates a sequence of rewards. The sequential decision problem is to choose, prior to the first decision epoch, a policy to maximize a function of this reward sequence. We choose this function to reflect the decision maker's intertemporal tradeoffs. Possible choices for these functions include the expected total discounted reward or the long-run average reward.

This book focuses on a particular sequential decision model which we refer to as a *Markov decision process* model. In it, the set of available actions, the rewards, and the transition probabilities depend only on the current state and action and not on states occupied and actions chosen in the past. The model is sufficiently broad to allow modeling most realistic sequential decision-making problems.

We address the following questions in this book.

1. When does an optimal policy exist?
2. When does it have a particular form?
3. How do we determine or compute an optimal policy efficiently?

We will see that the choice of the optimality criterion and the form of the basic model elements has significant impact on the answers to these questions.

Often you see these models referred to as *dynamic programming models* or *dynamic programs*. We reserve the expression “dynamic programming” to describe an approach for solving sequential decision models based on inductive computation.

In the remainder of this chapter, we illustrate these concepts with significant and colorful applications from several disciplines. The Bibliographic Remarks section provides a brief historical review.

## 1.2 INVENTORY MANAGEMENT

Sequential decision models have been widely applied to inventory control problems and represent one of the earliest areas of application. The scope of these applications ranges from determining reorder points for a single product to controlling a complex multiproduct multicenter supply network. Some of the earliest and most noteworthy results in stochastic operations research concern the form of the optimal policy under various assumptions about the economic parameters. We describe an application of a model of this type.

Through local dealerships, Canadian Tire, Inc. operates a chain of automotive supply stores throughout Canada. The 21 stores in the Pacific region are operated by a single management group. Backup inventory for these 21 stores is maintained at a central warehouse in Burnaby, British Columbia. It stocks roughly 29,000 products. Periodically, inventory is delivered from the central warehouse to each of its stores to maintain target stock levels.

The timing of inventory replenishment varies with store size. At stores designated as “small,” the inventory position of each product is reviewed once a week. For each product the inventory position (stock on hand) at the time of review determines the quantity, if any, to order. Orders arrive in about three days. Associated with an order for a particular product is a fixed charge associated with the time spent locating the item in the warehouse and shelving the item at the store. In addition to the fixed charge for filling the order, there is a daily carrying charge for keeping an item in inventory at a store. Management policy also dictates that at least 97.5% of demand be satisfied from stock on hand.

We now describe a sequential decision model for determining optimal reorder points and reorder levels for a single product at a single store. Decision epochs are the weekly review periods, and the system state is the product inventory at the store at the time of review. In a given state, actions correspond to the amount of stock to order from the warehouse for delivery at the store. Transition probabilities depend on the quantity ordered and the random customer demand for the product throughout the week. A decision rule specifies the quantity to be ordered as a function of the stock on hand at the time of review, and a policy consists of a sequence of such restocking functions. Management seeks a reordering policy which minimizes long-run average ordering and inventory carrying costs subject to the above constraint on the probability of being unable to satisfy customer demand.

Desirable properties for optimal policies in this setting are that they be simple to implement and not vary with time. Without the constraint on the probability of satisfying customer demand, the optimal policy may be shown to be of the following type: when the stock level falls below a certain threshold, order up to a target level;

otherwise do not order. With the inclusion of such a constraint, a policy of this form may not be optimal.

The importance of effective inventory control to effective cost management cannot be overemphasized. Sir Graham Day, chairman of Britain's Cadbury-Schweppes PLC notes (*The Globe and Mail*, October 20, 1992, p. C24):

"I believe that the easiest money any business having any inventory can save lies with the minimization of that inventory."

The roots of sequential decision making lie in this discipline. The book by Arrow, Karlin, and Scarf (1958) provides a good overview of the foundations of mathematical inventory theory; Porteus (1991) provides a recent review.

### 1.3 BUS ENGINE REPLACEMENT

Markov decision process models have been applied to a wide range of equipment maintenance and replacement problems. In these settings, a decision maker periodically inspects the condition of the equipment, and based on its age or condition decides on the extent of maintenance, if any, to carry out. Choices may vary from routine maintenance to replacement. Costs are associated with maintenance and operating the equipment in its current status. The objective is to balance these two cost components to minimize a measure of long-term operating costs.

Howard (1960) provided a prototype for such models with his "automobile replacement problem." In it, an individual periodically decides whether or not to trade in an automobile and, if so, with what age automobile to replace it. Subsequently, many variants of this model have been studied and analyzed. In this section and the next, we describe two applications of such models.

Rust (1987) formulates and analyzes the following problem. Harold Zurcher, superintendent of maintenance at the Madison (Wisconsin) Metropolitan Bus Company, has the responsibility of keeping a fleet of buses in good working condition. One aspect of the job is deciding when to replace the bus engines.

Zurcher's replacement problem may be formulated as a Markov decision process model as follows. Replacement decisions are made monthly and the system state represents the accumulated engine mileage since the last replacement. Costs include an age-dependent monthly operating cost and a replacement cost. The monthly operating costs include a routine operating and maintenance cost component and an unexpected failure cost component. The failure cost accounts for the probability of breakdown for a bus of a given age and costs associated with towing, repair, and lost goodwill. If Zurcher decides to replace an engine, then the company incurs a (large) replacement cost and, subsequently, the routine maintenance and operating cost associated with the replacement engine. Transition probabilities describe changes in accumulated mileage and the chance of an unplanned failure for a bus engine of a particular age. For each engine, Zurcher seeks an age-dependent replacement policy to minimize expected total discounted or long-run average costs.

The algorithms in Chaps. 4, 6, or 9 can be used to compute such an optimal policy for Harold Zurcher. However, the theory shows that, under reasonable assumptions, an optimal policy has a particularly simple and appealing form; at the first monthly



inspection at which the mileage exceeds a certain level, referred to as a *control limit*, the engine must be replaced; otherwise it is not. Rust examines whether Zurcher adopts such a policy using data from the Madison Metropolitan Bus Company.

Operating, maintenance, and replacement costs vary with engine type. The table below summarizes Zurcher's data on replacement costs and average mileage at replacement for two main engine types.

| Engine Type     | Replacement Cost | Average<br>Mileage at Replacement |
|-----------------|------------------|-----------------------------------|
| 1979 GMC T8H203 | \$9499           | 199,733                           |
| 1975 GMC 5308A  | \$7513           | 257,336                           |

This data shows that, although the replacement cost for a 1979 engine exceeded that of a 1975 engine by \$2000, Zurcher decided to replace the 1979 engines 57,600 miles and 14 months earlier than the 1975 engines. This suggests that routine maintenance and operating costs differ for these two engine types and that they increase faster with mileage in the 1979 engines. Rust's analysis of the data suggests that these costs may be modeled by linear or "square-root" functions of age.

Further data suggests that Zurcher's decisions departed from a simple control limit policy. Between 1974 and 1985, 27 T8H203 engines and 33 5308A engines were replaced. The mileage at replacement varied from 124,800 to 273,400 for the T8H203 engine and between 121,200 and 387,300 for the 5308A engine. Thus we might infer that Zurcher is making his decisions suboptimally. Rust adopts a different viewpoint. He hypothesizes that Zurcher's decisions coincide with an optimal policy of a Markov decision process model; however, Zurcher takes into account many measurements and intangibles that are not known by the problem solver. In his extensive paper, Rust (1987) provides an approach for accounting for these factors, estimating model parameters, and testing this hypothesis. He concludes that, after taking these unobservables into account, Zurcher's behavior is consistent with minimizing long-run average operating cost.

1.4 HIGHWAY PAVEMENT MAINTENANCE

The Arizona Department of Transportation (ADOT) manages a 7,400 mile road network. Up to the mid 1970s its primary activity was construction of new roadways. As the Arizona roadway system neared completion, and because of changing federal guidelines, ADOT's emphasis shifted in the late 1970's to maintaining existing roads. Between 1975 and 1979, highway preservation expenditures doubled from \$25 million to \$52 million, and evidence suggested that such an increase would continue. By this time it was evident to ADOT management that a systematic centralized procedure for allocation of these funds was needed. In 1978, in conjunction with Woodward-Clyde Consultants of San Francisco, ADOT developed a pavement management system based on a Markov decision process model to improve allocation of its limited resources while ensuring that the quality of its roadways was preserved. In 1980, the first year of implementation, this system saved \$14 million, nearly a third of Arizona's maintenance budget, with no decline in road quality. Cost savings over the next four years were predicted to be \$101 million. Subsequently, this model was modified for

use in Kansas, Finland, and Saudi Arabia. Related models have been developed for bridge and pipeline management. In this section, we describe the Arizona pavement management model. We base our presentation on Golabi, Kulkarni, and Way (1982), and additional information provided by Golabi in a personal communication.

The pavement management system relies on a dynamic long-term model to identify maintenance policies which minimize long-run average costs subject to constraints on road quality. To apply the model, the Arizona highway network was divided into 7,400 one-mile sections and nine subnetworks on the basis of road type, traffic density, and regional environment. For each category, a dynamic model was developed that specified the conditions of road segments, maintenance actions that could be used under each condition, and the expected yearly deterioration or improvement in pavement conditions resulting from each such action. In addition, costs associated with each maintenance action were determined. Developing categories for system states, actions, costs, and the state-to-state dynamics under different actions was a nontrivial task requiring data, models of road conditions, statistical analysis, and subject matter expertise.

We describe the management model for asphalt concrete highways; that for Portland cement concrete roadways had different states and actions. Decisions were made annually. The system state characterized the pavement condition of a one-mile segment by its roughness (three levels), its percentage of cracking (three levels), the change in cracking from the previous year (three levels), and an index which measured the time since the last maintenance operation and the nature of the operation (five levels). Consequently, a road segment could be described by one of 135 ( $3 \times 3 \times 3 \times 5$ ) possible states, but, since some combinations were not possible, 120 states were used.

Actions corresponded to available pavement rehabilitation activities. These ranged from relatively inexpensive routine maintenance to costly actions such as thick resurfacing or recycling of the entire roadway. A list of possible actions and associated construction costs appear in Table 1.4.1 below. For each state, however, only about six of the actions were considered feasible.

Costs consisted of the action-dependent construction costs (Table 1.4.1) and annual routine maintenance costs (Table 1.4.2). Annual routine maintenance costs varied with the road condition and rehabilitation action. When only routine maintenance was carried out, these costs varied with the roughness and degree of cracking of the road segment; when a seal coat was applied, these costs varied only with roughness; and if any other rehabilitation action was taken, maintenance costs were independent of previous road condition. These costs were determined through a regression model based on existing data.

Transition probabilities specify the likelihood of yearly changes in road condition under the various maintenance actions. These were estimated using existing data, under the assumption that each dimension of the state description varied independently. Since in each state only a limited number of subsequent states could occur, most of the transition probabilities (97%) were zero.

The performance criteria was cost minimization subject to constraints on the proportion of roads in acceptable and unacceptable states. For example, ADOT policy requires that at least 80% of high traffic roadways must have a roughness level not exceeding 165 inches/mile, while at most 5% of these roads could have roughness exceeding 256 inches/mile. Similar constraints applied to levels of cracking.

**Table 1.4.1 Rehabilitation Actions and Construction Costs**

| Action Index | Action Description <sup>a</sup>     | Construction Cost<br>\$/yd <sup>2</sup> |
|--------------|-------------------------------------|-----------------------------------------|
| 1            | Routine Maintenance                 | 0                                       |
| 2            | Seal Coat                           | 0.55                                    |
| 3            | ACFC                                | 0.75                                    |
| 4            | ACFC + AR                           | 2.05                                    |
| 5            | ACFC + HS                           | 1.75                                    |
| 6            | 1.5 inch AC                         | 1.575                                   |
| 7            | 1.5 inch AC + AR                    | 2.875                                   |
| 8            | 1.5 inch AC + HS                    | 2.575                                   |
| 9            | 2.5 inch AC                         | 2.625                                   |
| 10           | 2.5 inch AC + AR                    | 3.925                                   |
| 11           | 2.5 inch AC + HS                    | 3.625                                   |
| 12           | 3.5 inch AC                         | 3.675                                   |
| 13           | 3.5 inch AC + AR                    | 4.975                                   |
| 14           | 3.5 inch AC + HS                    | 4.675                                   |
| 15           | 4.5 inch AC                         | 4.725                                   |
| 16           | 5.5 inch AC                         | 5.775                                   |
| 17           | Recycling (equivalent to 6 inch AC) | 6.3                                     |

<sup>a</sup>Abbreviations used in table: ACFC-Asphalt concrete fine coat, AR-Asphalt Rubber, HS-Heater Scarifier, AC-Asphalt concrete

**Table 1.4.2 Annual Routine Maintenance Costs**

| State After Rehabilitation Action |                        | Rehabilitation<br>Action <sup>a</sup> | Cost<br>\$/yd <sup>2</sup> |
|-----------------------------------|------------------------|---------------------------------------|----------------------------|
| Roughness (in/mile)               | Percentage of Cracking |                                       |                            |
| 120 (±45)                         | 5 (±5)                 | RM                                    | 0.066                      |
| 120 (±45)                         | 20 (±10)               | RM                                    | 0.158                      |
| 120 (±45)                         | 45 (±15)               | RM                                    | 0.310                      |
| 120 (±45)                         | Any                    | SC                                    | 0.036                      |
| 210 (±45)                         | 5 (±5)                 | RM                                    | 0.087                      |
| 210 (±45)                         | 20 (±10)               | RM                                    | 0.179                      |
| 210 (±45)                         | 45 (±15)               | RM                                    | 0.332                      |
| 210 (±45)                         | Any                    | SC                                    | 0.057                      |
| 300 (±45)                         | 5 (±5)                 | RM                                    | 0.102                      |
| 300 (±45)                         | 20 (±10)               | RM                                    | 0.193                      |
| 300 (±45)                         | 45 (±15)               | RM                                    | 0.346                      |
| 300 (±45)                         | Any                    | SC                                    | 0.071                      |
| Any                               | Any                    | OT                                    | 0.036                      |

<sup>a</sup>Action Abbreviations; RM-routine maintenance, SC-seal coat, OT-any other

This model is an example of a *constrained* average reward Markov decision process model and can be solved using the linear programming methodology in Chaps. 8 and 9. This model was designed not only to yield a single solution but also to interactively examine the consequences of regulatory policies and budget changes. Examples of solutions are too lengthy to be presented here, but one aspect of the solution is worth noting. Because of the addition of constraints, the optimal policy may be *randomized*. This means that in some states, it may be optimal to use a chance mechanism to determine the course of action. For example, if the road segment is cracked, 40% of the time it should be resurfaced with one inch of asphalt concrete (AC) and 60% of the time with two inches of AC. This caused no difficulty because the model was applied to individual one-mile road segments so that this randomized policy could be implemented by repairing 40% of them with one inch of AC and 60% with two inches of AC. Also, in a few instances, the model recommended applying a different maintenance action to a road segment than to its two adjacent segments. In such cases the solution was modified to simplify implementation yet maintain the same level of overall cost and satisfy road quality constraints.

In addition to producing significant cost reductions, the model showed that

“...corrective actions in the past were too conservative; it was common to resurface a road with five inches of asphalt concrete... The policies recommended by the pavement management system... are less conservative; for example, a recommendation of three inches of overlay is rather rare and is reserved for the worst conditions. (Golabi, Kulkarni, and Way, 1982, p. 16).”

Observations such as this are consistent with the findings of many operations research studies. For example, preliminary results in the inventory control study described in Sect. 1.2 suggest that current in store inventory levels are 50% too high.

## 1.5 COMMUNICATIONS MODELS

A wide range of computer, manufacturing, and communications systems can be modeled by networks of interrelated queues (waiting lines) and servers. Efficient operation of these systems leads to a wide range of dynamic optimization problems. Control actions for these systems include rejecting arrivals, choosing routings, and varying service rates. These decisions are made frequently and must take into account the likelihood of future events to avoid congestion.

These models are widely applied and have had significant impact as noted by the following article in *The New York Times*, May 12, 1992, p. C2.

### “More Dial Mom Than Expected”

Even greater numbers of people called their mothers on Mother’s Day than AT&T had expected.

...A call-routing computer technique enabled the American Telephone and Telegraph Company to complete more calls than last year, when it logged 93.4 million calls.

On Sunday, there were about 1.5 million uncompleted calls, where customers got a recorded announcement advising them to call later, compared with 3.9 million last year.

A new computer technique called real-time network routing helped AT&T shepherd a larger number of calls through the labyrinth of telephone computers known as switches. By creative zigzagging around the country, AT&T could direct calls so that they were more likely to avoid congestion, especially in suburbs, which do not have the high-capacity telephone lines that big cities do.

We now describe a sequential decision process model for a particular communication system. Many packet communications systems are configured so that multiple terminals generating low rate, bursty traffic and must share a single channel to communicate with each other or with a central hub (Fig. 1.5.1).

This system architecture is typical of satellite broadcast networks where multiple earth stations communicate over the same radio frequency, and computer local area networks (LAN's) where many computers send job requests to a central file server over a single coaxial cable. Since a single channel may only carry one stream of traffic, the problem arises as to how to coordinate the traffic from the terminals to make the most efficient use of the channel.

The *Slotted ALOHA Protocol* is a popular and especially simple technique for providing such coordination. We describe the slotted ALOHA channel model and the mechanism by which it controls channel access. Stations communicate over a slotted ALOHA channel through equal-length packets of data. Time on the channel is divided into slots of the same length as the packets, and all terminals are synchronized so that packet transmissions always begin at the leading edge of a time slot and occupy exactly one slot. New packets are randomly generated at any idle terminal during a slot, and are transmitted in the following slot. If no other stations transmit a packet in that slot, the transmission is considered successful and the terminal returns to idle mode. If more than one terminal generates a packet, a collision occurs, the data become garbled, and the station goes into retransmission mode and must retransmit the packet in a future slot. If a collision occurs and all involved terminals always retransmit in the next slot, collisions will continue endlessly. To avoid this situation, the slotted ALOHA protocol specifies that stations in retransmission mode transmit in the next slot with a specified retransmission probability, thus achieving a random backoff between retransmission attempts. When a terminal successfully retransmits the packet, it returns to idle mode and waits for a new packet to be

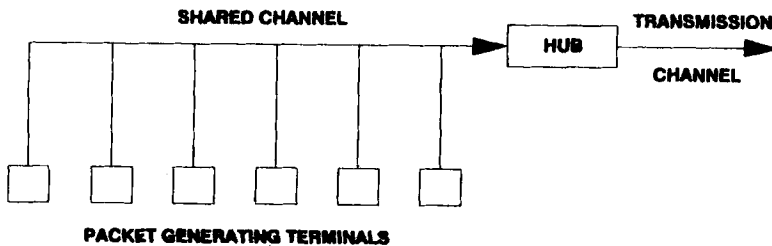


Figure 1.5.1 Multiple access channel configuration.

generated. We see then that, although the slotted ALOHA protocol does not avoid collisions on the channel, the use of a random retransmission backoff provides a scheme for effective contention resolution among terminals.

Since the message generating probability is fixed, the only means available to control channel access within this model is by regulating the retransmission probability. If it is held constant and the number of terminals in retransmission mode becomes large, the probability of a collision in the next slot will also become large. As the collisions become more frequent, newly arriving packets tend to become backlogged, increasing the number of terminals in retransmission mode. Thus, with a fixed retransmission probability, the system is prone to become highly congested, reducing the chance of a successful transmission to close to zero. This instability may be alleviated (for certain values of the packet generation probability) by taking into account the current number of terminals in retransmission mode when choosing a retransmission probability.

We now describe a Markov decision process model for this control problem. Decision epochs correspond to time slots, and the system state is the number of terminals in retransmission mode. Actions correspond to choosing a retransmission probability. The system generates a reward of one unit for each packet successfully transmitted, and transition probabilities combine the probabilities that new packets are generated in a time slot and a successful packet transmission occurs when the retransmission probability has been set at a particular level. The objective is to choose a retransmission probability-setting policy which maximizes the long-run average expected channel throughput (rate of successful packets per slot).

Feinberg, Kogan, and Smirnov (1985) show that the optimal retransmission probability is a monotonically decreasing function of the system state whenever the mean packet arrival rate (number of terminals times packet generation probability) is less than one. If this rate exceeds one, the system will become congested and other optimality criteria may be used. This control policy agrees with intuition in that the system will react to increasing congestion by decreasing retransmission probabilities and thus maintaining a reasonable probability of successful packet transmission.

In practical applications, the number of stations in retransmission mode and the packet generation probability are rarely known. They must be estimated on the basis of the history of channel observations (idle, successful, and collision slots). In this case, incorporation of both state and parameter estimation into the Markov decision process model is necessary to find the optimal retransmission policy. We provide references for models of this type in the Afterword.

## 1.6 MATE DESERTION IN COOPER'S HAWKS

Markov decision process models are becoming increasingly popular in behavioral ecology. They have been used in a wide range of contexts to gain insight into factors influencing animal behavior. Examples include models of social and hunting behavior of lions (Clark, 1987; Mangel and Clark, 1988), site selection and number of eggs laid by apple maggots and medflies (Mangel, 1987), daily vertical migration of sockeye salmon and zooplankton (Levy and Clark, 1988; Mangel and Clark 1988), changes in mobility of spiders in different habitats (Gallepie and Caraco, 1987), and singing versus foraging tradeoffs in birds (Houston and McNamara, 1986).

The theory of natural selection suggests that organisms predisposed to behavioral characteristics that allow them to adapt most efficiently to their environment have the greatest chance of reproduction and survival. Since any organism alive today has a substantial evolutionary history, we might infer that this organism has adopted optimal or near-optimal survival strategies which can be observed in day-to-day activity.

Models have been based on regarding the behavior of an organism as its reaction or response to its environment, conditional on its state of well being. Throughout its life, it makes behavioral choices which affect its chances of survival and successful reproduction. Investigators have used probabilistic sequential decision process models to determine state- and time-dependent strategies which maximizes a function of its survival and reproductive success probabilities and then compared model results to observed behavior. If there is "reasonable agreement," then the derived optimal policy may provide insight into the behavioral strategy of the organism.

We describe Kelly and Kennedy's (1993) use of this methodology in their study of mate desertion in Cooper's hawks (*Accipiter cooperii*). Over a five-year period, they studied nesting behavior of several birds near Los Alamos National Laboratory in north-central New Mexico (Fig. 1.6.1.) They observed that more than 50% of the females deserted their nests before the young reached independence, and noted that the male of this species continued to feed the young regardless of whether or not a female was present. At issue was determining factors that influenced the female's decision to desert and the female's tradeoffs between her survival and that of her offspring.

In the study, the physical conditions of both the nestlings (young birds) and the female were monitored, assisted by the use of radiotelemetry. Females were trapped and tagged early in the breeding season, providing an opportunity to assess the initial health of the females. Birds with greater body mass had larger energy reserves and were considered healthier. Rather than disturb the nestlings, their health was determined by assuming that nestlings were initially healthy. A developmental model was



**Figure 1.6.1** A female Cooper's hawk and her brood. (Photograph courtesy of Patricia L. Kennedy.)

used to account for parental hunting behavior, captures, and nestling growth rates.

Kelly and Kennedy developed a model for a single nesting season based on the assumption that behavioral choices of the female hawk maximized a weighted average of the probability of nestling survival and the probability of the female's survival to the next breeding season. The sequential decision model was used to determine an optimal behavioral strategy.

The nesting season was divided into four periods representing basic stages of development of the young.

1. Early nestling period.
2. Late nestling period.
3. Early fledgling dependence period.
4. Late fledgling dependence period.

The end of the late fledgling period marked the point at which the brood reaches independence.

The system state is a two-dimensional health index representing female and brood energy reserves. The states were constrained to lie between lower levels which represented the minimum physical condition for survival, and upper levels corresponding to limiting physical attributes of the birds.

Three basic behavioral strategies were observed for the female.

1. Stay at the nest to protect the young.
2. Hunt to supplement the food supplied by the male.
3. Desert the nest.

Decisions were assumed to have been made at the start of each of the above periods.

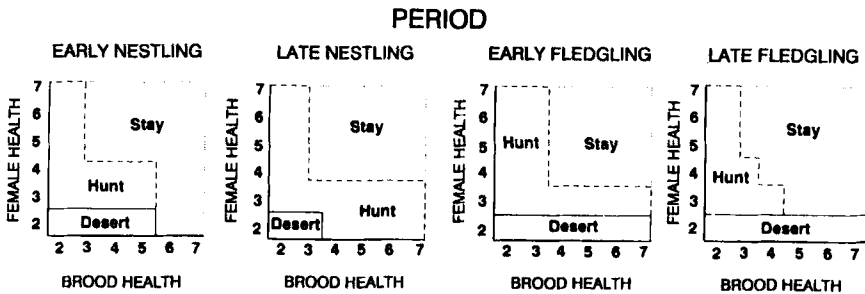
From one developmental stage to the next, the change in energy reserves of both the female and the young depends on the female's behavioral strategy and the amount of food captured, a random quantity. At the time of independence, the female's and brood's states of health depend on their initial energy reserves, the female's behavior, and the availability of food. The respective health indices at the end of the decision making period determine the probability of survival of the female and of the brood to the subsequent nesting period.

Using data estimated from the five-year study, results in the literature, and some intelligent guesswork, Kelly and Kennedy determined transition probabilities for the above model. They then solved the model to determine the optimal policy using inductive methods we describe in Chap. 4. Figure 1.6.2, which we adopt from their paper, shows the optimal policy under a specified degree of tradeoff between female and brood survival.

The four graphs show the optimal behavioral action as a function of the health of the female and the brood at each of the four decision periods. The vertical axis represents the female's health index and the horizontal axis represents the brood's health index. Low values indicate states of low-energy reserve.

Observe that, in all periods, if both the female's and the brood's health index exceed 4, the optimal strategy for the female is to stay at the nest and protect the young. At the other extreme, if the health index of both the female and the brood is at its lowest value, the optimal strategy for the female is to desert the nest.





**Figure 1.6.2** Symbolic representation of optimal desertion strategy under a specific tradeoff parameter choice. Quantities on axes denote health indices. (Adapted from Kelly and Kennedy, 1993.)

There are other patterns to this optimal strategy. For a fixed value of the brood health index, as the female's health index increases the optimal strategy changes from desert to hunt to stay. Similarly, if the female's health index is fixed, as the brood's energy reserves increase the strategy changes from desert to hunt to stay. Thus there is a form of monotonicity in the optimal strategy. One might conjecture that the behavioral strategy of the female will have this form under any parameter values. Observing such patterns can sometimes yield insight into theoretical results beyond a specific numerical scenario. In subsequent chapters, we will provide methods for identifying models in which optimal strategies have a particular form.

Kelly and Kennedy (p. 360–361) conclude

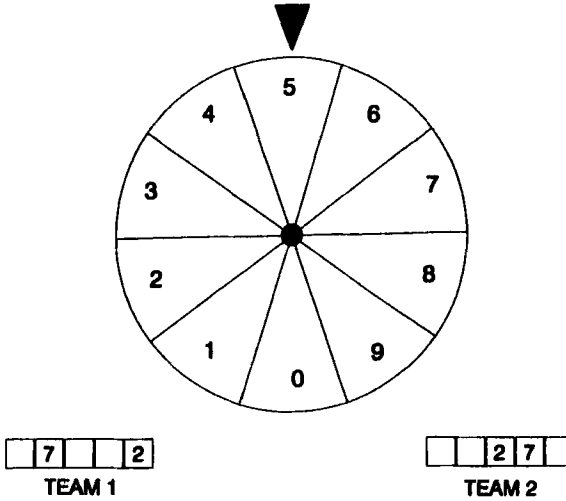
“The agreement of model predictions and observed strategies supported, but did not prove, the modelling hypotheses that:

1. a female's strategy during brood rearing maximizes the weighted average of the expected probability of survival of her current offspring and her future reproductive potential, and
2. the female's strategy choices were influenced by multiple factors including her state, the state of her brood, the risks to nestlings associated with each strategy, and the male's and female's foraging capabilities.

... dynamic state variable models are powerful tools for studying the complexities of animal behavior from an evolutionary standpoint because they lead to quantitative testable predictions about behavioral strategies.”

## 1.7 SO WHO'S COUNTING

Games of chance and strategy provide natural settings for applying sequential decision models. Dubins and Savage (1965) in their monograph *How to Gamble if You Must* developed a model for gambling, not unlike the sequential decision model herein, and developed a rich mathematical theory for analyzing it. Their basic observation was that, even in an unfair game, some betting strategies might be better than others. Markov decision process models apply to such games of chance and also to a wide range of board and computer games. In this section we show how such a



**Figure 1.7.1** Spinner for “So Who’s Counting.”

model can be used to determine an optimal strategy in a challenging yet easy to describe television game show.

The mock game show *But Who’s Counting* appeared on *Square One*, a mathematically oriented educational television program on the U.S. public broadcasting network. The game is played as follows. There are two teams of players. At each of five consecutive rounds of the game, a spinner (Fig. 1.7.1) produces a number between 0 and 9, each with equal probability. After each spin, the teams select an available digit of a five-digit number to place the number produced by the spinner. The team which generates the largest number wins the game.

Figure 1.7.1 illustrates the status of the game immediately following the third spin. At the two previous spins, the numbers 2 and 7 had appeared. At this point in time, the order in which they previously appeared is immaterial. Team 1 placed the 7 in the 1,000s place and the 2 in 1s place, while team 2 placed the 2 in the 100s place and the 7 in the 10s place. Each team must now decide where to place the 5. What would you do in this case if you were on team 1? On team 2?

Now ignore the competitive aspect of this game and suppose you were in the game alone with the objective of obtaining the highest five-digit number. Reflect on what strategy you would use. If you observed a 9, surely you would want to get the best out of it, and, thus, place it in the highest digit available. However, what would you do if you had a 5 or a 6?

We formulate the single-player game as a sequential decision problem. Decision epochs correspond to the instant immediately after the spinner identifies a number. We take as the system state the locations of the unoccupied digits and the number which has appeared on the spinner. Actions correspond to placing the number into one of the available digits and the reward equals the number times the place value for that digit. The objective is to choose a digit-placing strategy which maximizes the expected value of the five-digit number.

We can use the methods of Chap. 4 directly to derive an optimal policy. It has the property that the decision into which unoccupied digit to place the observed number

Table 1.7.1 Optimal Policy for “But Who’s Counting.”

| Observed<br>Number | Optimal Digit Locations |        |        |        |        |
|--------------------|-------------------------|--------|--------|--------|--------|
|                    | Spin 1                  | Spin 2 | Spin 3 | Spin 4 | Spin 5 |
| 0                  | 5                       | 4      | 3      | 2      | 1      |
| 1                  | 5                       | 4      | 3      | 2      | 1      |
| 2                  | 5                       | 4      | 3      | 2      | 1      |
| 3                  | 4                       | 3      | 3      | 2      | 1      |
| 4                  | 3                       | 3      | 2      | 2      | 1      |
| 5                  | 3                       | 2      | 2      | 1      | 1      |
| 6                  | 2                       | 2      | 1      | 1      | 1      |
| 7                  | 1                       | 1      | 1      | 1      | 1      |
| 8                  | 1                       | 1      | 1      | 1      | 1      |
| 9                  | 1                       | 1      | 1      | 1      | 1      |

should be based on the number of unoccupied positions remaining, and not on their place values or the values of the previously placed digits. This observation enables us to summarize succinctly the optimal policy as in Table 1.7.1.

In this table, the entries represent the location of the unoccupied digit (counting from the left) into which to place the observed number. For example, consider the decision faced by a player on team 1 in Fig. 1.7.1 after observing a 5 on spin 3. Table 1.7.1 prescribes that the 5 should be placed in the 100’s position. To proceed optimally, a player on team 2 should place the 5 in the 1000’s position.

Furthermore, using methods in Chap. 4, we can show that using this policy yields an expected score of 78,734.12, compared to that of a random-digit choice policy which would result in an expected score of 49,999.5. Of course, in a particular game, this strategy may not always yield the greatest score, but in the long run, it will do best on average.

This problem is a special case of a *sequential assignment problem*. Ross (1983, p. 124) provides a clever approach for solving these problems in general. He establishes existence of and gives a method for computing a set of *critical levels* which in this context determine the optimal placement of the number. If the number is above the highest critical level, then it should be placed in the leftmost digit available. If it is between the highest and second highest, it should be placed in the second-leftmost unoccupied digit, and so on. His approach shows that the optimal policy would still be the same if we had any other increasing values for the contribution of digits to the total reward instead of 1, 10, 100, 1000, or 10,000.

It is not hard to think of variations of this problem. We may view it from a game theoretic point of view in which the objective is to derive a strategy which maximizes the probability of winning the game, or we may consider a single-person game in which the numbers have unequal probabilities.

HISTORICAL BACKGROUND

The books by Bellman (1957) and Howard (1960) popularized the study of sequential decision processes; however, this subject had earlier roots. Certainly some of the basic

concepts date back to the calculus of variations problems of the 17th century. Cayley's paper (Cayley, 1875), which did not resurface until the 1960s, proposed an interesting problem which contains many of the key ingredients of a stochastic sequential decision problem. We describe and analyze this problem in detail in Chaps. 3 and 4.

The modern study of stochastic sequential decision problems began with Wald's work on sequential statistical problems during the Second World War. Wald embarked on this research in the early 1940's, but did not publish until later because of wartime security requirements. His book (1947) presents the essence of this theory.

Pierre Massé, director of 17 French electric companies and minister in charge of French electrical planning, introduced many of the basic concepts in his extensive analysis of water resource management models (1946). Statistician Lucien Le Cam (1990), reflecting on his early days at Electricité de France, noted

"Massé had developed a lot of mathematics about programming for the future. What had become known in this country (the United States) as "dynamic programming," invented by Richard Bellman, was very much alive in Massé's work, long before Bellman had a go at it."

A description of Massé's reservoir management model appears in Gessford and Karlin (1958).

Arrow (1958, p. 13), in his colorful description of the economic roots of the dynamic stochastic inventory model, comments

"... it was Wald's work (rather than Massé's, which was unknown in this country at the time) which directly led to later work in multi-period inventory."

A precise time line with proper antecedents is difficult to construct. Heyman and Sobel (1984, p. 192) note

"The modern foundations were laid between 1949 and 1953 by people who spent at least part of that period as staff members at the RAND Corporation in Santa Monica, California. Dates of actual publication are not reliable guides to the order in which ideas were discovered during this period."

Investigators associated with this path breaking work include Arrow, Bellman, Blackwell, Dvoretzky, Girschik, Isaacs, Karlin, Kiefer, LaSalle, Robbins, Shapley, and Wolfowitz. Their work on games (Bellman and Blackwell, 1949; Bellman and LaSalle, 1949; Shapley, 1953), stochastic inventory models (Arrow, Harris, and Marschak, 1951; Dvoretzky, Kiefer, and Wolfowitz, 1952), pursuit problems (Isaacs, 1955, 1965) and sequential statistical problems (Arrow, Blackwell, and Girschik, 1949; Robbins, 1952; Kiefer, 1953) laid the groundwork for subsequent developments.

Bellman in numerous papers identified common ingredients to these problems and through his work on functional equations, dynamic programming, and the principle of optimality, became the first major player. Bellman (1954) contains a concise presentation of many of his main ideas and a good bibliography of early work. His 1957 book contains numerous references to his own and other early research and is must reading for all investigators in the field. Karlin (1955) recognized and began studying the rich mathematical foundations of this subject.

## CHAPTER 2

# Model Formulation

This chapter introduces the basic components of a Markov decision process and discusses some mathematical and notational subtleties. Chapters 1 and 3 contain many examples of Markov decision processes. We encourage you to refer to those examples often to gain a clear understanding of the Markov decision process model. Section 2.2 illustrates these concepts and their interrelationship in the context of a one-period model.

A Markov decision process model consists of five elements: decision epochs, states, actions, transition probabilities, and rewards. We describe these in detail below.

### 2.1 PROBLEM DEFINITION AND NOTATION

A decision maker, agent, or controller (who we refer to as he with no sexist overtones intended) is faced with the problem, or some might say, the opportunity, of influencing the behavior of a probabilistic system as it evolves through time. He does this by making decisions or choosing actions. His goal is to choose a sequence of actions which causes the system to perform optimally with respect to some predetermined performance criterion. Since the system we model is ongoing, the state of the system prior to tomorrow's decision depends on today's decision. Consequently, decisions must not be made myopically, but must anticipate the opportunities and costs (or rewards) associated with future system states.

#### 2.1.1 Decision Epochs and Periods

Decisions are made at points of time referred to as *decision epochs*. Let  $T$  denote the set of decision epochs. This subset of the non-negative real line may be classified in two ways: as either a discrete set or a continuum, and as either a finite or an infinite set. When discrete, decisions are made at all decision epochs. When a continuum, decisions may be made at

1. all decision epochs (continuously),
2. random points of time when certain events occur, such as arrivals to a queueing system, or
3. opportune times chosen by the decision maker.

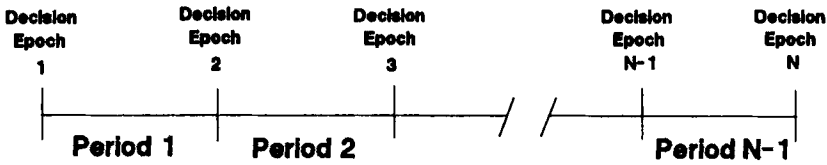


Figure 2.1.1 Decision Epochs and Periods.

When decisions are made continuously, the sequential decision problems are best analyzed using control theory methods based on dynamic system equations.

In discrete time problems, time is divided into *periods* or *stages*. We formulate models so that a decision epoch corresponds to the beginning of a period (see Fig. 2.1.1). The set of decision epochs is either finite, in which case  $T \equiv \{1, 2, \dots, N\}$  for some integer  $N < \infty$ , or infinite, in which case  $T \equiv \{1, 2, \dots\}$ . We write  $T = \{1, 2, \dots, N\}$ ,  $N \leq \infty$  to include both cases. When  $T$  is an interval, we denote it by either  $T = [0, N]$  or  $T = [0, \infty)$ . Elements of  $T$  (decision epochs) will be denoted by  $t$  and usually referred to as “time  $t$ .” When  $N$  is finite, the decision problem will be called a *finite horizon* problem; otherwise it will be called an *infinite horizon* problem. Most of this book will focus on infinite horizon models. We adopt the convention that, in finite horizon problems, decisions are *not* made at decision epoch  $N$ : we include it for evaluation of the final system state. Consequently, the last decision is made at decision epoch  $N-1$ . Frequently we refer to this as an  $N-1$  period problem.

The primary focus of this book will be models with discrete  $T$ . A particular continuous time model (a semi-Markov decision process) will be discussed (Chapter 11).

## 2.1.2 State and Action Sets

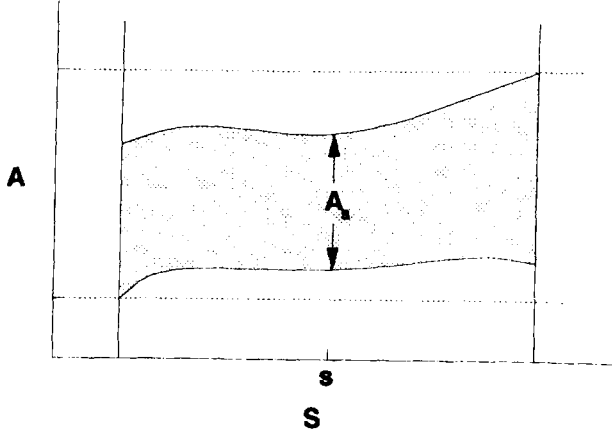
At each decision epoch, the system occupies a *state*. We denote the set of possible system states by  $S$ . If, at some decision epoch, the decision maker observes the system in state  $s \in S$ , he may choose action  $a$  from the set of allowable actions in state  $s$ ,  $A_s$ . Let  $A = \bigcup_{s \in S} A_s$  (Fig. 2.1.2.) Note we assume that  $S$  and  $A_s$  do not vary with  $t$ . We expand on this point below.

The sets  $S$  and  $A_s$  may each be either

1. arbitrary finite sets,
2. arbitrary countably infinite sets,
3. compact subsets of finite dimensional Euclidean space, or
4. non-empty Borel subsets of complete, separable metric spaces.

In nondiscrete settings, many subtle mathematical issues arise which, while interesting, detract from the main ideas of Markov decision process theory. We expand on such issues in Section 2.3 and other sections of this book. These more technical sections will be indicated by asterisks. Otherwise, we assume that  $S$  and  $A_s$  are *discrete* (finite or countably infinite) unless explicitly noted.

Action may be chosen either randomly or deterministically. Denote by  $\mathcal{P}(A_s)$  the collection of probability distributions on (Borel) subsets of  $A_s$  and by  $\mathcal{P}(A)$  the set of probability distributions on (Borel) subsets of  $A$ . (We may regard  $q(\cdot) \in \mathcal{P}(A_s)$  as an



**Figure 2.1.2** Relationship between action sets when  $S$  is a continuum.

element of  $\mathcal{P}(A)$  with support  $A_s$ .) Choosing actions randomly means selecting a probability distribution  $q(\cdot) \in \mathcal{P}(A_s)$ , in which case action  $a$  is selected with probability  $q(a)$ . Degenerate probability distributions correspond to deterministic action choice.

This model may be generalized by allowing both or either of  $S$  and  $A_s$  to depend explicitly on  $t$ . Such generality is unnecessary for most applications and has little effect on theory. It can be included in the formulation above by setting  $S = \bigcup_{t \in T} S_t$ , where  $S_t$  denotes the set of possible states at time  $t$  and  $A_s = \bigcup_{t \in T} A_{s,t}$ , where  $A_{s,t}$  denotes the set of allowable actions in state  $s$  at time  $t$ . Transition probabilities and rewards (defined below) must be modified accordingly.

We might restrict the model by requiring that  $A_s = A$  for all  $s \in S$ . This simplifies notation and doesn't impact theory but limits the applicability of the model since applications as simple as those in Sections 3.1 and 3.3 are not easily included. (Problem 2.7)

Sometimes a distinction is made between a model in which decisions are made at every decision epoch, and one in which the system is uncontrolled and interventions or decisions are made only when necessary. We take the view that this distinction is unnecessary; the latter model is a special case of that model proposed here, in which, for each  $s \in S$ ,  $A_s$  includes an action "do not intervene" (Problem 2.3).

### 2.1.3 Rewards and Transition Probabilities

As a result of choosing action  $a \in A_s$  in state  $s$  at decision epoch  $t$ ,

1. the decision maker receives a reward,  $r_t(s, a)$  and
2. the system state at the next decision epoch is determined by the probability distribution  $p_t(\cdot | s, a)$ .

Let the real-valued function  $r_t(s, a)$  defined for  $s \in S$  and  $a \in A_s$  denote the value at time  $t$  of the reward received in period  $t$ . When positive,  $r_t(s, a)$  may be regarded as income, and when negative as cost. From the perspective of the models

studied herein, it is immaterial how the reward is accrued during the period. We only require that its value or expected value be known before choosing an action, and that it not be effected by future actions. The reward might be

1. a lump sum received at a fixed or random time prior to the next decision epoch,
2. accrued continuously throughout the current period,
3. a random quantity that depends on the system state at the subsequent decision epoch, or
4. a combination of the above.

When the reward depends on the state of the system at the next decision epoch, we let  $r_t(s, a, j)$  denote the value at time  $t$  of the reward received when the state of the system at decision epoch  $t$  is  $s$ , action  $a \in A_s$  is selected, and the system occupies state  $j$  at decision epoch  $t + 1$ . Its expected value at decision epoch  $t$  may be evaluated by computing

$$r_t(s, a) = \sum_{j \in S} r_t(s, a, j) p_t(j|s, a). \quad (2.1.1)$$

In (2.1.1), the non-negative function  $p_t(j|s, a)$  denotes the probability that the system is in state  $j \in S$  at time  $t + 1$ , when the decision maker chooses action  $a \in A_s$  in state  $s$  at time  $t$ . The function  $p_t(j|s, a)$  is called a *transition probability function*. Note that many system transitions might occur in the time period between decision epoch  $t$  and decision epoch  $t + 1$ . We formulate the model so that transitions which occur between decision epochs do not influence the decision maker. Under most notions of optimality, all of the information necessary to make a decision at time  $t$  is summarized in  $r_t(s, a)$  and  $p_t(j|s, a)$ ; however, under some criteria we must use  $r_t(s, a, j)$  instead of  $r_t(s, a)$ .

We usually assume that

$$\sum_{j \in S} p_t(j|s, a) = 1. \quad (2.1.2)$$

Not requiring equality in (2.1.2) increases the range of systems which we may model (cf. Rothblum and Whittle, 1982 and Chapter 11).

In finite horizon Markov decision processes, no decision is made at decision epoch  $N$ . Consequently, the reward at this time point is only a function of the state. We denote it by  $r_N(s)$  and sometimes refer to it as a *salvage value* or *scrap value*.

We refer to the collection of objects

$$\{T, S, A_s, p_t(\cdot|s, a), r_t(s, a)\}$$

as a *Markov decision process*. The qualifier “Markov” is used because the transition probability and reward functions depend on the past only through the current state of the system and the action selected by the decision maker in that state. Under certain policies, the induced stochastic process need not be Markov. Some authors refer to the above collection of objects as a *Markov decision problem*; here that expression is reserved for a Markov decision process together with an optimality criterion.



### 2.1.4 Decision Rules

A *decision rule* prescribes a procedure for action selection in each state at a specified decision epoch. Decision rules range in generality from deterministic Markovian to randomized history dependent, depending on how they incorporate past information and how they select actions. Our primary focus will be on deterministic Markovian decision rules because they are easiest to implement and evaluate. Such decision rules are functions  $d_t: S \rightarrow A_s$ , which specify the action choice when the system occupies state  $s$  at decision epoch  $t$ . For each  $s \in S$ ,  $d_t(s) \in A_s$ . This decision rule is said to be *Markovian* (memoryless) because it depends on previous system states and actions only through the current state of the system, and *deterministic* because it chooses an action with certainty.

We call a deterministic decision rule *history dependent* if it depends on the past history of the system as represented by the sequence of previous states and actions. That is,  $d_t$  is a function of the history  $h_t = (s_1, a_1, \dots, s_{t-1}, a_{t-1}, s_t)$  where  $s_u$  and  $a_u$  denote the state and action of the system at decision epoch  $u$ . *As convention, when the history is  $h_t$ ,  $s_t$  denotes the state at decision epoch  $t$ .* The history  $h_t$  follows the recursion  $h_t = (h_{t-1}, a_{t-1}, s_t)$ . When using a history-dependent decision rule and observing  $h_t$ , the decision maker chooses actions from  $A_s$ . We let  $H_t$  denote the set of all histories  $h_t$ . Note that  $H_1 = S$ ,  $H_2 = S \times A \times S$  and  $H_n = S \times A \times S \times \dots \times S$ , and that  $H_t$  satisfies the recursion  $H_t = H_{t-1} \times A \times S$ . For sets  $C$  and  $D$ ,  $C \times D$  denotes the *Cartesian product* of  $C$  and  $D$ . That is,  $C \times D = \{(c, d): c \in C, d \in D\}$ . A deterministic history-dependent decision rule  $d_t$  maps  $H_t$  into  $A$ , subject to the restriction that  $d_t(h_t) \in A_{s_t}$ .

A *randomized* decision rule  $d_t$  specifies a probability distribution  $q_{d_t}(\cdot)$  on the set of actions. Randomized Markovian decision rules map the set of states into the set of probability distributions on the action space, that is  $d_t: S \rightarrow \mathcal{P}(A)$ , and randomized history-dependent decision rules according to  $d_t: H_t \rightarrow \mathcal{P}(A)$ . When Markovian,  $q_{d_t(s_t)}(\cdot) \in \mathcal{P}(A_{s_t})$ , and when history dependent,  $q_{d_t(h_t)}(\cdot) \in \mathcal{P}(A_{s_t})$  for all  $h_t \in H_t$ . A deterministic decision rule may be regarded as a special case of a randomized decision rule in which the probability distribution on the set of actions is degenerate, that is,  $q_{d_t(s_t)}(a) = 1$  or  $q_{d_t(h_t)}(a) = 1$  for some  $a \in A_{s_t}$ .

We classify decision rules as history dependent and randomized (HR), history dependent and deterministic (HD), Markovian and randomized (MR), or Markovian and deterministic (MD) depending on their degree of dependence on past information and on their method of action selection. We denote the set of decision rules at time  $t$  by  $D_t^K$ , where  $K$  designates a class of decision rules ( $K = \text{HR, HD, MR, MD}$ );  $D_t^K$  is called a *decision rule set*. A summary of this classification scheme appears in Table 2.1.1.

A fundamental question in Markov decision problem theory is

*For a given optimality criterion, under what conditions is it optimal to use a deterministic Markovian decision rule at each stage?*

The rewards and transition probabilities become functions on  $S$  or  $H_t$  after specifying decision rules. For  $d_t \in D_t^{\text{MD}}$ , the reward equals  $r_t(s, d_t(s))$  and the transition probability equals  $p_t(j|s, d_t(s))$  and for  $d_t \in D_t^{\text{HD}}$ , they equal  $r_t(s, d_t(h_t))$  and  $p_t(j|s, d_t(h_t))$  whenever  $h_t = (h_{t-1}, a_{t-1}, s)$ . If  $d_t$  is a randomized Markov

**Table 2.1.1 Classification of Decision Rules and Decision Rule Sets**

| History Dependence | Action Choice                               |                                                                     |
|--------------------|---------------------------------------------|---------------------------------------------------------------------|
|                    | Deterministic                               | Randomized                                                          |
| Markovian          | $d_t(s_t) \in A_{s_t}$<br>$D_t^{\text{MD}}$ | $q_{d_t(s_t)}(\cdot) \in \mathcal{P}(A_{s_t})$<br>$D_t^{\text{MR}}$ |
| History Dependent  | $d_t(h_t) \in A_{s_t}$<br>$D_t^{\text{HD}}$ | $q_{d_t(h_t)}(\cdot) \in \mathcal{P}(A_{s_t})$<br>$D_t^{\text{HR}}$ |

decision rule the *expected* reward satisfies

$$r_t(s, d_t(s)) = \sum_{a \in A_s} r_t(s, a) q_{d_t(s)}(a) \quad (2.1.3)$$

and the transition probability satisfies

$$p_t(j|s, d_t(s)) = \sum_{a \in A_s} p_t(j|s, a) q_{d_t(s)}(a). \quad (2.1.4)$$

Analogous constructions apply to randomized history-dependent rules.

### 2.1.5 Policies

A *policy*, *contingency plan*, *plan*, or *strategy* specifies the decision rule to be used at all decision epoch. It provides the decision maker with a prescription for action selection under any possible future system state or history. A policy  $\pi$  is a sequence of decision rules, i.e.,  $\pi = (d_1, d_2, \dots, d_{N-1})$  where  $d_t \in D_t^K$  for  $t = 1, 2, \dots, N-1$  for  $N \leq \infty$ , with  $K$  representing any of the above classes. Let  $\Pi^K$  denote the set of all policies of class  $K$  ( $K = \text{HR}, \text{HD}, \text{MR}, \text{MD}$ ); that is,  $\Pi^K = D_1^K \times D_2^K \times \dots \times D_{N-1}^K$ ,  $N \leq \infty$ .

We call a policy *stationary* if  $d_t = d$  for all  $t \in T$ . A stationary policy has the form  $\pi = (d, d, \dots)$ ; we denote it by  $d^\infty$ . We let  $\Pi^{\text{SD}}$  be the set of stationary deterministic policies and  $\Pi^{\text{SR}}$  be the set of stationary randomized policies. We sometimes refer to stationary deterministic policies as *pure* policies. Stationary policies are fundamental to the theory of infinite horizon Markov decision processes.

The relationship between the various classes of policies is as follows:  $\Pi^{\text{SD}} \subset \Pi^{\text{SR}} \subset \Pi^{\text{MR}} \subset \Pi^{\text{HR}}$ ,  $\Pi^{\text{SD}} \subset \Pi^{\text{MD}} \subset \Pi^{\text{MR}} \subset \Pi^{\text{HR}}$ , and  $\Pi^{\text{SD}} \subset \Pi^{\text{MD}} \subset \Pi^{\text{HD}} \subset \Pi^{\text{HR}}$ . Thus randomized, history-dependent policies are most general and stationary deterministic policies most specific.

### \*2.1.6 Induced Stochastic Processes, Conditional Probabilities, and Expectations

In this section we provide a formal model for the stochastic process generated by a Markov decision process. *To simplify presentation we assume discrete  $S$  and  $A$ .* Section 2.3.2 discusses extension of this construction to a more general model.

A probability model consists of three elements: a sample space  $\Omega$ , a  $\sigma$ -algebra of (Borel) measurable subsets of  $\Omega$ ,  $B(\Omega)$ , and a probability measure  $P$  on  $B(\Omega)$ . An elaborate discussion of these concepts is beyond the scope of this book but note that when  $\Omega$  is a *finite* set  $B(\Omega)$  equals the set of all subsets of  $\Omega$  and a probability measure is a probability mass function.

In a finite horizon Markov decision process, we choose

$$\Omega = S \times A \times S \times A \times \cdots \times A \times S = \{S \times A\}^{N-1} \times S,$$

and in an infinite horizon model,  $\Omega = \{S \times A\}^\infty$ . A typical element  $\omega \in \Omega$  consists of a sequence of states and actions, that is

$$\omega = (s_1, a_1, s_2, a_2, \dots, a_{N-1}, s_N),$$

and, in an infinite horizon model,

$$\omega = (s_1, a_1, s_2, a_2, \dots).$$

We refer to  $\omega$  as a *sample path*. Given a set  $C$ , let  $\{C\}^k$  denote the Cartesian product of  $C$  with itself  $k$  times. In finite horizon models, with horizon length  $N$ ,  $B(\Omega) = B(\{S \times A\}^{N-1} \times S)$  and in infinite horizon models,  $B(\Omega) = B(\{S \times A\}^\infty)$ .

We define the random variables  $X_t$  and  $Y_t$  which take values in  $S$  and  $A$ , respectively, by

$$X_t(\omega) = s_t \text{ and } Y_t(\omega) = a_t \quad (2.1.5)$$

for  $t = 1, 2, \dots, N$ ,  $N \leq \infty$ . This means that when the observed sequence of states and actions is  $\omega$ , the random variable  $X_t$  denotes the state at time  $t$ , and  $Y_t$  denotes the action at time  $t$ . Define the history process  $Z_t$  by

$$Z_1(\omega) = s_1 \quad \text{and} \quad Z_t(\omega) = (s_1, a_1, \dots, s_t) \text{ for } 1 \leq t \leq N; N \leq \infty.$$

Let the probability distribution  $P_1(\cdot)$  denote the *initial distribution* of the system state. In most applications, we assume degenerate  $P_1(\cdot)$ , that is,  $P_1(s_1) = 1$  for some  $s_1 \in S$ .

A randomized history-dependent policy  $\pi = (d_1, d_2, \dots, d_{N-1})$ ,  $N \leq \infty$ , induces a probability  $P^\pi$  on  $(\Omega, B(\Omega))$  through

$$P^\pi\{X_1 = s\} = P_1(s), \quad (2.1.6)$$

$$P^\pi\{Y_t = a | Z_t = h_t\} = q_{d_t(h_t)}(a), \quad (2.1.7)$$

$$P^\pi\{X_{t+1} = s | Z_t = (h_{t-1}, a_{t-1}, s_t), Y_t = a_t\} = p_t(s | s_t, a_t) \quad (2.1.8)$$

so that the probability of a sample path  $\omega = (s_1, a_1, s_2, \dots, s_N)$  is given by

$$\begin{aligned} P^\pi(s_1, a_1, s_2, \dots, s_N) &= P_1(s_1) q_{d_1(s_1)}(a_1) p_1(s_2 | s_1, a_1) q_{d_2(h_2)}(a_2) \\ &\quad \cdots q_{d_{N-1}(h_{N-1})}(a_{N-1}) p_{N-1}(s_N | s_{N-1}, a_{N-1}). \end{aligned} \quad (2.1.9)$$

For  $\pi$  in  $\Pi^{\text{HD}}$  or  $\Pi^{\text{MD}}$ , (2.1.9) simplifies to

$$P^\pi(s_1, a_1, s_2, \dots, s_N) = P_1(s_1)p_1(s_2|s_1, a_1) \cdots p(s_N|s_{N-1}, a_{N-1}).$$

Note that the policy determines  $P^\pi$  explicitly through (2.1.7) and implicitly through (2.1.8). The Markovian nature of the model is reflected through (2.1.8).

For computation, we require conditional probabilities of the process from  $t$  onward conditional on the history at time  $t$ . Under the discreteness assumptions, we compute these probabilities as follows:

$$P^\pi(a_t, s_{t+1}, \dots, s_N | s_1, a_1, \dots, s_t) = \frac{P^\pi(s_1, a_1, \dots, s_N)}{P^\pi(s_1, a_1, \dots, s_t)} \quad (2.1.10)$$

provided the quantity in the denominator is nonzero; otherwise the probability equals zero. We evaluate the denominator of (2.1.10) by summing  $P^\pi(s_1, a_1, \dots, s_N)$  over all sample paths which equal  $s_1, a_1, \dots, s_t$  in the first  $2t - 1$  components. This reduces to

$$\begin{aligned} P^\pi(s_1, a_1, s_2, \dots, s_t) \\ = P_1(s_1)q_{d_1(s_1)}(a_1)p_1(s_2|s_1, a_1)q_{d_2(h_2)}(a_2) \cdots q_{d_{t-1}(h_{t-1})}(a_{t-1})p_{t-1}(s_t|s_{t-1}, a_{t-1}) \end{aligned}$$

so that upon substitution (2.1.10) simplifies to

$$\begin{aligned} P^\pi(a_t, s_{t+1}, \dots, s_N | s_1, a_1, \dots, s_t) \\ = q_{d_t(h_t)}(a_t)p(s_{t+1}|s_t, a_t) \cdots q_{d_{N-1}(h_{N-1})}(a_{N-1})p(s_N|s_{N-1}, a_{N-1}). \end{aligned} \quad (2.1.11)$$

By similar arguments,

$$\begin{aligned} P^\pi(s_{t+1}, a_{t+1}, \dots, s_N | s_1, \dots, s_t, a_t) \\ = p(s_{t+1}|s_t, a_t)q_{d_{t+1}(h_{t+1})}(a_{t+1}) \cdots q_{d_{N-1}(h_{N-1})}(a_{N-1})p(s_N|s_{N-1}, a_{N-1}) \end{aligned} \quad (2.1.12)$$

Observe that  $P^\pi\{X_{t+1} = s | Z_t = (h_{t-1}, a_{t-1}, s_t), Y_t = a_t\}$  is a function of  $Z_t$  only through  $X_t = s_t$ ; however, because of the dependence on the history through (2.1.7), the process need not be a Markov chain. For Markovian  $\pi$  (i.e., for  $\pi \in \Pi^{\text{MD}}$  or  $\pi \in \Pi^{\text{MR}}$ ),  $d_t$  depends on the history only through the current state of the process, so that (2.1.7) becomes

$$P^\pi\{Y_t = a | Z_t = (h_{t-1}, a_{t-1}, s_t)\} = P\{Y_t = a | X_t = s_t\} = q_{d_t(s_t)}(a).$$

Consequently,

$$P^\pi(a_t, s_{t+1}, \dots, s_N | s_1, a_1, \dots, s_t) = P^\pi(a_t, s_{t+1}, \dots, s_N | s_t),$$

so that the induced stochastic processes  $\{X_t; t \in T\}$  is a discrete time Markov chain.

When  $\pi$  is Markovian, we refer to the bivariate stochastic process  $\{(X_t, r_t(X_t, Y_t)); t \in T\}$  as a *Markov reward process* (a Markov chain together with a real-valued function defined on its state space). This process represents the sequence of system states and stream of rewards received by the decision maker when using policy  $\pi$ .

Let  $W$  denote a (real-valued) random variable defined on  $\{\Omega, \mathcal{B}(\Omega), P^\pi\}$ . When  $n$  is finite we define the *expected value* of  $W$  with respect to policy  $\pi$  by

$$E^\pi\{W\} = \sum_{\omega \in \Omega} W(\omega) P^\pi\{\omega\} = \sum_{w \in R^1} w P^\pi\{\omega: W(\omega) = w\}, \quad (2.1.13)$$

where, as above,  $\omega = (s_1, a_1, s_2, a_2, \dots, a_{N-1}, s_N)$ . When  $N$  is infinite integrals replace sums in (2.1.13). Usually we will evaluate the expectation of quantities of the form

$$W(s_1, a_1, \dots, s_N) = \sum_{t=1}^{N-1} r_t(s_t, a_t) + r_N(s_N). \quad (2.1.14)$$

Expression (2.1.13) does not provide an efficient procedure for computing such expectations and ignores the dynamic aspects of the problem. When  $\{X_t; t \in T\}$  is a Markov chain, such an expectation can be calculated by standard matrix methods (Appendix A); however, induction methods are more suitable for history-dependent  $\pi$ .

Suppose we observe  $h_t = (s_1, a_1, \dots, s_t)$ , and  $W$  is a function of  $s_t, a_t, \dots, s_N$ . Then

$$E_{h_t}^\pi\{W(X_t, Y_t, \dots, X_N)\} = \sum W(s_t, a_t, \dots, s_N) P^\pi(a_t, s_{t+1}, \dots, s_N | s_1, a_1, \dots, s_t), \quad (2.1.15)$$

where the summation ranges over  $(a_t, s_{t+1}, \dots, s_N) \in \mathcal{A} \times S \times \dots \times S$  and the conditional probability  $P^\pi\{\cdot | s_1, a_1, \dots, s_t\}$  is evaluated according to (2.1.11). For Markovian  $\pi$ ,

$$E_{s_t}^\pi\{W(X_t, Y_t, \dots, X_N)\} \equiv \sum W(s_t, a_t, \dots, s_N) P^\pi(a_t, s_{t+1}, \dots, s_N | s_t). \quad (2.1.16)$$

Note that  $E^\pi\{W\} = \sum_{s \in S} P_1(s) E_s^\pi\{W\}$  for any  $\pi \in \Pi^{\text{HR}}$ .

## 2.2 A ONE-PERIOD MARKOV DECISION PROBLEM

We now describe a one-period Markov decision problem (MDP) in detail to illustrate notation, indicate the flavor of analyses to follow, and provide a focus for discussing some technical issues in Sec. 2.3. We assume *finite*  $S$  and  $A_s$  for each  $s \in S$ .

In a one-period model  $N = 2$  so that  $T = \{1, 2\}$ . We assume that whenever the system occupies state  $s'$  at the end of period 1, the decision maker receives a terminal reward  $v(s')$ , where  $v$  is a specified real-valued function on  $S$ . Suppose the decision maker finds the system in state  $s$  at the start of stage 1 and his objective is to choose an action  $a \in A_s$  to maximize the sum of the immediate reward,  $r_1(s, a)$ , and the expected terminal reward. If the decision maker chooses a deterministic policy

$\pi = (d_1)$ , which selects action  $a' \in A_s$  at decision epoch 1, the total expected reward equals

$$r_1(s, a') + E_s^\pi\{v(X_2)\} = r_1(s, a') + \sum_{j \in S} p_1(j|s, a')v(j). \quad (2.2.1)$$

To achieve his objective, the decision maker will choose an  $a' \in A_s$  to make the expression in (2.2.1) as large as possible. To find such an action, evaluate (2.2.1) for each  $a' \in A_s$  and select any action that achieves this maximum. Denote such an action by  $a^*$  and the maximum value of the expected total reward by

$$\max_{a' \in A_s} \left\{ r_1(s, a') + \sum_{j \in S} p_1(j|s, a')v(j) \right\}. \quad (2.2.2)$$

Any maximizing action satisfies

$$r_1(s, a^*) + \sum_{j \in S} p_1(j|s, a^*)v(j) = \max_{a' \in A_s} \left\{ r_1(s, a') + \sum_{j \in S} p_1(j|s, a')v(j) \right\}. \quad (2.2.3)$$

The following additional notation simplifies describing maximizing actions. Let  $X$  be an arbitrary set, and  $g(x)$  a real-valued function on  $X$ . Define

$$\arg \max_{x \in X} g(x) \equiv \{x' \in X: g(x') \geq g(x) \text{ for all } x \in X\}.$$

Using this notation we see that any  $a_s^*$  which satisfies (2.2.3) may be represented by

$$a_s^* \in \arg \max_{a' \in A_s} \left\{ r_1(s, a') + \sum_{j \in S} p_1(j|s, a')v(j) \right\}. \quad (2.2.4)$$

Because  $A_s$  is finite, at least one maximizing action exists. If it is unique, it is the optimal decision in state  $s$ . If not, choosing any maximizing action yields the maximum expected reward.

We stress that the operation “max” results in a value and “arg max” in a set. The following simple example illustrates this point.

**Example 2.2.1.** Suppose  $X = \{a, b, c, d\}$ ,  $g(a) = 5$ ,  $g(b) = 7$ ,  $g(c) = 3$ , and  $g(d) = 7$ . Then

$$\max_{x \in X} \{g(x)\} = 7 \quad \text{and} \quad \arg \max_{x \in X} \{g(x)\} = \{b, d\}.$$

Note that when the maximum of a function is not attained, we instead seek a supremum. We write this as “sup.” In this case the “arg max” is the empty set.

**Example 2.2.2.** Suppose  $X = \{1, 2, \dots\}$  and  $g(x) = 1 - 1/x$ . Then  $\max_{x \in X} \{g(x)\}$  does not exist,  $\arg \max_{x \in X} \{g(x)\} = \emptyset$  but  $\sup_{x \in X} \{g(x)\} = 1$ . Further, for any  $\varepsilon > 0$ , there exists a subset of  $X$ , say  $X_\varepsilon$ , for which  $g(x') > 1 - \varepsilon$  for all  $x' \in X_\varepsilon$ .

It is natural to ask whether the decision maker can obtain a larger reward by using a random mechanism to select actions in state  $s$ . If action  $a \in A_s$  is selected with probability  $q(a)$ , the expected total reward equals

$$\sum_{a \in A_s} q(a) \left[ r_1(s, a) + \sum_{j \in S} p_1(j|s, a) v(j) \right],$$

where  $\sum_{a \in A_s} q(a) = 1$  and  $q(a) \geq 0$  for  $a \in A_s$ .

Since

$$\begin{aligned} & \max_{q \in \mathcal{P}(A_s)} \left\{ \sum_{a \in A_s} q(a) \left[ r_1(s, a) + \sum_{j \in S} p_1(j|s, a) v(j) \right] \right\} \\ &= \max_{a' \in A_s} \left\{ r_1(s, a') + \sum_{j \in S} p_1(j|s, a') v(j) \right\}, \end{aligned}$$

we cannot obtain a larger expected reward in state  $s$  by randomization. Note, however, that any decision rule which randomizes over actions satisfying (2.2.4) obtains the maximum expected reward.

If  $s$  is not known prior to determining the optimal action, for example, if it is determined by an initial distribution  $P_1$ , then the decision maker must choose an action for each possible  $s \in S$ , that is, he must specify a decision rule. To find an optimal decision rule, he maximizes (2.2.2) independently for each  $s \in S$ . Therefore any decision rule  $d^*(s) = a_s^*$  where  $a_s^*$  satisfies (2.2.4), would yield the largest total expected reward. In this case, the deterministic policy  $\pi^* = (d^*)$  maximizes the total expected reward within the class of all Markov randomized policies. Since  $h_1 = s_1$  in this one-period model, we need not distinguish between Markov and history-dependent policies.

Thus we have provided a method for finding a  $\pi^* = \Pi^{\text{MD}}$  for which

$$E^{\pi^*} \{r_1(X_1, Y_1) + v(X_2)\} \geq E^{\pi} \{r_1(X_1, Y_1) + v(X_2)\}$$

for all  $\pi \in \Pi^{\text{HR}}$ . Furthermore, the same choice of  $\pi^*$  satisfies

$$E_s^{\pi^*} \{r_1(X_1, Y_1) + v(X_2)\} \geq E_s^{\pi} \{r_1(X_1, Y_1) + v(X_2)\}$$

for all  $\pi \in \Pi^{\text{HR}}$  and each  $s \in S$ .

## \*2.3 TECHNICAL CONSIDERATIONS

This section discusses some technical issues and provides a more general formulation of a Markov decision process model.

### 2.3.1 The Role of Model Assumptions

The model in Sec. 2.2 provides a focus for discussing the role of assumptions about  $S$  and  $A_s$ . Suppose first that  $S$  is a Borel measurable subset of Euclidean space. Let  $\pi = (d_1) \in \Pi^{\text{MD}}$  and suppose  $d_1(s) = a$ . In this case (2.2.1) becomes

$$r_1(s, a) + E_s^\pi\{v(X_2)\} = r_1(s, a) + \int_S v(u) p_1(u|s, a) du \quad (2.3.1)$$

if the density  $p_1(u|s, a)$  exists. In greater generality the integral in (2.3.1) may be represented by the Lebesgue-Stieltjes integral  $\int_S v(u) p_1(du|s, a)$ . For the above expressions to be meaningful requires that  $v(\cdot) p_1(\cdot|s, a)$  be Lebesgue integrable, or  $v(\cdot)$  be Lebesgue-Stieltjes integrable with respect to  $p_1(du|s, a)$  for each  $s \in S$  and  $a \in A_s$ .

We now consider (2.3.1) as a function on  $S$ . Upon substitution of  $d_1$ , it becomes

$$r_1(s, d_1(s)) + \int_S v(u) p_1(u|s, d_1(s)) du.$$

For analyzing multi-period models we require this to be a measurable and integrable function on  $S$ . This necessitates imposing assumptions on  $r_1(s, \cdot)$ ,  $p_1(u|s, \cdot)$ , and  $d_1(\cdot)$ . At a minimum we require  $d_1(s)$  to be a measurable function from  $S$  to  $A$ , which means that we must restrict the set of admissible decision rules to include only measurable functions. To identify measurable functions requires a topology on  $A$ .

In Sec. 2.2, we constructed an optimal decision rule  $d^*$  by setting  $d^*(s)$  equal to  $a_s^*$  for each  $s \in S$  where  $a_s^*$  satisfied (2.2.4). That is, we solved a separate maximization problem for each  $s \in S$ . For discrete  $S$ ,  $d^* \in D_1$ , since  $D_1$  consisted of the set of all functions from  $S$  to  $A$ , but, as discussed above, we require that  $d^*$  be measurable. To ensure this we need a *selection theorem* which ensures that  $\times_{s \in S} A_s^*$  contains a measurable function from  $S$  to  $A$  where

$$A_s^* \equiv \arg \max_{a \in A_s} \left\{ r_1(s, a) + \int_S v(u) p_1(u|s, a) du \right\}.$$

Such a result does not hold without further assumptions on  $r_1$ ,  $p_1$ , and  $A_s$ , so even in a one-period model an optimal policy need not exist (Sect. 6.3.5).

Assumptions about the form of the action sets affect the maximization in (2.2.3). When  $A_s$  is finite, a maximizing action always exists. If  $A_s$  is a compact subset of Euclidean space and the expression inside the brackets in (2.2.3) is upper semicontinuous (Appendix B), then the maximum is attained, but when  $A_s$  is countable the maximum need not exist. In this latter case, we replace “max” by “sup” (supremum) and seek actions or decision rules that yield values within some prespecified small  $\varepsilon > 0$  of the supremum.

### 2.3.2 The Borel Model

A formulation in greater generality follows. We refer to  $\{X, B(X)\}$  as a *Borel space* or *Polish space* if  $X$  is a Borel subset of a complete separable metric space and  $B(X)$  its family of Borel subsets. For Borel spaces  $\{X, B(X)\}$  and  $\{Y, B(Y)\}$ , we refer to  $q$



as a *conditional probability on  $Y$  given  $X$*  or a *transition kernel* if for  $x \in X$ ,  $q(\cdot|x)$  is a probability measure on  $B(Y)$  and for  $G \in B(Y)$ ,  $q(G|\cdot)$  is a Borel measurable function from  $X$  into  $[0, 1]$ .

In this generality a Markov decision process consists of

1. A Borel space  $\{S, B(S)\}$ .
2. A Borel space  $(A, B(A))$  and a collection of sets  $A_s$ , for which  $A_s \in B(A)$  for each  $s \in S$ . Let  $B(A_s)$  denote the induced collection of Borel subsets of  $A_s$ . Furthermore, we require existence of a measurable function  $\delta$  mapping  $S$  into  $A$  with  $\delta(s) \in A_s$  for each  $s \in S$ .
3. A family  $\mathcal{P}(A_s)$  of probability measures on  $B(A_s)$  for each  $s \in S$ .
4. Real-valued reward functions  $r_t(s, a)$  which for each  $t \in T$  satisfy
  - a.  $r_t(\cdot, \cdot)$  is measurable with respect to  $B(S \times A_s)$ ,
  - b.  $r_t(s, \cdot)$  is integrable with respect to all  $q \in \mathcal{P}(A_s)$  for all  $s \in S$ .
5. Conditional probabilities  $p_t(\cdot|s, a)$  which for  $t \in T$  satisfy
  - a.  $p_t(G|\cdot, \cdot)$  is measurable with respect to  $B(S \times A_s)$  for  $G \in B(S)$ ,
  - b.  $p_t(G|s, \cdot)$  is integrable with respect to each  $q \in \mathcal{P}(A_s)$  for each  $s \in S$  and  $G \in B(S)$ .

We now define sets of decision rules. As before, let  $H_t = S \times A \times \cdots \times S$  denote the set of histories up to time  $t$  and now let  $B(H_t)$  denote the derived family of Borel subsets on  $H_t$ . Then

$$\begin{aligned} D_t^{\text{HD}} &= \{\delta: H_t \rightarrow A: \delta \text{ is measurable and } \delta(h_{t-1}, a_{t-1}, s) \in A_s \text{ for all } s \in S\} \\ D_t^{\text{MD}} &= \{\delta: S \rightarrow A: \delta \text{ is measurable and } \delta(s) \in A_s \text{ for all } s \in S\}, \\ D_t^{\text{HR}} &= \{\text{conditional probabilities } q \text{ on } A \text{ given } H_t: q(A_s|h_{t-1}, a_{t-1}, s) = 1 \text{ for} \\ &\quad \text{each } s \in S\}, \\ D_t^{\text{MR}} &= \{\text{conditional probabilities } q \text{ on } A \text{ given } S: q(A_s|s) = 1 \\ &\quad \text{for each } s \in S\}. \end{aligned}$$

The policy sets  $\Pi^K$ ,  $K = \text{HR, MR, HD, MD}$ , are the Cartesian products of the corresponding decision sets.

Given  $\delta \in D_t^{\text{HR}}$  for  $h_t = (h_{t-1}, a_t, s_t) = H_t$  we generalize (2.1.3) to

$$r_t(s_t, \delta(h_t)) = \int_{A_{s_t}} r_t(s_t, a) q_{\delta(h_t)}(da|h_t). \quad (2.3.2)$$

Since  $r_t$  is measurable in  $S$  and  $q_{\delta(h_t)}$  is a conditional probability,  $r_t(\cdot, \delta(\cdot))$  is a measurable function on  $H_t$ . By a similar argument, the above assumptions imply that  $p_t(G|\cdot, \delta(\cdot))$  is a measurable function on  $H_t$  for every Borel subset  $G$  of  $S$ . Likewise for  $\delta \in D_t^K$ ,  $r_t(\cdot, \delta(\cdot))$  and  $p_t(G|\cdot, \delta(\cdot))$  are measurable functions on  $S$  for  $K = \text{MR, MD, HD}$ .

The construction of induced stochastic processes in Section 2.1.6 applies with the following modifications

1.  $B(S)$  and  $B(A)$  are the respective Borel sets of  $S$  and  $A$ .
2.  $\Omega$  is endowed with the product  $\sigma$ -algebra.
3. The coordinate mappings  $X_i$  and  $Y_i$  in (2.1.5) are measurable mappings from  $\Omega$  to  $S$  and  $A$ , respectively.
4. The expressions in (2.1.7) and (2.1.8) are conditional probabilities.
5. For  $\pi \in \Pi^{\text{HR}}$  the probability measure  $P^\pi$  satisfies

$$\begin{aligned} P^\pi(ds_1 da_1 ds_2 \cdots ds_n) \\ = P_1(ds_1) q_{d_1(s_1)}(da_1) p(ds_2 | s_1, a_2) \cdots p(ds_N | s_{N-1}, a_{N-1}), \end{aligned}$$

where its existence and uniqueness is guaranteed by a result of Ionescu-Tulcea (Hinderer, 1970; p. 149, Bertsekas and Shreve, 1978, pp. 140–144).

We regard the theory of Markov decision problems as sufficiently rich and complex without having to confront these additional mathematical subtleties. Several excellent books including Hinderer (1970), Bertsekas and Shreve (1978), Dynkin and Yushkevich (1979), and Hernandez-Lerma (1989) develop results at this level of generality. Our main focus will be models in which  $S$  and  $A_s$  are finite or countable. Results which hold in greater generality will be noted.

## BIBLIOGRAPHIC REMARKS

Bellman (1954) coined the expression “Markov decision process.” His early papers, many of which are summarized in his classic book (1957), provide the basis for the Markov decision process model. In this book, he introduces the concepts of states, actions, and transitions, and develops functional equations for finding optimal policies. Karlin (1955) elaborates on these concepts and provides a more abstract mathematical basis for these models. Shapley (1953) introduces and analyzes a two-person stochastic game which is very close in spirit to a Markov decision process. Howard’s monograph (1960) played a major role in establishing the importance of the Markov decision process model.

Rigorous foundations for the Markov decision model were established by Dubins and Savage (1965) and Blackwell (1965). Blackwell’s classic paper (1965) provides the model we analyze in this book. It defines state and action sets and rewards and transition probabilities in considerable generality, distinguishes randomized, deterministic (degenerate), Markov, and history-dependent policies and emphasizes the importance of stationary policies.

*How to Gamble if You Must* by Dubins and Savage (1965), which first appeared in mimeograph form in 1960, analyzes a gambling model which possesses a similar mathematical structure to the Markov decision model. Their development raises and addresses many of the mathematical subtleties that are still the subject of research efforts. Their terminology is quite different, so, when referring to their book, note the

equivalence between fortunes and states, gambling houses and action sets, and gambles and transition probabilities. The gambling model does not explicitly contain a reward function, and the system state represents the gambler's current wealth.

Hinderer (1970) addresses foundational questions for both countable and arbitrary state spaces models. His model is more general than ours in that rewards and transition probabilities may depend on the entire history of the process. The books by Bertsekas and Shreve (1978) and Dynkin and Yushkevich (1979) provide a detailed analysis of many of the deeper measurability questions. These two books also provide considerable insight into the underlying probabilistic structure of the Markov decision process model.

Fleming and Rishel (1975) and Bertsekas (1987) provide nice treatments of dynamic optimization problems formulated in terms of stochastic control models. Fleming and Rishel's focus is on continuous time problems with continuous state spaces while Bertsekas presents results for discrete time problems with continuous state spaces. We discuss the relationship between these models and the Markov decision process model in Sec. 3.5.

Royden (1963) is a suitable reference for the topological foundations of this chapter and Breiman (1968) and Lamperti (1966) for the more advanced probabilistic concepts.

## PROBLEMS

- 2.1. Derive  $r_t(s, a)$  for a discrete-time Markov decision process in which a reward  $g_t(s, a)$  arrives in a lump sum at a random point  $\tau$  between two decision epochs according to a probability distribution  $q_t(s, a)$ .
- 2.2. Consider the discrete-time Markov decision process model of this chapter but suppose instead that decisions are made at every second period and are used for two consecutive periods. That is, if the decision maker chooses decision rule  $d_t$  at time  $t$ , he will use the identical decision rule at time  $t + 1$ . The next decision epoch is  $t + 2$ .  
Reformulate this as a Markov decision process model. Clearly identify the reward and transition functions. Be sure to account for the possibility of a randomized decision rule.
- 2.3. Suppose a discrete-time Markov chain evolves according to a fixed transition law  $p_t(s'|s)$  and it generates rewards  $r_t(s)$  if the system occupies state  $s$  at time  $t$ . At any stage, the decision maker may either let the system evolve uninterrupted or instead intervene and choose an action from a state-dependent set  $A_s$  which determines a one-period transition law  $p_t(s'|s, a)$  and reward  $r_t(s, a)$ .
  - a. Formulate this as a Markov decision process. Clearly identify the action sets, rewards, and transition probabilities.
  - b. Explicitly provide expression (2.2.2) for this model.
- 2.4. Consider a finite horizon Markov decision problem in which the only reward is received at termination. Define the reward functions for such a problem.

- 2.5.** Consider a one-period model with  $S = \{s_1, s_2\}$ ,  $A_{s_1} = \{a_{1,1}, a_{1,2}\}$  and  $A_{s_2} = \{a_{2,1}, a_{2,2}\}$ ;  $r_1(s_1, a_{1,1}) = 5$ ,  $r_1(s_1, a_{1,2}) = 10$ ,  $r_1(s_2, a_{2,1}) = -1$ , and  $r_1(s_2, a_{2,2}) = 2$ ; and  $p_1(s_1|s_1, a_{1,1}) = p_1(s_2|s_1, a_{1,1}) = 0.5$ ,  $p_1(s_1|s_1, a_{1,2}) = 0$ ,  $p_1(s_2|s_1, a_{1,2}) = 1$ ,  $p_1(s_1|s_2, a_{2,1}) = 0.8$ ,  $p_1(s_2|s_2, a_{2,1}) = 0.2$ ,  $p_1(s_1|s_2, a_{2,2}) = 0.1$ , and  $p_1(s_2|s_2, a_{2,2}) = 0.9$ .
- Find the deterministic policy that maximizes the total one-period expected reward provided that the terminal reward  $v(s)$  is identically 0.
  - Find the deterministic policy that maximizes the total one-period expected reward provided that the terminal reward  $v(s_1) = d$  and  $v(s_2) = e$ . Investigate the sensitivity of this policy to the values of  $d$  and  $e$ .
  - Compute  $r_1(s, d_1(s))$  and  $p_1(j|s, d_1(s))$  for the randomized decision rule  $d_1$  which in state  $s_1$  chooses action  $a_{1,1}$  with probability  $q$  and action  $a_{1,2}$  with probability  $1 - q$ , and in state  $s_2$  uses action  $a_{2,1}$  with certainty.
  - Find the randomized policy that maximizes the total one-period expected reward provided that the terminal reward  $v(s_1) = 5$  and  $v(s_2) = -5$ .
- 2.6.** Consider the model in Problem 2.5. Suppose the initial state is determined by the distribution  $P_1(s_1) = P_1(s_2) = 0.5$ . Let  $\pi = (d_1)$  be the randomized policy which in  $s_1$  uses action  $a_{1,1}$  with probability 0.6 and in  $s_2$  uses action  $a_{2,1}$  with probability 0.3. Let  $v(s_1) = 0$  and  $v(s_2) = 1$ .
- Compute  $P^\pi$  as defined by (2.1.9) for all possible sample paths.
  - Compute the expected total reward using this policy by using the representation for  $W$  in (2.1.14) and then computing the expectation using each formula in (2.1.13). Compare these to values computed using (2.2.1).
- 2.7.** Consider the model in Problem 2.5. Reformulate it so that the set of actions is *not* state dependent. To do this, let  $A = A_{s_1} \cup A_{s_2}$  and redefine the rewards and transition probabilities. Note that this necessitates adding many superfluous components.
- 2.8.** Prove for a one-period Markov decision problem that there always exists a deterministic policy with a reward at least as great as that of any randomized policy.
- 2.9.** Suppose  $P^\pi$  has been given for each sample path of a deterministic Markov policy  $\pi$  in a two-period finite state and action Markov decision model. Find conditions under which you can recover the transition probabilities of the model and show how to compute them.

## CHAPTER 3

# Examples

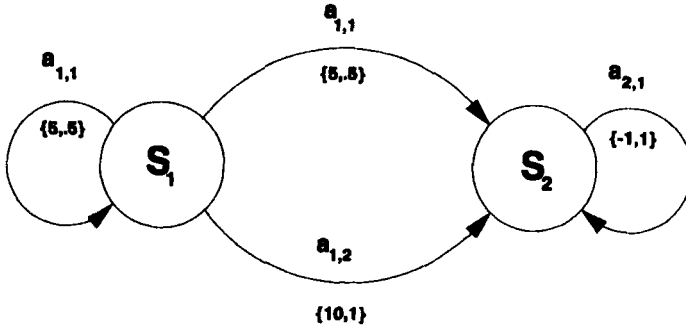
In Chap. 1, we described some significant applications of Markov decision process models. In this chapter we provide further applications and, through them, illustrate notation and concepts from Chap. 2. With the exception of Secs. 3.1 and 3.2, in which the models are quite specific and used to illustrate many points throughout this book, we have chosen to present Markov decision process formulations for classes of models that have wide applicability. We delay discussing optimality criteria until Chaps. 4 and 5 because in most cases, especially those which have an economic basis, they do not influence the identification of states and actions. An exception is the “secretary problem” of Sec. 3.4.3. To place models in an optimization context, assume for now that the decision maker seeks a policy which maximizes the expected total reward.

### 3.1 A TWO-STATE MARKOV DECISION PROCESS

The following simple abstract example illustrates the basic components of a Markov decision process. We refer to it often in later chapters.

*We assume stationary rewards and transition probabilities; that is the rewards and transition probabilities are the same at every decision epoch.* At each decision epoch the system occupies either state  $s_1$  or  $s_2$ . In state  $s_1$ , the decision maker chooses either action  $a_{1,1}$  or action  $a_{1,2}$ ; in state  $s_2$ , only action  $a_{2,1}$  is available. Choosing action  $a_{1,1}$  in  $s_1$  provides the decision maker with an immediate reward of five units, and at the next decision epoch the system is in state  $s_1$  with probability 0.5 and state  $s_2$  with probability 0.5. If instead he chooses action  $a_{1,2}$  in state  $s_1$ , he receives an immediate reward of ten units and at the next decision epoch the system moves to state  $s_2$  with probability 1. In state  $s_2$ , the decision maker must choose action  $a_{2,1}$ . As a consequence of this choice, the decision maker incurs a cost of one unit and the system occupies state  $s_2$  at the next decision epoch with certainty.

Figure 3.1.1, which illustrates this specific model, provides a convenient symbolic representation for any *stationary* model. In it, circles represent states and directed arcs represent possible transitions corresponding to actions. The first expression in brackets below the arc is the reward achieved when that action is selected and a transition between states connected by that arc occurs. The second quantity in brackets is the probability that the transition occurs. For example, if action  $a_{1,1}$  is selected, the process travels the arc between  $s_1$  and  $s_2$  with probability 0.5 and, when



**Figure 3.1.1** Symbolic representation of the two-state Markov decision process in this section.

it does so, the decision maker receives a reward of five units. Note that this representation allows the reward to depend on the subsequent state as well as the present state.

A formal description of this problem in terms of the notation of Chap. 2 follows.

**Decision Epochs:**

$$T = \{1, 2, \dots, N\}, \quad N \leq \infty.$$

**States:**

$$S = \{s_1, s_2\}.$$

**Actions:**

$$A_{s_1} = \{a_{1,1}, a_{1,2}\}, \quad A_{s_2} = \{a_{2,1}\}.$$

**Rewards:**

$$\begin{array}{lll} r_t(s_1, a_{1,1}) = 5 & r_t(s_1, a_{1,2}) = 10 & r_t(s_2, a_{2,1}) = -1 \\ r_N(s_1) = 0 & r_N(s_2) = 0 & \text{if } N < \infty \end{array}$$

**Transition Probabilities:**

$$\begin{array}{ll} p_t(s_1|s_1, a_{1,1}) = 0.5, & p_t(s_2|s_1, a_{1,1}) = 0.5, \\ p_t(s_1|s_1, a_{1,2}) = 0, & p_t(s_2|s_1, a_{1,2}) = 1, \\ p_t(s_1|s_2, a_{2,1}) = 0, & p_t(s_2|s_2, a_{2,1}) = 1. \end{array}$$

Suppose instead that the quantities below the arcs corresponding to action  $a_{1,1}$  were  $\{3, 0.5\}$  and  $\{7, 0.5\}$ . Then using the notation of (2.1.1)

$$r_t(s_1, a_{1,1}, s_1) = 3 \quad \text{and} \quad r_t(s_1, a_{1,1}, s_2) = 7.$$

In this case, the reward received when action  $a_{1,1}$  is chosen in state  $s_1$  depends on the state at the next decision epoch; if it is  $s_1$ , the reward is 3, and if it is  $s_2$ , the reward is

7. As before, the expected reward  $r_t(s, a)$  is 5. This is because from (2.1.1)

$$\begin{aligned} r_t(s_1, a_{1,1}) &= r_t(s_1, a_{1,1}, s_1)p_t(s_1|s_1, a_{1,1}) + r_t(s_1, a_{1,1}, s_2)p_t(s_2|s_1, a_{1,1}) \\ &= 3 \times 0.5 + 7 \times 0.5 = 5. \end{aligned}$$

We now provide examples of the policies distinguished in Sec. 2.1.5. We assume that  $N = 3$ , that is, decisions are made at decision epochs 1 and 2. These policies may be represented by  $\pi^K = (d_1^K, d_2^K)$  with  $K = \text{MD, MR, HD, or HR}$ .

#### A deterministic Markov policy $\pi^{\text{MD}}$ :

**Decision epoch 1:**

$$d_1^{\text{MD}}(s_1) = a_{1,1}, \quad d_1^{\text{MD}}(s_2) = a_{2,1}.$$

**Decision epoch 2:**

$$d_2^{\text{MD}}(s_1) = a_{1,2}, \quad d_2^{\text{MD}}(s_2) = a_{2,1}.$$

#### A randomized Markov policy $\pi^{\text{MR}}$ ,

**Decision epoch 1:**

$$\begin{aligned} q_{d_1^{\text{MR}}(s_1)}(a_{1,1}) &= 0.7, & q_{d_1^{\text{MR}}(s_1)}(a_{1,2}) &= 0.3 \\ q_{d_1^{\text{MR}}(s_2)}(a_{2,1}) &= 1. \end{aligned}$$

**Decision epoch 2:**

$$\begin{aligned} q_{d_2^{\text{MR}}(s_1)}(a_{1,1}) &= 0.4, & q_{d_2^{\text{MR}}(s_1)}(a_{1,2}) &= 0.6 \\ q_{d_2^{\text{MR}}(s_2)}(a_{2,1}) &= 1. \end{aligned}$$

Because of the transition structure of the above model, the set of history-dependent policies equals the set of Markov policies. To see this, note that if the system is in state  $s_1$  at any decision epoch, then the history must have been  $(s_1, a_{1,1}, s_1, a_{1,1}, \dots, s_1, a_{1,1})$  so that knowing the history would be redundant to the decision maker. In state  $s_2$ , the situation is different. Many possible histories could result in the process being in state  $s_2$ , but in  $s_2$ , there is only one available action  $a_{2,1}$ , so that the decision maker must choose it regardless of the history.

We modify the model to illustrate history-dependent policies (Figure 3.1.2). In  $s_1$ , we add an action  $a_{1,3}$  which causes the system to remain in state  $s_1$  with certainty, that is,  $p_t(s_1|s_1, a_{1,3}) = 1$ . We assume also that  $r_t(s_1, a_{1,3}) = 0$ . Note that  $\mathcal{A}_{s_1} = \{a_{1,1}, a_{1,2}, a_{1,3}\}$ . We now provide an example of a history-dependent policy for this modified model.

#### A deterministic history-dependent-policy $\pi^{\text{HD}}$ .

**Decision epoch 1:**

$$d_1^{\text{HD}}(s_1) = a_{1,1}, \quad d_1^{\text{HD}}(s_2) = a_{2,1}.$$

**Decision epoch 2:**

| Decision epoch 1 |                              |                              |
|------------------|------------------------------|------------------------------|
| $(s, a)$         | $d_2^{\text{HD}}(s, a, s_1)$ | $d_2^{\text{HD}}(s, a, s_2)$ |
| $(s_1, a_{1,1})$ | $a_{1,3}$                    | $a_{2,1}$                    |
| $(s_1, a_{1,2})$ | infeasible                   | $a_{2,1}$                    |
| $(s_1, a_{1,3})$ | $a_{1,1}$                    | infeasible                   |
| $(s_2, a_{2,1})$ | infeasible                   | $a_{2,1}$                    |

**A randomized history-dependent policy  $\pi^{\text{HR}}$ :****Decision epoch 1:**

$$q_{d_1^{\text{HR}}(s_1)}(a_{1,1}) = 0.6, \quad q_{d_1^{\text{HR}}(s_1)}(a_{1,2}) = 0.3, \quad q_{d_1^{\text{HR}}(s_1)}(a_{1,3}) = 0.1,$$

$$q_{d_1^{\text{HR}}(s_2)}(a_{2,1}) = 1.$$

**Decision epoch 2:**

When the system is in state  $s_1$  at decision epoch 2, probabilities are chosen according to the following probability distribution:

| Decision epoch 1<br>$(s, a)$ | $q_{d_2^{\text{HR}}(s,a,s_1)}(a)$ |               |               |
|------------------------------|-----------------------------------|---------------|---------------|
|                              | $a = a_{1,1}$                     | $a = a_{1,2}$ | $a = a_{1,3}$ |
| $(s_1, a_{1,1})$             | 0.4                               | 0.3           | 0.3           |
| $(s_1, a_{1,2})$             | infeasible                        | infeasible    | infeasible    |
| $(s_1, a_{1,3})$             | 0.8                               | 0.1           | 0.1           |
| $(s_2, a_{2,1})$             | infeasible                        | infeasible    | infeasible    |

and, for every  $(s, a)$ ,

$$q_{d_2^{\text{HR}}(s,a,s_2)}(a_{2,1}) = 1.$$

Some comments regarding history-dependent decision rules follow. Since there is only one action in  $s_2, a_{2,1}$ , both deterministic and randomized policies choose it. In the history-dependent case, specifying actions for  $h_2 = (s_1, a_{1,2}, s_1)$  or  $h_2 = (s_2, a_{2,1}, s_1)$  is unnecessary since neither of these histories can occur. However, if action  $a_{1,1}$  or  $a_{1,3}$  were chosen at decision epoch 1, then several distinct histories are possible. For example, if we adopt  $\pi^{\text{HR}}$  for the model in Fig. 3.1.2, and the history at decision epoch 2 is  $(s_1, a_{1,1}, s_1)$ , then the randomized decision rule  $d_2^{\text{HR}}$  chooses action  $a_{1,1}$  with probability 0.4, action  $a_{1,2}$  with probability 0.3, and action  $a_{1,3}$  with probability 0.3.

We return to the model in Fig. 3.1.1. Suppose the decision maker uses the deterministic Markov policy  $\pi^{\text{MD}}$ . Then the system evolves as a nonstationary two-period Markov chain with transition probability matrices

$$P_{d_1^{\text{MD}}} = \begin{bmatrix} 0.5 & 0.5 \\ 0 & 1 \end{bmatrix}, \quad P_{d_2^{\text{MD}}} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$



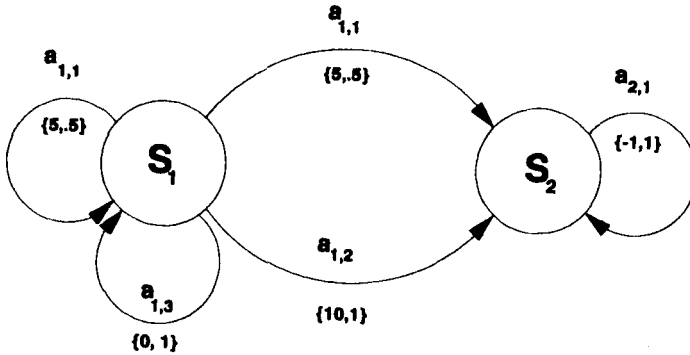


Figure 3.1.2 Symbolic representation of the modified model.

The corresponding rewards are

$$r_1(s_1, d_1^{\text{MD}}(s_1)) = 5, \quad r_1(s_2, d_1^{\text{MD}}(s_2)) = -1,$$

and

$$r_2(s_1, d_2^{\text{MD}}(s_1)) = 10, \quad r_2(s_2, d_2^{\text{MD}}(s_2)) = -1.$$

We leave it as an exercise (Problem 3.1) to derive the analogous expressions for the randomized Markov policy  $\pi^{\text{MR}}$ .

## 3.2 SINGLE-PRODUCT STOCHASTIC INVENTORY CONTROL

In Sec. 1.3, we described how Markov decision processes may be used to model inventory control systems. This section presents a simplified version of that model. We also include a numerical example which we refer to in later chapters. Inventory models were among the first problems solved using Markov decision problem methods, and their study has motivated many theoretical developments. These models are widely applicable; they can also be used to manage fisheries, forests, and cash balances. We expand on the model herein in Sec. 3.5.2.

### 3.2.1 Model Formulation

Each month, the manager of a warehouse determines current inventory (stock on hand) of a single product. Based on this information, he decides whether or not to order additional stock from a supplier. In doing so, he is faced with a tradeoff between the costs associated with keeping inventory and the lost sales or penalties associated with being unable to satisfy customer demand for the product. The manager's objective is to maximize some measure of profit (sales revenue less inventory holding and ordering costs) over the decision-making horizon. Demand for the product is random with a known probability distribution.

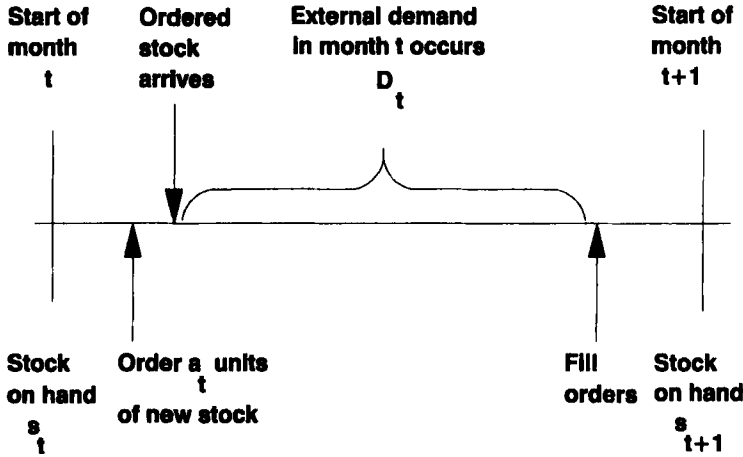


Figure 3.2.1 Timing of events in the inventory model of Sec. 3.2.1.

We formulate a model under the following set of simplifying assumptions. Generalizations which make the model more realistic are explored through problems at the end of the chapter.

1. The decision to order additional stock is made at the beginning of each month and delivery occurs instantaneously.
2. Demand for the product arrives throughout the month but all orders are filled on the last day of the month.
3. If demand exceeds inventory, the customer goes elsewhere to purchase the product; that is, there is no backlogging of unfilled orders so that excess demand is lost.
4. The revenues, costs, and the demand distribution do not vary from month to month.
5. The product is sold only in whole units.
6. The warehouse has capacity of  $M$  units.

Figure 3.2.1 illustrates the timing of arrival of orders and demands implied by the above assumptions.

Let  $s_t$  denote the inventory on hand at the beginning of month  $t$ ,  $a_t$  the number of units ordered by the inventory manager in month  $t$  and  $D_t$  the random demand in month  $t$ . We assume that the demand has a known time-homogeneous probability distribution  $p_j = P\{D_t = j\}$ ,  $j = 0, 1, 2, \dots$ . The inventory at decision epoch  $t + 1$ ,  $s_{t+1}$ , is related to the inventory at decision epoch  $t$ ,  $s_t$ , through the system equation

$$s_{t+1} = \max\{s_t + a_t - D_t, 0\} \equiv [s_t + a_t - D_t]^+. \quad (3.2.1)$$

Because backlogging is not permitted, the inventory level cannot be negative. Thus whenever  $s_t + a_t - D_t < 0$ , the inventory level at the subsequent decision epoch is 0.

We now describe the economic parameters of this model. We express them as values at the start of the month so that we are implicitly considering the time value of money when defining these quantities. We refer to them as *present values* to

emphasize this point. The present value of the cost of ordering  $u$  units in any month is  $O(u)$ . We assume it is composed of a fixed cost  $K > 0$  for placing orders and a variable cost  $c(u)$  that increases with quantity ordered. Hence

$$O(u) = \begin{cases} K + c(u) & \text{if } u > 0 \\ 0 & \text{if } u = 0 \end{cases} \quad (3.2.2)$$

The present value of the cost of maintaining an inventory of  $u$  units for a month (between receipt of the order and releasing inventory to meet demand) is represented by the nondecreasing function  $h(u)$ . In finite-horizon problems, the remaining inventory at the last decision epoch has value  $g(u)$ . Finally, if the demand is  $j$  units and sufficient inventory is available to meet demand, the manager receives revenue with present value  $f(j)$ . Assume  $f(0) = 0$ .

In this model the reward depends on the state of the system at the subsequent decision epoch; that is,

$$r_t(s_t, a_t, s_{t+1}) = -O(a_t) - h(s_t + a_t) + f(s_t + a_t - s_{t+1}).$$

For subsequent calculations, it is more convenient to work with  $r_t(s_t, a_t)$ , which we evaluate using (2.1.1). To this end we compute  $F_t(u)$ , the expected present value (at the start of month  $t$ ) of the revenue received in month  $t$  when the inventory prior to receipt of customer orders is  $u$  units. It is derived as follows. If inventory  $u$  exceeds demand  $j$ , the present value of the revenue is  $f(j)$ . This occurs with probability  $p_j$ . If demand exceeds inventory, the present value of the revenue equals  $f(u)$ . This occurs with probability  $q_u = \sum_{j=u}^{\infty} P_j$ . Thus

$$F(u) = \sum_{j=0}^{u-1} f(j)p_j + f(u)q_u.$$

A Markov decision process formulation follows.

**Decision epochs:**

$$T = \{1, 2, \dots, N\}, \quad N \leq \infty.$$

**States** (the amount of inventory on hand at the start of a month):

$$S = \{0, 1, 2, \dots, M\}.$$

**Actions** (the amount of additional stock to order in month  $t$ ):

$$A_s = \{0, 1, 2, \dots, M - s\}.$$

**Expected rewards** (expected revenue less ordering and holding costs):

$$r_t(s, a) = F(s + a) - O(a) - h(s + a), \quad t = 1, 2, \dots, N - 1 \quad (3.2.3)$$

(the value of terminal inventory)

$$r_N(s) = g(s), \quad t = N.$$

**Transition probabilities:**

$$p_i(j|s, a) = \begin{cases} 0 & \text{if } M \geq j > s + a \\ p_{s+a-j} & \text{if } M \geq s + a \geq j > 0 \\ q_{s+a} & \text{if } M \geq s + a \text{ and } j = 0, \end{cases}$$

where  $q_{s+a}$  is defined above.

A brief explanation of the derivation of the transition probabilities follows. If the inventory on hand at the beginning of period  $t$  is  $s$  units and an order is placed for  $a$  units, the inventory prior to external demand is  $s + a$  units (Assumption 1). An inventory level of  $j > 0$  at the start of period  $t + 1$  requires a demand of  $s + a - j$  units in period  $t$ . This occurs with probability  $p_{s+a-j}$ . Because backlogging is not permitted (Assumption 3), if the demand in period  $t$  exceeds  $s + a$  units, then the inventory at the start of period  $t + 1$  is 0 units. This occurs with probability  $q_{s+a}$ . The probability that the inventory level exceeds  $s + a$  units is 0, since demand is non-negative. Assumption 6 constrains the inventory always to be less than or equal to  $M$ .

Assumption 2 implies that the inventory throughout a month is  $s + a$ , so that the total monthly holding cost is  $h(s + a)$ . If instead, the demand  $D_t$  arrives at the beginning of a month, the expected holding cost  $E\{h([s + a - D_t]^+)\}$  replaces the holding cost in (3.2.3).

Deterministic Markov decision rules assign the quantity of inventory to be ordered each month to each possible starting inventory position. A policy is a sequence of such ordering rules. An example of such a decision rule is:

*Order sufficient stock to raise the inventory to  $\Sigma$  units whenever the inventory level at the beginning of a month is less than  $\sigma$  units. When the inventory level at the beginning of a month is  $\sigma$  units or greater, do not place an order.*

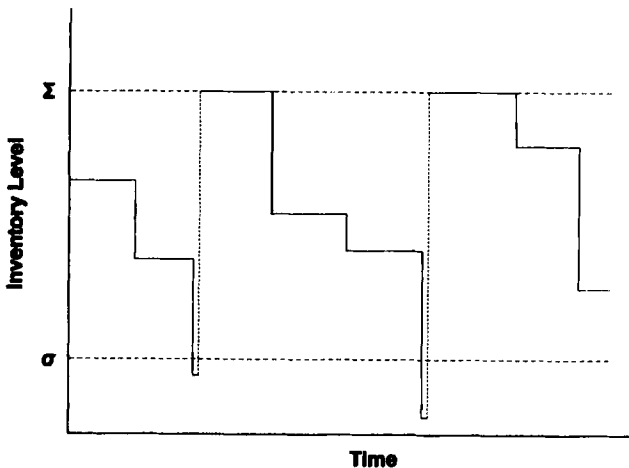


Figure 3.2.2 Realization of an inventory model under a  $(\sigma, \Sigma)$  policy.

This decision rule may be represented by

$$d_t(s) = \begin{cases} \Sigma - s & s < \sigma \\ 0 & s \geq \sigma. \end{cases}$$

We refer to a policy which uses decision rules of the above form as a  $(\sigma, \Sigma)$  policy. Figure 3.2.2 illustrates a realization of an inventory process under this policy. Such policies have practical appeal and are optimal under various assumptions on the model components. In practice,  $\Sigma$  is referred to as the *target stock* and  $\Sigma - \sigma$  as the *minimum fill*.

### 3.2.2 A Numerical Example

We provide values for the parameters of the above model. We choose  $K = 4$ ,  $c(u) = 2u$ ,  $g(u) = 0$ ,  $h(u) = u$ ,  $M = 3$ ,  $N = 3$ ,  $f(u) = 8u$ , and

$$p_j = \begin{cases} \frac{1}{4} & \text{if } j = 0 \\ \frac{1}{2} & \text{if } j = 1 \\ \frac{1}{4} & \text{if } j = 2. \end{cases}$$

This model can be interpreted as follows. The inventory is constrained to be 3 or fewer units. All costs and revenues are linear. This means that for each unit ordered the per unit cost is 2, for each unit held in inventory for one month, the per unit cost is 1 and for each unit sold the per unit revenue is 8. The expected revenue when  $u$  units of stock are on hand prior to receipt of an order is given by

| $u$ | $F(u)$                                                                    |
|-----|---------------------------------------------------------------------------|
| 0   | 0                                                                         |
| 1   | $0 \times \frac{1}{4} + 8 \times \frac{3}{4} = 6$                         |
| 2   | $0 \times \frac{1}{4} + 8 \times \frac{1}{2} + 16 \times \frac{1}{4} = 8$ |
| 3   | $0 \times \frac{1}{4} + 8 \times \frac{1}{2} + 16 \times \frac{1}{4} = 8$ |

Combining the expected revenue with the ordering and holding costs gives the expected profit in period  $t$  if the inventory level is  $s$  at the start of the period and an order for  $a$  units is placed. If  $a = 0$ , the ordering and holding cost equals  $s$ , and if  $a$  is positive, it equals  $4 + s + 3a$ . It is summarized in the table below where  $\times$  denotes an infeasible action. Transition probabilities only depend on the total inventory on hand prior to receipt of orders. They are the same for any  $s$  and  $a$  which have the same value for  $s + a$  and are functions of  $s + a$  only. The information in the following tables defines this problem completely.

|     |  | $r_t(s, a)$ |          |          |          |   |
|-----|--|-------------|----------|----------|----------|---|
|     |  | $a$         | 0        | 1        | 2        | 3 |
| $s$ |  |             |          |          |          |   |
| 0   |  | 0           | -1       | -2       | -5       |   |
| 1   |  | 5           | 0        | -3       | $\times$ |   |
| 2   |  | 6           | -1       | $\times$ | $\times$ |   |
| 3   |  | 5           | $\times$ | $\times$ | $\times$ |   |

|         |  | $p_i(j s, a)$ |               |               |               |   |
|---------|--|---------------|---------------|---------------|---------------|---|
|         |  | $j$           | 0             | 1             | 2             | 3 |
| $s + a$ |  |               |               |               |               |   |
| 0       |  | 1             | 0             | 0             | 0             | 0 |
| 1       |  | $\frac{3}{4}$ | $\frac{1}{4}$ | 0             | 0             | 0 |
| 2       |  | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ | 0             | 0 |
| 3       |  | 0             | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ | 0 |

### 3.3 DETERMINISTIC DYNAMIC PROGRAMS

Deterministic dynamic programs (DDPs) constitute an important and widely studied class of Markov decision processes. Applications include finding the shortest route in a network, critical path analysis, sequential allocation, and inventory control with known demands.

#### 3.3.1 Problem Formulation

In a deterministic dynamic program, choice of an action determines the subsequent state with *certainty*. Standard formulations take this into account by using a transfer function instead of a transition probability to specify the next state. By a *transfer function*, we mean a mapping,  $\tau_t(s, a)$ , from  $S \times A_s$  to  $S$ , which specifies the system state at time  $t + 1$  when the decision maker chooses action  $a \in A_s$  in state  $s$  at time  $t$ . To formulate a DDP as a Markov decision process, define the transition probability function as

$$p_t(j|s, a) = \begin{cases} 1 & \text{if } \tau_t(s, a) = j \\ 0 & \text{if } \tau_t(s, a) \neq j. \end{cases} \quad (3.3.1)$$

The rewards are given by  $r_t(s, a)$ .

#### 3.3.2 Shortest Route and Critical Path Models

When the total reward is used to compare policies, every DDP with finite  $S$ ,  $A$ , and  $T$  is equivalent to a shortest or longest route problem. We introduce such a problem and discuss this equivalence.

A *finite directed graph* consists of a set of nodes and directed arcs. By a *path* we mean a sequence of arcs that connects one node to another node. We call such a graph *acyclic* whenever there are no paths which begin and end at the same node. An example of an acyclic directed graph appears in Fig. 3.3.1. In it, nodes are labeled by integers  $\{1, \dots, 8\}$  and directed arcs are represented by lines with arrows. Node 1 is called the *origin* and node 8 the *destination*. The value above the arc give the “distance,” “cost,” or “time” associated with traversing the arc. A “shortest route problem” corresponds to finding a path or sequence of arcs from the origin to the

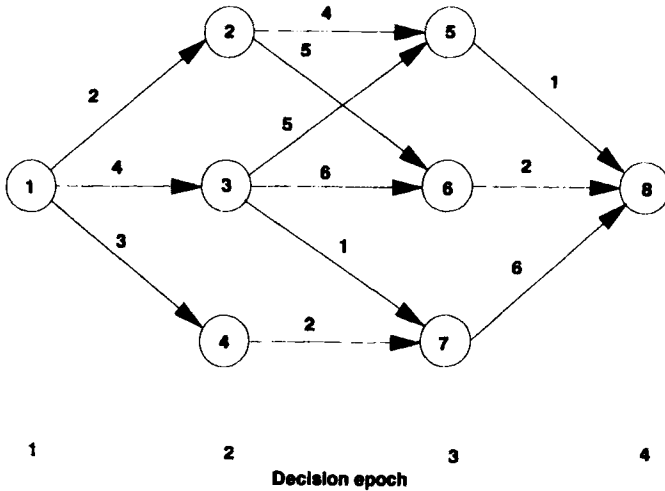


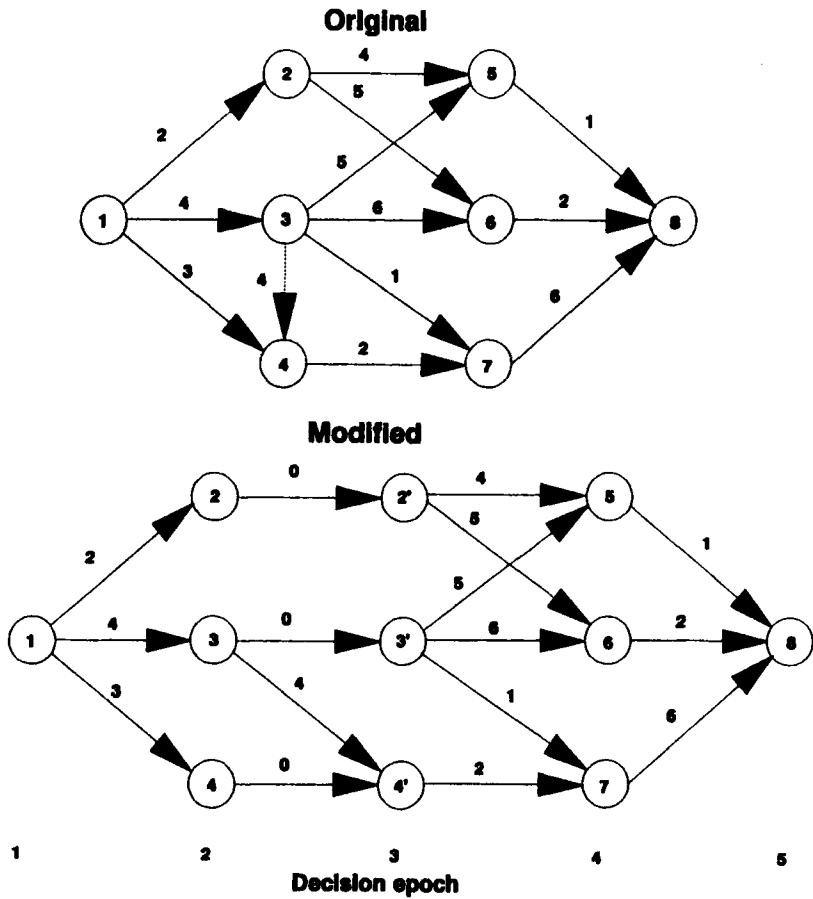
Figure 3.3.1 Example of a routing problem.

destination (node 1 to node 8) of minimal total value, while a “longest route problem” seeks a path of greatest total value.

Applications of shortest route problems include optimization of shipping and communication networks. Longest route problems are the basis for *critical path analysis*. In critical path analysis, each arc represents a task and its length the time to complete the task. The task cannot begin until completion of all tasks which immediately precede it (those with arcs ending at the node from which the designated arc originates). The length of the longest path from the first node to the last node gives the minimum time to complete all tasks and hence the entire project. It is the “critical path” because if any task on it is delayed, the time to complete the project increases. In addition to the critical path, earliest and latest start dates for each task may be easily obtained from a solution to this problem.

Formulation of a shortest route problem as a deterministic dynamic program involves identifying nodes with states, arcs with actions, transfer functions with transition probabilities and values with rewards. However direct identification of decision epochs or stages is not always possible and not necessary for solution. In the example in Fig. 3.3.1, node 1 can be identified with decision epoch 1, nodes 2, 3, and 4 with decision epoch 2, nodes 5, 6, and 7 with decision epoch 3, and node 8 with decision epoch 4. However, if the example was modified to contain an arc connecting nodes 3 and 4 (Fig. 3.3.2), an obvious identification of decision epochs would not be available. The lower portion of Fig. 3.3.2 shows that adding nodes to the graph and modifying arc lengths allows identification of decision epochs.

Figure 3.3.3 provides a graphical representation for an arbitrary DDP. At each decision epoch, there is a node corresponding to each state. Arcs originate at each state, one for each action, and end at the next stage at the node determined by the corresponding transfer function. Arc lengths give rewards. We add a “dummy” origin “0” and destination “D.” Arc lengths from nodes at stage  $N$  to the destination give terminal rewards  $r_N(s)$ . If a solution is sought for a particular initial state  $s'$  and the problem is one of maximization, then we give the arc from the origin to  $s'$  a large



**Figure 3.3.2** Identification of decision epochs in a modification of the routine problem of Figure 3.3.1.

positive length, say  $L$ , and set lengths of all other arcs starting from the origin equal to the zero as in Figure 3.3.3.

**3.3.3 Sequential Allocation Models**

A decision maker has  $M$  units of resource to utilize or consume over  $N$  periods. Denote by  $x_t$  the quantity of resource used or consumed in period  $t$  and by  $f(x_1, \dots, x_N)$  the utility of this allocation pattern to the decision maker. Regard negative  $f(x_1, \dots, x_N)$  as disutility or cost. The problem of choosing an optimal consumption pattern is given by the following mathematical program.

$$\text{maximize } f(x_1, \dots, x_N),$$



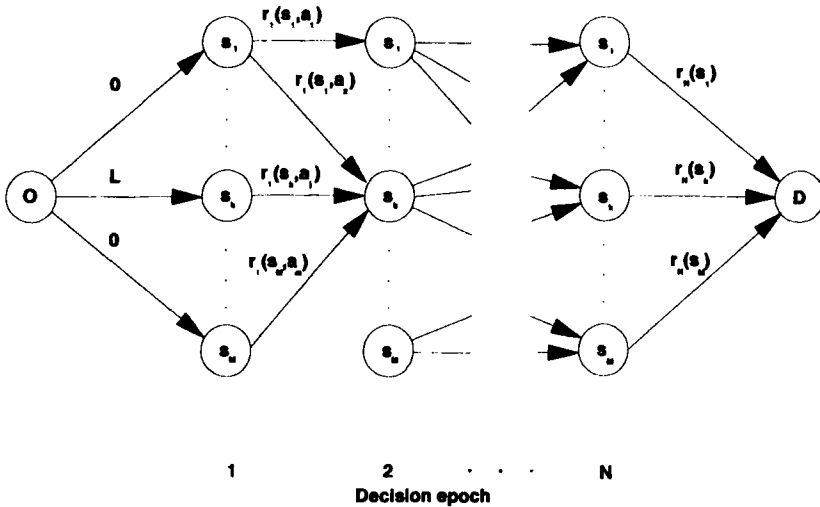


Figure 3.3.3 Graphical representation of an arbitrary DDP.

subject to

$$x_1 + \cdots + x_N = M$$

and

$$x_t \geq 0, \quad t = 1, \dots, N.$$

Direct formulation as a DDP or a Markov decision process is easiest for *separable* objective functions  $f(x_1, \dots, x_N)$ , that is those which satisfy

$$f(x_1, \dots, x_N) = \sum_{t=1}^N g_t(x_t).$$

To define this as a Markov decision process model, let the state at time  $t$  represent the total quantity of resources available for consumption in periods  $t, t+1, \dots, N$  and the action represent the quantity chosen to be consumed in period  $t$ . If the state at time  $t$  is  $s$ , and  $a$  units are consumed in period  $t$ , then  $s-a$  units are available for consumptions in periods  $t+1, t+2, \dots, N$ . No decisions are made in period  $N$  because specifying the allocations in periods  $1, \dots, N-1$  fixes the allocation in period  $N$ . Note that, in this model, the sets of states and actions are not discrete; they are compact intervals of the real line.

The Markov decision process formulation for this problem follows.

**Decision epochs:**

$$T = \{1, \dots, N\}, \quad N < \infty.$$

**States:**

$$S = [0, M], \quad M < \infty.$$

**Actions:**

$$A_s = [0, s], \quad 0 \leq s \leq M.$$

**Rewards:**

$$r_t(s, a) = g_t(a), \quad s \in S, a \in A_s, t = 1, \dots, N-1,$$

and

$$r_N(s) = g_N(s), \quad s \in S.$$

**Transition probabilities:**

$$p_t(j|s, a) = \begin{cases} 1 & j = s - a \\ 0 & \text{otherwise,} \end{cases} \quad s \in S, a \in A_s, t = 1, \dots, N-1.$$

### 3.3.4 Constrained Maximum Likelihood Estimation

We now formulate the problem of finding maximum likelihood estimators of the parameters of a multinomial distribution as a sequential allocation model. The multinomial probability function may be represented as

$$p(y_1, y_2, \dots, y_N) = \frac{K!}{y_1! y_2! \dots y_N!} q_1^{y_1} q_2^{y_2} \dots q_N^{y_N},$$

where  $y_i$  is the number of observations of type  $i$ ,  $q_i$  is the probability of an observation of type  $i$ ,  $K$  is total number of observations, and  $N$  is the number of observation types. Note that  $\sum_{i=1}^N y_i = K$ . Since  $q_i$  is a probability, it is non-negative and satisfies

$$\sum_{i=1}^N q_i = 1. \quad (3.3.2)$$

Maximum likelihood estimators are found by regarding the observations  $(y_1, \dots, y_N)$  as fixed and choosing values of the parameters  $(q_1, \dots, q_N)$  to maximize the multinomial log likelihood

$$L(q_1, \dots, q_N) = \log(K!) - \sum_{i=1}^N \log(y_i!) + \sum_{i=1}^N y_i \log(q_i). \quad (3.3.3)$$

Since the first two terms in (3.3.3) do not effect the choice of the maximizing values of  $q_i$ , the problem of finding a maximum likelihood estimator becomes the following.

Maximize

$$\sum_{i=1}^N y_i \log(q_i) \quad (3.3.4)$$

subject to (3.3.2) and  $q_i \geq 0$  for  $i = 1, \dots, N$ .

This is a sequential allocation problem with  $S = [0, 1]$  and  $g_i(x_i) = y_i \log(x_i)$ .

### 3.4 OPTIMAL STOPPING

This section formulates and provides examples of optimal stopping problems. These represent a colorful collection of problems that often have elegant structured solutions.

#### 3.4.1 Problem Formulation

In an optimal stopping problem, the system evolves as an uncontrolled, possibly nonstationary, Markov chain with state space  $S'$ . At each decision epoch, the decision maker has two actions available in each state: to stop or to continue. If he decides to stop in state  $s$  at time  $t$ , he receives reward  $g_t(s)$ , and if he decides to continue, he incurs a cost  $f_t(s)$  and the system evolves until the next decision epoch according to the transition probability matrix of the Markov chain. The problem may be finite horizon, in which case a function  $h(s)$  represents the reward if the system reaches decision epoch  $N$  without stopping. The objective is to choose a policy to maximize the expected total reward. When stopped, it remains in the stopped state, yielding zero reward.

Applications of these models include selling an asset, choosing a candidate for a job (the secretary problem), exercising and valuing a financial option, sequential hypothesis testing, and gambling. A MDP formulation of the optimal stopping problem follows.

**Decision epochs:**

$$T = \{1, 2, \dots, N\}, \quad N \leq \infty.$$

**States:**

$$S = S' \cup \{\Delta\}.$$

**Actions:**

$$A_s = \begin{cases} \{C, Q\} & s \in S' \\ \{C\} & s = \Delta. \end{cases}$$

**Rewards:**

$$r_t(s, a) = \begin{cases} -f_t(s) & s \in S', \quad a = C \\ g_t(s) & s \in S', \quad a = Q \quad t < N, \\ 0 & s = \Delta, \end{cases}$$

$$r_N(s) = h(s).$$

**Transition probabilities:**

$$p_t(j|s, a) = \begin{cases} p_t(j|s) & s \in S', \quad j \in S', \quad a = C \\ 1 & s \in S', \quad j = \Delta, \quad a = Q \quad \text{or } s = j = \Delta, \quad a = C \quad t < N \\ 0 & \text{otherwise.} \end{cases}$$

The state space consists of  $S'$ , the state space for an uncontrolled Markov chain with transition probabilities  $p_t(j|s)$ , in addition to a distinguished absorbing state  $\Delta$ . In  $S'$ , two actions are available in each state: continue,  $C$ , or quit,  $Q$ . Continuation causes the process to move to state  $j$  in  $S'$  with probability  $p_t(j|s)$ . Termination moves the system to state  $\Delta$ , where it remains forever and receives reward zero. Usually, for some states  $g_t(s)$  is large, so that the decision maker's objective is to terminate the process in these states.

**3.4.2 Selling an Asset**

An "investor" owns an expensive property or asset which he expects to appreciate in value over time. At the end of each week, the investor decides whether to accept the best offer he received during the preceding week and sell the property, or else decline this offer and solicit new offers in the next week. We assume he must sell the property within one year, although we also consider infinite-horizon variants of this problem in Chap. 7.

We formulate this as a finite-horizon optimal stopping problem as follows. Decision epochs correspond to the times at which decisions are made. Number them 1 to 53 corresponding to the weeks of the year. The state is the best offer available at the end of a week (supposedly the market value of the property). The state space  $S'$  represents the set of all possible offers over the decision-making horizon. It can be either discrete or a continuum but it is usually bounded. The best available offer at decision epoch  $t + 1$  is related to the best available offer at decision epoch  $t$  through the nonstationary transition probability matrix or function  $p_t(j|s)$ . Nonstationarity reflects changing market conditions. If the investor accepts an offer  $s$ , his reward is  $s - K(s)$ , where  $K(s)$  includes fixed and variable costs associated with selling a property of value  $s$ . The continuation cost  $f_t(s)$  includes advertising expenses, interest charges, and taxes associated with carrying the property for one week. If he holds the property for the entire year without sale, he must sell it at the end of the year and receive a terminal reward of  $s - K(s)$ .

Good policies for this problem use decision rules of the form

$$d_t(s) = \begin{cases} Q & s \geq B_t \\ C & s < B_t, \end{cases} \quad s \in S'.$$

We refer to this as a *control limit* policy; the number  $B_t$  is the control limit. Solving the problem reduces to determining the boundary  $B_t$  between the continuation and stopping regions. Of course, the solution depends on the forecasts of  $p_t(j|s)$  for  $t = 1, \dots, 52$ .

### 3.4.3 The Secretary Problem

This classic dynamic programming problem was first proposed by Cayley (1875) in the context of finding an optimal policy for playing a lottery. We provide a more colorful description. An employer seeks to hire an individual to fill a vacancy for a secretarial position. There are  $N$  candidates or applicants for this job, with  $N$  fixed and known by the employer. Candidates are interviewed sequentially. Upon completion of each interview, the employer decides whether or not to offer the job to the *current* candidate. If he does not offer the job to this candidate, that individual seeks employment elsewhere and is no longer eligible to receive an offer.

Formulation of this problem depends on the decision maker's objective. Here we assume that the employer wishes to maximize the probability of giving an offer to the best candidate.

A more rigorous statement of the problem follows. A collection of  $N$  objects is ranked from 1 to  $N$ , with that ranked 1 being the most desirable. The true rankings are unknown to the decision maker. He observes the objects one at a time in random order. He can either select the current object and terminate the search, or discard it and choose the next object. His objective is to maximize the probability of choosing the object ranked number 1. We assume that the decision maker's relative rankings are consistent with the absolute rankings. That is, if object  $A$  has a lower numerical ranking than object  $B$ , the decision maker will prefer object  $A$  to object  $B$ .

Decisions are made after observing each object, so that the horizon  $N$  equals the number of candidates. The state space  $S' = \{0, 1\}$ ; 1 denotes that the current object is the best object (rank closest to 1) seen so far, and 0 that a previous object was better.

In either state, the action  $Q$  means select the current object (give an offer to the current candidate) and  $C$  means do not select the current object and continue the search (interview the next candidate). Rewards are received only when stopping; that is, choosing action  $Q$ . They correspond to the probability of choosing the best candidate. In the notation above, the continuation cost  $f_t(s) = 0$ ,  $s = 0, 1$ ; the rewards at stopping,  $g_t(0) = 0$  and  $g_t(1) = t/N$ ; and the terminal rewards,  $h(0) = 0$  and  $h(1) = 1$ . A derivation of  $g_t$  and  $h$  follows.

Suppose after observing  $t$  objects that the decision maker ranks the current object as the best of those seen so far. Then  $g_t(1)$ , the probability that object is the best among all objects, is determined as follows.

$$\begin{aligned}
 P \{ \text{Best object is in first } t \} \\
 &= \frac{\text{Number of subsets of } \{1, 2, \dots, N\} \text{ of size } t \text{ containing } 1}{\text{Number of subsets of } \{1, 2, \dots, N\} \text{ of size } t} \\
 &= \frac{\binom{N-1}{t-1}}{\binom{N}{t}} = \frac{t}{N}
 \end{aligned}$$

where  $\binom{n}{r}$  denotes a binomial coefficient. If all objects have been seen, the last object

must be chosen. If that object is best, that is  $s = 1$ , then the probability of choosing the best object is 1 so that  $h(1) = 1$ ; otherwise,  $h(0) = 0$ .

Transition probabilities for the uncontrolled system are independent of the system state, that is, the status of the current object. Therefore  $p_t(j|s) = p_t(j)$  for  $s = 0, 1$ . The probability that the subsequent candidate is the best among the first  $t + 1$ ,  $p_t(1|s) = 1/(t + 1)$ , and the probability that the subsequent candidate is not the best among the first  $t + 1$ ,  $p_t(0|s) = t/(t + 1)$  for  $s = 0, 1$ .

Variants of this model include random numbers of candidates, maximization of the probability of choosing one of the  $k$  best candidates, minimization of the expected rank of the candidate who receives the offer, or a candidate selection rule in which, after interviewing the current candidate, the employer may make an offer to the current candidate and some designated subset of previous candidates whose availability depends on a specified probability distribution.

### 3.4.4 Call Options

In financial markets, a *call option* gives the owner the right to purchase an asset such as shares of common stock, at a fixed price  $W$  (the strike price) at any time prior to a specified expiry date. For example we might purchase a call option to buy 100 shares of Sun Microsystems at \$40 per share at or before June 30. Suppose, at some day prior to June 30, the stock price reached \$45 per share. Then we might exercise the option and buy 100 shares of Sun Microsystems stock for \$4000 and immediately sell them for \$4500 to make a profit of \$500 less transaction costs (commissions). If the stock price lies below \$40 per share, we would not exercise the option.

Options are traded in financial markets and values vary on a day-to-day basis in relationship to the price of the underlying stock. A fundamental problem in investment theory is how to determine the value of an option. Clearly its value should depend on using it as efficiently as possible. A sequential decision process formulation provides insight into how to do this.

We formulate a simplified form of the option holder's decision problem as an optimal stopping problem by ignoring the impact of dividends and the real life possibilities of selling the options on any day. At the start of each trading day prior to the expiry date, the option holder decides whether or not to exercise the option. Decision epochs correspond to the times when the option holder decides whether or not to exercise the option, and the state represents the market price of the underlying asset (stock). The choice of actions is to exercise the option (stop) or not exercise the option (continue). After the expiry date, the option has no value. By exercising an option for 100 shares with strike price  $W$  when the security has value  $s$ , the option holder receives  $100(s - W) - K$ , where  $K$  represents the transaction costs. In the above notation,  $h(s) = g_t(s) = 100(s - W) - K$  for all  $t$ . Since not exercising the option has no financial implications,  $f_t(s) = 0$ . The transition probability function  $p_t(j|s)$  gives the probability that the next day's stock price equals  $j$  given that the stock price on day  $t$  equals  $s$ . Clearly it is not affected by the action of the option holder.

There are many intricate uses of call options ranging from the conservative to the highly speculative. We refer the interested reader to references noted in the Bibliographic Remarks section.

### 3.5 CONTROLLED DISCRETE-TIME DYNAMIC SYSTEMS

In many settings, especially when the states represent real-valued quantities such as wealth or inventory levels, it is natural to describe the evolution of the system in terms of sample paths and system equations instead of in terms of transition probabilities. We refer to this as the *control theory* approach. In this section we introduce the control theory model, show how it may be reformulated as a Markov decision process, and provide illustrations from operations research, economics, and engineering.

#### 3.5.1 Model Formulation

Let the state space  $S$ , control set  $U$ , and disturbance set  $W$  be subsets of  $R^k$ ,  $R^m$ , and  $R^n$ , respectively. We allow, but do not require, that  $k = m = n$ , or even that they all equal 1. The state of the system at time  $t + 1$ ,  $s_{t+1}$ , is related to the state of the system at time  $t$ ,  $s_t$ , the control used at time  $t$ ,  $u_t$ , and the disturbance at time  $t$ ,  $w_t$ , through the system equation or “law of motion”

$$s_{t+1} = f_t(s_t, u_t, w_t) \quad (3.5.1)$$

where  $f_t$  is a  $R^k$ -valued function mapping  $S \times U \times W$  into  $S$ . (Note an inconsistency in notation in this section; in contrast to Sec. 3.1 where we distinguished states as  $s_1$  and  $s_2$ , here  $s_t$  denotes the state at decision epoch  $t$ .) The unobservable disturbance  $w_t$  is the realization of a random variable  $W_t$  that assumes values in  $W$ , is independent of  $w_\tau$  for all  $\tau < t$ , and has a distribution that does not depend on  $s_t$  or  $u_t$ . We assume that the distribution of  $W_t$  is expressed in terms of a density or probability function  $q_t(\cdot)$ .

When the system is in state  $s_t$  and the decision maker, who in this context is frequently referred to as “the controller,” chooses control  $u_t$  from the set of admissible controls in state  $s_t$ ,  $U_{s_t} \subset U$ , the system receives a reward of  $g_t(s_t, u_t)$  (or incurs a cost  $-g_t(s_t, u_t)$ ). The externally generated disturbance effects this evolution by interacting with the state,  $s_t$  and control  $u_t$ , to determine the state  $s_{t+1}$  through (3.5.1). When the horizon is finite and the system terminates in state  $s_N$  at time  $N$ , it receives a terminal reward  $g(s_N)$ . The time evolution of this system is depicted in Fig. 3.5.1.

An equivalent Markov decision process formulation follows.

**Decision epochs:**

$$T = \{1, 2, \dots, N\}, \quad N \leq \infty.$$

**States:**

$$S \subset R^k.$$

**Actions:**

$$A_s = U_s.$$

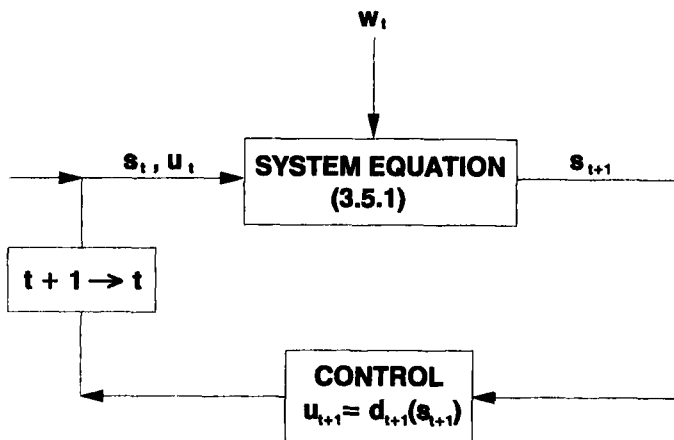


Figure 3.5.1 Time evolution of the discrete-time control theory model.

#### Rewards:

$$r_t(s, a) = g_t(s, a), \quad t < N, \quad r_N(s) = g_N(s), \quad t = N \text{ when } N < \infty.$$

#### Transition probabilities:

$$p_t(j|s, a) = P[W_t \in \{w \in W: j = f_t(s, a, w)\}] \quad (3.5.2)$$

$$= \sum_{\{w \in W: j = f_t(s, a, w)\}} q_t(w). \quad (3.5.3)$$

We see from the above that the only substantial difference in these two approaches to modeling stochastic sequential control problems is the way in which transition probabilities are incorporated. In the Markov decision process formulation, the transition probabilities are fundamental, while in the stochastic control formulation they must be derived. Equations (3.5.2) and (3.5.3) appear quite formidable, but in applications their evaluation is usually straightforward.

When  $S$  and  $W$  are continuous,  $p_t(j|s, a)$  in (3.5.2) and  $q_t(w)$  in (3.5.3) are densities, and integration replaces summation. In either case, the state space is usually discretized prior to numerical solution.

In control theory language, decision rules are referred to as *feedback* controls because they use the current state information to feedback into the system equation  $f_t$  and influence future states of the system as in Fig. 3.5.1. An *open loop* control is a decision rule  $d_t$  in which  $d_t(s) = a$  for all  $s \in S$ . Since it uses the same action regardless of the state, it does not require observation of the system for implementation. In practice, open loop controls are easier to implement and evaluate but are usually suboptimal.

The system equation (3.5.1) may be generalized to include controls, states, and disturbances from some or all previous periods. In such generality it may be represented by

$$s_{t+1} = f_t(s_t, s_{t-1}, \dots, s_1, u_t, u_{t-1}, \dots, u_1, w_t, w_{t-1}, \dots, w_1). \quad (3.5.4)$$



This increases the flexibility of the model by allowing correlated disturbances and delays in the effect of control actions. Equation (3.5.4) can usually be reduced to (3.5.1) by reparameterizing the model in terms of vectors which account for previous systems states, controls, or disturbances.

### 3.5.2 The Inventory Control Model Revisited

We reformulate the single-product inventory model of Sec. 3.2 as a controlled discrete-time dynamic system and discuss some generalizations.

The state space  $S = \{0, 1, \dots, M\}$ , the control set  $U = \{0, 1, \dots, M\}$ , and the disturbance set  $W = \{0, 1, \dots\}$  are all subsets of  $R^1$ . In state  $s$ , the set of admissible controls is  $U_s = \{0, 1, \dots, M - s\} \subset U$ . The system equation (3.5.1) is given by (3.2.1); that is,

$$s_{t+1} = [s_t + u_t - w_t]^+$$

where  $u_t$  is identified with  $a_t$  and  $w_t$  with  $D_t$  in (3.2.1). The rewards are as derived in Sec. 3.2.

We now illustrate the calculation of transition probabilities based on (3.5.2). Since

$$\{w \in W: j = [s + a - w]^+\} = \begin{cases} \emptyset & j > s + a \\ s + a - j & 0 < j \leq s + a \\ \{s + a, s + a + 1, \dots\} & j = 0 \end{cases}$$

where  $\emptyset$  represents the empty set, the transition probabilities satisfy

$$p_t(j|s, a) = \begin{cases} 0 & j > s + a \\ p_{s+a-j} & 0 < j \leq s + a \\ q_{s+a} & j = 0 \end{cases}$$

for  $a \in \{0, 1, \dots, M - s\}$  where  $q_u = \sum_{j=u}^{\infty} p_j$ .

We now describe some modifications of this model.

#### Backlogging

Suppose that unfilled orders may be backlogged, that is, orders which cannot be filled from current inventory are filled when sufficient stock arrives. We modify the above model as follows.

- The state space  $S = \{\dots, -1, 0, 1, \dots, M\}$ , where negative states correspond to backlogged orders.
- The holding cost  $h(x)$  is also defined for negative values of  $x$ . When  $x < 0$ , it may be regarded as a penalty cost associated with unsatisfied demand. A typical form for  $h(x)$  is

$$h(x) = \begin{cases} -\phi x & x < 0 \\ \eta x & x \geq 0 \end{cases}$$

where  $\phi$  and  $\eta$  are positive constants.

c. The system equation is

$$s_{t+1} = s_t + u_t - w_t$$

### ***Lag in Delivery of Orders (Lead Times)***

Assume an  $L$ -month lag ( $L > 0$ ) between the time of placing an order to replenish inventory and its delivery to the supply facility. That is, an order placed at the beginning of month  $t$  is not available to meet demand until the start of month  $L + t$ . We assume that demand that cannot be satisfied from stock on hand is never filled.

From system equation (3.5.4), we obtain

$$s_{t+1} = [s_t + u_{t-L} - w_t]^+,$$

where  $u_{t-L}$  is the quantity ordered in month  $t - L$  for delivery to meet demand in month  $t$ .

Since the stock level at time  $t$  may exceed  $M$ , it is more natural to formulate this model with  $S = \{0, 1, \dots\}$  and  $U_t = \{0, 1, \dots\}$ .

### ***Correlated Demands***

In practice, demands in successive periods may be correlated. The simplest representative scheme is one in which we assume a correlation between demand in month  $t + 1$  and month  $t$  of  $\rho_t$ ,  $-1 \leq \rho_t \leq 1$ . When  $\rho_t$  is negative, high demands are usually followed by low demands, and when  $\rho_t$  is positive, high demands usually follow high demands. We refer to such a correlation scheme as *first-order autocorrelation*.

This situation may be modeled directly using (3.5.4); however, we formulate it in terms of (3.5.1) by using a two-dimensional state vector in which the first component represents the inventory level and the second component the demand level. Let  $s_t^i$  represent the inventory level at the start of period  $t$ , and  $s_t^d$  represent the demand in period  $t$ . Assuming no backlogging and no restriction on warehouse capacity,  $S = \{0, 1, \dots\} \times \{0, 1, \dots\} \subset R^2$ . In this model,  $w_t$  denotes an externally generated disturbance that is uncorrelated with  $w_{t-1}$  and  $s_t^d$ . This model may be represented by the system of dynamic equations

$$s_{t+1}^i = [s_t^i + u_t - s_t^d]^+, \quad (3.5.5)$$

$$s_{t+1}^d = \rho_t s_t^d + w_t. \quad (3.5.6)$$

Equation (3.5.5) is the familiar inventory balance equation while equation (3.5.6) provides a representation for a first-order autocorrelated system commonly used in statistical modeling.

When  $\rho_t = \rho$  for all  $t$ , we show that (3.5.6) yields a sequence of demands with approximate correlation  $\rho$ . For a pair of random variables  $X$  and  $Y$ , denote the covariance of  $X$  and  $Y$  by  $\text{cov}(X, Y)$  and the variance of  $X$  by  $\text{var}(X)$ . Recall that

the correlation between  $X$  and  $Y$ , denoted  $\text{corr}(X, Y) \equiv \text{cov}(X, Y) / [\text{var}(X)\text{var}(Y)]^{1/2}$ . Using (3.5.6),

$$\begin{aligned}\text{cov}(s_{t+1}^d, s_t^d) &= E\{(s_{t+1}^d - E\{s_{t+1}^d\})(s_t^d - E\{s_t^d\})\} \\ &= E\{(\rho s_t^d + w_t - E\{\rho s_t^d + w_t\})(s_t^d - E\{s_t^d\})\} \\ &= \rho \text{var}(s_t^d) + \text{cov}(s_t^d, w_t) = \rho \text{var}(s_t^d).\end{aligned}$$

Since for large  $t$ ,  $\text{var}(s_t^d) \approx \text{var}(s_{t+1}^d)$ , the conclusion follows.

When applying such a model, we often assume that  $\rho$  is unknown. Since  $\rho$  cannot be observed, the controller must estimate  $\rho$  prior to deciding on the order level. The model may also be generalized with  $\rho_t$  described by a dynamic equation.

### ***Multiproduct Inventory, No Backlogging, and Unlimited Warehouse Capacity***

Assume that the inventory manager controls the level of  $k$  products. They are managed together because of correlated demands and a fixed cost component associated with placing an order for *any* product. Let  $S = S^1 \times S^2 \times \cdots \times S^k \subset R^k$ , where  $S^i = \{0, 1, \dots\}$  is the set of possible stock levels of the  $i$ th product and  $U = U^1 \times U^2 \times \cdots \times U^k \subset R^k$ , where  $U^i = \{0, 1, \dots\}$  is the set of possible order quantities of product  $i$ ,  $i = 1, 2, \dots, k$ .

In this model, the dynamic equation holds for each product individually; that is

$$s_{t+1}^i = [s_t^i + u_t^i - w_t^i]^+, \quad i = 1, 2, \dots, k,$$

where  $w_t^i$  is the realized demand for product  $i$  in month  $t$ . Transition probabilities are derived from the joint demand distribution  $P(W_t^1 = w_t^1, \dots, W_t^k = w_t^k)$ . Under the assumed timing of placement and arrival of orders in Sec. 3.2, the holding cost can be represented by  $h(s_t^1 + u_t^1, \dots, s_t^k + u_t^k)$ , and the ordering cost by

$$0(u^1, \dots, u^k) = \begin{cases} 0, & u^i = 0, \\ K + c(u^1, \dots, u^k), & u^i > 0 \end{cases} \quad \begin{matrix} i = 1, \dots, k \\ \text{for some } i, \end{matrix}$$

where  $c$  is non-decreasing in each component of  $U$ .

### **3.5.3 Economic Growth Models**

We formulate a dynamic model for optimal consumption and investment in a single-good planned economy. The good, which we refer to as capital, may be either invested or consumed. When invested, it produces capital according to a production function  $F_t$ , and when consumed it yields “utility”  $\Psi_t$ . The economic planner chooses a sequence of investments to maximize total utility of consumption. When most consumption occurs early in the planning period, little capital is available to finance subsequent growth. Conversely, little early consumption permits a big “splurge” at the end of the planning horizon.

More formally, let  $s_t$  denote the capital available for consumption in period  $t$ ;  $s_t \in S = [0, \infty)$ . By restricting  $s_t$  to be non-negative, we are assuming the economy to

be debt free. After observing the level of capital at decision epoch  $t$ , the planner chooses a level of consumption  $u_t$  from  $U_{s_t} = [0, s_t]$ , and invests the remaining capital  $s_t - u_t$ . The consumption generates immediate utility  $\Psi_t(u_t)$  and the investment produces capital at the next decision epoch according to the dynamic equation

$$s_{t+1} = w_t F_t(s_t - u_t), \quad (3.5.7)$$

where the function  $F_t$  represents existing technology and anticipated inflation, and the quantity  $w_t$  denotes a random disturbance which accounts for unanticipated inflation and random shocks to the system. We regard  $w_t$  as the realization of a non-negative random variable with a mean of one.

We model the dynamics in (3.5.7) using a multiplicative disturbance instead of an additive disturbance. This implicitly assumes that the disturbance produces a constant percentage change over the range of possible capital levels; a value of 1.1 for  $w_t$  corresponds to a 10% appreciation due to unforeseen events, and a  $w_t$  of 0.9 corresponds to a 10% decrease. This percentage change effects all levels of capital similarly.

Taking logarithms of the expression in (3.5.7) yields the transformed system equation

$$\log(s_{t+1}) = \log[F_t(s_t - u_t)] + \log(w_t).$$

In this equation, the error enters additively on the logarithmic scale. For empirical investigations, we often assume that  $\log(w_t)$  has a normal distribution with mean 0 and variance  $\sigma^2$ ; consequently,  $w_t$  has a lognormal distribution.

Generalization of this model include multiple-goods economies and dynamic competitive equilibrium models.

### 3.5.4 Linear Quadratic Control

Linear systems with quadratic costs play a central role in control engineering. Such problems arise when a controller attempts to guide an observable system along a predetermined trajectory or maintain a system near some desirable operating level. Departures from this trajectory are penalized proportionally to the square of the distance. We refer to this as the *linear regulator* or *linear quadratic control* problem.

The state of the system  $s_t \in S = R^k$  follows the linear dynamic equation

$$s_{t+1} = A_t s_t + B_t u_t + C_t w_t, \quad (3.5.8)$$

where the control  $u_t \in U_{s_t} = R^m$ , the disturbance  $w_t$  is the realization of an  $R^n$ -valued random variable with mean 0, and covariance matrix  $\Sigma_t$ ,  $A_t$  is a  $k \times k$  matrix,  $B_t$  is a  $k \times m$  matrix, and  $C_t$  is a  $k \times n$  matrix.

Normalizing the model so that the desirable trajectory is the origin (the zero vector in  $R^k$ ), the cost components consist of a quadratic penalty for deviations from the origin and a quadratic penalty for using control  $u_t$ . In guidance applications, we may think of the  $u_t$  as fuel utilization at decision epoch  $t$ . We represent the combined cost

in any period by

$$r_t(s_t, u_t) = s_t^T Q_t s_t + u_t^T R_t u_t, \quad (3.5.9)$$

where  $Q_t$  is a  $k \times k$  positive definite matrix,  $R_t$  is an  $m \times m$  positive definite matrix, and the superscript  $T$  denotes transposition. In a finite-horizon problem, the ultimate deviation from the objective is penalized at cost  $s_N^T Q_N s_N$ . The objective is to choose a sequence of controls to minimize total expected cost.

### 3.6 BANDIT MODELS

The expression *bandit model* refers to a sequential decision model in which, at each decision epoch, the decision maker observes the state of each of  $K$  Markov reward processes and, based on the states, the transition probabilities, and rewards of each, selects a process to use in the current period. The selected process changes state according to its transition probabilities, and the state of all other processes remain fixed.

The effects of choosing process  $i$  when it is in state  $s^i \in S^i$ ,  $i = 1, 2, \dots, K$  are that

1. it changes state according to a transition law  $p_i'(j^i | s^i)$ ,
2. the decision maker receives a reward  $r_i'(s^i)$ , and
3. all other processes remain in their current state.

At each decision epoch, the state of process  $i$  changes while all others remain fixed, so that a process which is attractive to choose at the current decision epoch may be in an unattractive state at the subsequent decision epoch. The decision maker's objective is to choose a process selection sequence to maximize some function of the expected total reward over the planning horizon.

This model receives its colorful name because it applies (in a rather complicated fashion) to the decision problem facing a casino gambler when deciding whether or not to play a particular slot machine or "one-armed bandit" when the probability of winning is unknown. When  $K > 1$ , we refer to the model as a multiarmed bandit; the gambler must choose between  $K$  different slot machines with unknown win probabilities.

After formulating this model as a Markov decision process, we describe some applications.

#### 3.6.1 Markov Decision Problem Formulation

We formulate this model as a Markov decision problem.

**Decision epochs:**

$$T = \{1, 2, \dots, N\}, \quad N \leq \infty.$$

**States:**

$$S = S^1 \times S^2 \times \dots \times S^K.$$

**Actions** (process to be selected):

$$A_s = \{1, 2, \dots, K\}, \quad s \in S.$$

**Rewards:**

$$r_i((s^1, s^2, \dots, s^K), a) = r_i^i(s^i) \quad \text{if } a = i.$$

**Transition probabilities:**

$$p_i((u^1, \dots, u^k) | (s^1, \dots, s^k), i) = \begin{cases} p(j^i | s^i) & u^i = j^i \text{ and } u^m = s^m \text{ for } m \neq i \\ 0 & u^m \neq s^m \text{ for some } m \neq i \end{cases}$$

Actions correspond to the process chosen in the current period and rewards and transition probabilities reflect this choice. Observe that when action  $i$  is selected, only component  $i$  of the state vector changes. A Markovian decision rule assigns a specific process to each possible configuration of process states.

The Markov decision process formulation requires a very large state space, making direct computation prohibitive. The beauty of this model is that, under most optimality criteria, the optimal policy can be computed by decomposing it into  $K$  one-dimensional problems and analyzing each separately.

### 3.6.2 Applications

Applications of the bandit model include project selection, gambling, allocation of treatments in a clinical trial, random search, and job scheduling. We describe the details of the first three applications below.

#### *Project Selection*

In any period a decision maker (such as the author of this book) can choose to work on any of  $K$  available projects (such as writing a book, working on a research paper, responding to e-mail, or exercising). The state  $s^i$  represents the degree of completion, or status, of project  $i$ , and  $r^i(s^i)$  the expected reward for working on project  $i$  for one period. For example, when  $S^i = [0, 1]$ ,  $s^i \in S^i$  represents the fraction of the project that has been completed. Hopefully, working on a project for an additional period will draw us closer to completion, but, as we all know, there is a chance that we might not be any further along and in fact we may be further behind. If state  $C$  corresponds to completion of the project, and a reward of  $R^i$  is received only upon completion of the project, then  $r^i(s^i) = R^i p^i(C | s^i)$ . Since most of us can only do one thing at a time, by working on project  $i$ , its state changes but the status of all other projects remain unaltered.

#### *Gambling—The One-Armed Bandit*

This model is considerably more complicated than any model encountered so far because it requires the state space to be a set of functions. A gambler in a smoke-filled casino may either pay  $c$  units and pull the lever on a slot machine that pays one unit with probability  $q$  and zero units with probability  $1 - q$ , or decide not to play. Unfortunately, the gambler does not know  $q$ ; instead she summarizes her

beliefs regarding its values through a probability density  $f(q)$ , on  $[0, 1]$ . By playing the game several times, the gambler acquires information about the distribution of  $q$  and revises her assessed probability density accordingly.

We model this as a bandit model as follows. Let process 1 correspond to the “do not play” option;  $S^1 = \{1\}$ ,  $r^1(1) = 0$ , and  $p^1(1|1) = 1$ . Let process 2 correspond to the “play” option. We let the state of the system be the gambler’s current assessment of the density of  $q$ ; that is,

$$S^2 = \{f: f \text{ is a density with support on } [0, 1]\}.$$

The quantity  $f$  is referred to as a *prior distribution* in a Bayesian setting. Since the probability that the gambler wins one unit is  $q$ ,

$$r^2(f) = \int_0^1 qf(q) dq - c = E_f[Q] - c,$$

where  $Q$  denotes the random variable with density  $f$ , and  $E_f$  an expectation with respect to  $f$ .

Transitions in this model take place between densities according to Bayes’ rule. If the current estimate of the density is  $f$  and the gambler decides to play and wins, then the revised density  $f'$  (posterior probability) will put more weight on larger values of  $p$  than  $f$ . The transition probabilities between densities satisfy

$$p^2(f'|f) = \begin{cases} E_f[Q], & f' = \frac{qf(q)}{E_f[Q]} \\ 1 - E_f[Q], & f' = \frac{(1-q)f(q)}{1 - E_f[Q]}. \end{cases}$$

To see this, observe that the probability of the gambler winning in the current round is  $E_f[Q]$ , and, if she wins, she revises her probability assessment of the value of  $Q$  according to Bayes’ theorem as follows. Writing  $Q \approx q$  instead of  $Q \in [q, q + dq]$ , we have

$$\begin{aligned} P(Q \approx q | \text{win}) &= \frac{P(\text{win} | Q \approx q) P(Q \approx q)}{\int_0^1 P(\text{win} | Q \approx q) P(Q \approx q) dq} \\ &= \frac{qf(q)}{\int_0^1 qf(q) dq}. \end{aligned} \tag{3.6.1}$$

This formulation is quite different than those encountered previously in that the system state corresponds to a non-negative real-valued function on  $[0, 1]$ . This makes the problem computationally infeasible. In practice such generality is unnecessary; instead we may adopt either of the following two approaches:

1. Choose a parametric family of densities that is closed under the calculation in (3.6.1). Sometimes this is referred to as a *conjugate family* of distributions. For

example, if the initial state  $s_1^2 = f_1$  is a beta density, that is

$$f_1(q) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} q^{\alpha-1}(1-q)^{\beta-1}, \quad 0 < p < 1,$$

where  $\alpha > 0$  and  $\beta > 0$  and  $\Gamma(\cdot)$  denotes the gamma function ( $\Gamma(\alpha) = (\alpha - 1)!$  if  $\alpha$  is a non-negative integer), then  $s_n^2$  will be a beta density for all  $n$  and hence can be described completely by the number of trials, the number of wins, and the initial parameters. Suppose that after  $n$  plays of the game there have been  $k$  wins, then  $s_{n+1}^2$  will be a beta density with parameters  $\alpha + k$  and  $\beta + n - k$ . Therefore, the state space can be represented instead as either  $S^2 = (0, \infty) \times (0, \infty)$  corresponding to the parameters of the beta distribution or  $S^2 = \{0, 1, \dots\} \times \{0, 1, \dots\}$  corresponding to the number of wins and losses.

2. In general, represent the state space as  $S^2 = \{0, 1, \dots\} \times \{0, 1, \dots\}$  corresponding to the number of wins and losses and recompute  $f$  in state  $(k, n - k)$  according to

$$f(q) = \frac{q^k(1-q)^{n-k}f_1(q)}{\int_0^1 q^k(1-q)^{n-k}f_1(q) dq}$$

prior to each play of the game.

### Sequential Clinical Trials

Controlled clinical trials are the basis of most applied medical research. They date back to James Lind's studies of the use of citrus fruits for prevention of scurvy among British seamen. In a clinical trial, the investigator compares a new treatment with unknown properties to an existing treatment with known properties, or to an inactive treatment referred to as a placebo.

We provide a bandit model for a controlled clinical trial in which successive subjects are allocated to one of two treatments sequentially. We assume that there are two outcomes of the treatment, one of which can be designated as a "success" and the other as "failure," and, furthermore, that the treatment is fast acting so that its outcome is known prior to allocating the next subject to a treatment. This assumption may be valid in a trial comparing headache remedies, but not in oncology (cancer) trials, in which success is not determined until a subject has been in remission for several years.

Let process 1 represent the new treatment and process 2 the existing or placebo treatment, and let  $q^i$  represent the respective success probabilities. We assume that  $q^1$  is unknown with prior distribution  $f^1$ , and that  $q^2$  is unknown with prior distribution  $f^2$ . The prior distribution on the success probability of existing therapy may be derived from published studies or investigator experience. We assume a distribution for  $q^2$  instead of a fixed probability because responses vary considerably from subject to subject and study to study. Further, it might be expeditious to initially assume that  $f^1 = f^2$ .

Let  $c^i$  denote the cost of treating a subject with treatment  $i$ ,  $i = 1, 2$ , and  $R$  the reward associated with successful treatment. Note that instead of a reward for a



success, we might include the cost to society (and the individual) of a treatment failure. Also  $c^1$  and  $c^2$  might differ considerably.

We represent the state of each process as the density of success probabilities. The model for the one-armed bandit in Sec. 3.6.1 now applies to each treatment separately, as do the modifications discussed at the end of that section. Consequently this is a two-armed bandit problem.

This approach to designing clinical trials differs from actual practice since we do not address the problem from the hypothesis-testing perspective. Instead we view the trial as continuing indefinitely, with the objective being to determine an adaptive treatment allocation sequence to maximize total expected reward or minimize expected cost. Once the evidence is convincing that one treatment dominates the other, then the allocation rule will assign subjects to that treatment. Of course, in practice we must be sure to have ample numbers of subjects receiving each treatment to ensure that results are convincing to the medical community, not biased by erroneous prior information and generalizable to a larger population.

### 3.6.3 Modifications

The bandit model above may be modified in several ways.

#### *Restless Bandits*

This model relaxes Assumption 3 at the beginning of this section, by allowing the other processes to change states between decision epochs. The effect of this modification is that the transition probability structure of Sec. 3.6.1 must be made more general. This modification is appropriate for the project selection models, but not realistic for models in which the state space represents a probability density. In the restless bandit model, the decomposition approach may not lead to optimal policies but can be used to obtain policies that are close to optimal.

This formulation might provide a conceptual model for a sequential clinical trial in which there is a time lag between treatment and its outcome.

#### *Arm-Acquiring Bandits*

In this model, the number of processes does not remain fixed over the decision-making horizon. Instead new arms (projects) become available in each decision epoch according to a Poisson process. Therefore  $K_t$ , the number of projects available at time  $t$ , increases with  $t$ . Of course this model may be modified so that projects also disappear.

#### *The Tax Problem*

In this model, Assumption 2 is modified. Instead of receiving a reward for choosing process  $k$ , a tax is charged for all projects that remain idle. That is, if project  $j$  is not selected at decision epoch  $t$ , the system incurs a cost  $c^j(s^j)$  for that period. Therefore the cost of choosing project  $k$  when the state of the projects is  $[s^1, s^2, \dots, s^K]$  is  $\sum_{k \neq j} c^j(s^j)$ .

#### *Continuous-Time Bandits*

Instead of choosing processes periodically, the decision maker might instead be able to switch between processes at any time (preemptively) or at fixed random times

corresponding to state changes in a process (nonpreemptively). In the latter case, the individual processes may be better modeled by semi-Markov processes and methods (Chap. 11).

### 3.7 DISCRETE-TIME QUEUEING SYSTEMS

Queueing systems model a wide range of phenomenon including multi-user computer systems, communication networks, manufacturing systems, supermarket checkout lines, or even a waiting room at a physician's office. A controlled queueing system consists of three components: queues, servers, and controllers. In a single-server uncontrolled queueing system, jobs arrive, enter the queue, wait for service, receive service, and depart from the system. In the absence of a controller, system behavior can be quite erratic with periods of long queues followed by periods in which the server remains idle. Controllers may improve system utilization by reducing expected throughput time or queue length.

In this section we consider discrete-time queueing systems, even though continuous-time formulations are usually more natural. Chapter 11 provides some continuous-time queueing control models.

We assume the system is observed every  $\eta > 0$  units of time and that the controller makes decisions at times  $0, \eta, 2\eta, \dots$ . These are the decision epochs for the model. We sometimes refer to the period between two decision epochs as a *time slot* and refer to time as *slotted*.

In this section, we formulate models for two types of controlled single-server queueing systems; one in which the controller decides on the number of arriving jobs to admit to the eligible queue (admission control), and one in which the controller adjusts the service rate (service rate control). For both models, we assume independent probability distributions for the number of arrivals and the number of service completions during each time slot. Section 1.5 and the problems at the end of this chapter provide other examples of queueing control models.

#### 3.7.1 Admission Control

Jobs arrive for service and are placed in a "potential job queue." At each decision epoch, the controller observes the number of jobs in the system ("eligible queue" plus server), and on this basis decides how many jobs, if any, to admit from the potential job queue into the eligible queue. These jobs eventually receive service; those not admitted never enter the system. (Fig. 3.7.1).

We now provide a Markov decision process model for this system. Let  $X_t$  represent the number of jobs in the system immediately prior to decision epoch  $t$ , and  $Z_t$  the number of arrivals in period  $t$ . Arriving jobs during period  $t - 1$  enter the "potential job queue." At decision epoch  $t$ , the controller admits  $u_t$  jobs from the potential job queue into the system. Let  $Y_t$  denote the number of "possible service completions" during period  $t$ . We include the adjective "possible" because if  $Y_t$  exceeds  $X_t + u_t$ , then only  $X_t + u_t$  jobs are serviced. Table 3.7.1 represents the content of the components of the queueing system throughout period  $t$ . In it  $t +$  denotes a point in time immediately after the control has been implemented but prior to any service completions.

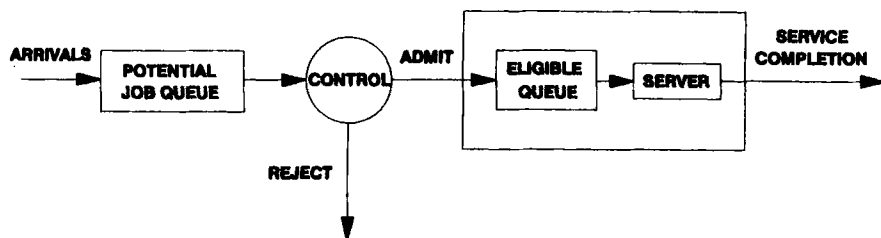


Figure 3.7.1 A queueing system with admission control.

We represent the system state by the pair  $(X_t, V_t)$ , where  $V_t$  denotes the content of the potential job at decision epoch  $t$ . Following the notation of Sec. 3.5, the system equation (3.5.1) for these two components may be written as

$$X_{t+1} = [X_t + u_t - Y_t]^+,$$

$$V_{t+1} = Z_t.$$

Note that we can only admit as many jobs as are present, so that  $0 \leq u_t \leq V_t$ . Because of this, we must retain  $V_t$  in the state description.

The random variable  $Y_t$  assumes non-negative integer values and follows a time invariant probability distribution  $f(n) = P(Y_t = n)$ ,  $t = 0, 1, \dots$ , and  $Z_t$  assumes non-negative values and follows a time invariant probability distribution  $g(n) = P(Z_t = n)$ ,  $t = 0, 1, \dots$ . We assume no partial service completions during any period and that all serviced jobs depart the system at the end of the period.

The stationary reward structure consists of two components: a constant reward of  $R$  units for every completed service, and an expected holding cost of  $h(x)$  per period when there are  $x$  jobs in the system.

A Markov decision process formulation follows.

**Decision epochs:**

$$T = \{0, \eta, 2\eta, \dots, N\eta\}, \quad N \leq \infty.$$

**States** ( $S_1$  is the number in the system, and  $S_2$  the number in the potential job queue):

$$S = \{0, 1, \dots\} \times \{0, 1, \dots\} = S_1 \times S_2.$$

Table 3.7.1

| Time    | Potential Job Queue | System                |
|---------|---------------------|-----------------------|
| $t$     | $Z_{t-1}$           | $X_t$                 |
| $t +$   | 0                   | $X_t + u_t$           |
| $t + 1$ | $Z_t$               | $[X_t + u_t - Y_t]^+$ |

**Actions** (number of jobs to admit):

$$A_{s_1, s_2} = \{0, 1, \dots, s_2\}.$$

**Rewards:**

$$r_t(s_1, s_2, a) = R \cdot E\{\min(Y_t, s_1 + a)\} - h(s_1 + a). \quad (3.7.1)$$

**Transition probabilities:**

$$p_t(s'_1, s'_2 | s_1, s_2, a) = \begin{cases} f(s_1 + a - s'_1)g(s'_2) & a + s_1 > s'_1 > 0 \\ \left[ \sum_{i=s_1+a}^{\infty} f(i) \right] g(s'_2) & s'_1 = 0, a + s_1 > 0 \\ g(s'_2) & s'_1 = a + s_1 = 0 \\ 0 & s'_1 > a + s_1 \geq 0. \end{cases}$$

Some comments on the model formulation follow. Note that the set of states need not be finite and that the reward is bounded above but may be unbounded below. It is natural to formulate the model as one in which the reward depends on the system state at the next decision epoch, as in (2.1.1), since it depends on the number of service completions in period  $t$ . That is the basis for the expression (3.7.1). The quantity  $\min(Y_t, s_1 + a)$  gives the number of jobs completed in period  $t$ . To see this, note that if the number of possible services  $Y_t$  exceeds the number of eligible jobs  $s_1 + a$ , only  $s_1 + a$  receive service. The expectation of this quantity satisfies

$$E\{\min(Y_t, s_1 + a)\} = \sum_{i=1}^{s_1+a-1} if(i) + (s_1 - a) \sum_{i=s_1+a}^{\infty} f(i).$$

Note that this is a concave, nondecreasing function of  $a$ . To derive the transition probabilities, we use the same argument and note that the number of arrivals is assumed to be independent of the number of service completions.

**Example 3.7.1.** We now consider a special case of this model with deterministic service rate of one job per period; that is,  $Y_t \equiv 1$ , so that  $f(1) = 1$  and  $f(j) = 0$  for  $j \neq 1$ . For this model (3.7.1) becomes

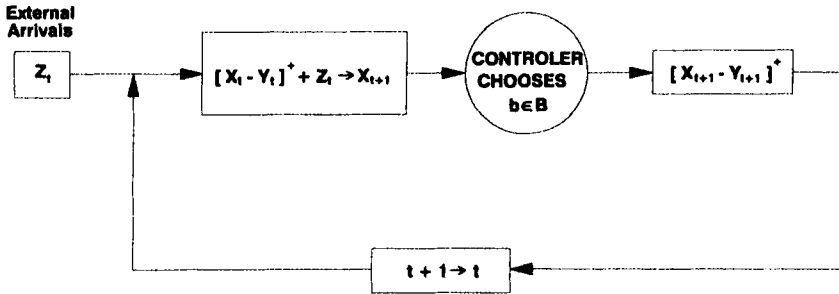
$$r_t(s_1, s_2, a) = \begin{cases} R - h(s_1 + a) & s_1 + a > 0 \\ 0 & s_1 + a = 0, \end{cases}$$

and the transition probabilities reduce to

$$p_t(s'_1, s'_2 | s_1, s_2, a) = \begin{cases} g(s'_2) & s'_1 = s_1 + a - 1 \geq 0 \text{ or } s'_1 = s_1 + a = 0 \\ 0 & \text{otherwise.} \end{cases}$$

### 3.7.2 Service Rate Control

In this model, all arriving jobs enter the eligible queue and are served on a first come first serve (FCFS) basis. At each decision epoch, the controller observes the number



**Figure 3.7.2** Dynamics in a service rate controlled queueing system.

of jobs in the system and selects the service rate from a set of probability distributions indexed by elements of a set  $B$ . In most applications, we model the service distribution by a parametric family such as the geometric or Poisson, so that  $B$  might represent the set of possible parameter values. For each  $b \in B$ , we let  $f_b(n)$  denote the corresponding probability function. As in Sec. 3.7.1, let  $Y_t$  represent the number of possible service completions in time slot  $t$ .

The reward structure for this model augments that of the model in Sec. 3.7.1 by including a fixed cost  $K$  for changing the service rate and a per period cost  $d(b)$  for using service distribution  $f_b(\cdot)$  for one period. Suppose the controller uses  $f_b(\cdot)$  in period  $t$  and uses  $f_{b'}(\cdot)$  in period  $t + 1$ ; then the extra cost component  $c(b, b')$  equals

$$c(b, b') = \begin{cases} K + d(b') & b' \neq b \\ d(b) & b' = b. \end{cases}$$

The Markov decision process formulation below assumes that any arrivals during a period do not receive service in that period. Consequently the system equation becomes

$$X_{t+1} = [X_t - Y_t]^+ + Z_t, \quad (3.7.2)$$

where the distribution of  $Y_t$  depends on  $u_t$ , the index of the distribution chosen in period  $t$  (Fig. 3.7.2). Otherwise the notation follows that of Sec. 3.7.1.

**Decision epochs:**

$$T = \{0, \eta, \dots, N\eta\}, \quad N \leq \infty.$$

**States** (number of jobs in the system and the current service rate):

$$S = \{0, 1, \dots\} \times B.$$

**Actions:**

$$A_{s,a} = B.$$

**Rewards:**

$$r_t((s, a), a') = R \cdot E\{\min(Y_t, s)\} - h(s) - c(a, a').$$

**Transition probabilities:**

$$p_i(s', a' | (s, a), a') = \sum_{n=0}^{s'-1} f_{a'}(n) g(s' - s + n) + \left[ \sum_{n=s}^{\infty} f_{a'}(n) \right] g(s').$$

We derive the transition probability by appealing to (3.7.2) and using the law of total probabilities. The first term in the expression for the transition probability is the sum over  $n$ , the number of service completions, of the product of the probability of  $n$  service completions and  $s' - s + n$  arrivals, while the second term gives the probability of at least  $s$  service completions and  $s'$  arrivals.

**BIBLIOGRAPHIC REMARKS**

Bellman's path breaking book (1957) contains many interesting and novel applications, especially in the problem sections. They include optimal search, sequential allocation, experimental design, optimal replacement, scheduling, and inventory control. Many of these appeared in his earlier publications. Howard (1960) provides some colorful examples including strategy choice in baseball and taxicab routing. His automobile replacement model has often been used as a test problem for computational algorithms.

Inventory models have their origins in work of Arrow, Harris, and Marschak (1951) and Dvoretzky, Kiefer, and Wolfowitz (1952). These papers were a driving force in popularizing dynamic programming methods. The former paper introduces the concept of a penalty cost as follows:

"The organization—whether commercial or noncommercial—has a general idea of the value it would attach to the damage that would be caused by the nonavailability of an item. It knows the cost and the poorer performance of emergency substitutes. The penalty for depleted stocks may be very high: 'A horse, a horse, my kingdom for a horse,' cried defeated Richard III."

Arrow, Karlin, and Scarf (1958) provide many interesting models and applications. Conditions for the optimality of  $(s, S)$  or  $(\sigma, \Sigma)$  policies were first provided by Scarf (1960). There exists a voluminous literature on the inventory model and its applications. Noteworthy surveys include Veinott (1966) and Porteus (1991).

Early papers on shortest-path models include Minty (1957), Dantzig (1957), and Dijkstra (1959). Ford and Fulkerson (1962) made important contributions to theory and methods for these models. A review article by Pollack and Wiebenson (1960) and papers by Dreyfus (1969) and Denardo and Fox (1979) provide extensive bibliographies on such models. Denardo's text (1982) serves as an excellent reference for many aspects of deterministic dynamic programming models. The critical path model has its origins of the PERT charts of the early days of operations research. Kelly (1961) and Bigelow (1962) summarize and review some of the earlier references on this model and its application. This technique has become more popular of late with many readily available PC software packages. Sequential allocation models are among the first applications of dynamic programming (Bellman, 1957). Apparently the application to constrained maximum likelihood estimation is new.

Optimal stopping problems have roots in the sequential hypothesis testing models of Wald (1947) and subsequent papers by Wald and Wolfowitz (1948), Arrow, Blackwell, and Girshick (1949). Karlin (1962) proposed and solved the asset selling problem. The secretary problem, or candidate problem, originates in Cayley (1875), where he formulates it in the following way:

"A lottery is arranged as follows:—There are  $n$  tickets representing  $a, b, c$  pounds respectively. A person draws once; looks at his ticket; and if he pleases, draws again (out of the remaining  $n - 1$  tickets); looks at his ticket, and if he pleases draws again (out of the remaining  $n - 2$  tickets); and so on, drawing in all not more than  $k$  times; and he receives the value of the last drawn ticket. Supposing he regulates his drawings in the manner most advantageous to him according to the theory of probabilities, what is the value of his expectation?"

The model presented in this chapter relates to those studied by Gilbert and Mosteller (1966) and Chow, Moriguti, Robbins, and Samuels (1964). This appealing problem has generated a voluminous literature which explores many variants; recent references include Hlynka and Sheahan (1988) and Reeves and Flack (1988). The book by Chow, Robbins, and Siegmund (1971) provides a comprehensive study of the optimal stopping problem.

Option pricing models have been studied extensively in the financial literature. The classical references are the papers by Black and Scholes (1973) and Merton (1973). Cox and Rubenstein (1985) is a comprehensive book on this topic, with an extensive bibliography.

The discrete-time dynamic equation approach has its origins in continuous-time control theory and the calculus of variations. Bellman (1961) provides a lively presentation discussing some of its origins and many applications. The advent of computers and Bellman's dynamic programming approach seem to have gotten the ball rolling on the discrete-time dynamic model. Its earliest applications were to inventory models as described above; Arrow, Harris, and Marschak (1951) provide a dynamic equation for the inventory state. Applications in economics are the subject of Stokey and Lucas (1990). Simon (1956) and Theil (1957) provide early results on the linear quadratic control model with the latter paper among the first to use matrix methods. A special issue of *IEEE Transactions of Automatic Control*, edited by Athans (1971) focuses on the linear quadratic control problem providing an overview of theory together with applications in chemical process control and satellite navigation. The excellent book by Bertsekas (1987) approaches dynamic programming from the system equation perspective.

Bandit problems originate in the work of Thompson (1933 and 1935). They seem to have evolved fairly independently in the statistical and engineering/operations research literature with different emphases. The former approach is reflected in the gambling model, the latter in the project selection model. Gittins and Jones (1974) demonstrate the optimality of dynamic allocation indices and hence the separability of the model. Whittle (1980c) refines and extends these results using Markov decision process methods. Bandit problems are the subject of the book by Berry and Fristedt (1985). It contains a thorough annotated bibliography. Colton (1963) provides an early reference to the application of bandit models in clinical trials.

Queueing control represents an active area of current research and one of the main areas of application of Markov decision process methodology. Early research in

queueing control primarily concerned single-server models; examples include Yadin and Naor (1967), Heyman (1968), Naor (1969), Sobel (1969), and Bell (1969). Sobel (1974) and Crabill, Gross, and Magazine (1977) survey this research. Stidham (1985) surveys admission control models paying particular attention to Markov decision process approaches. Most of the recent work on queueing control has been concerned with controlling networks of queues and allocating jobs of different classes to one of several servers. The book by Walrand (1988) and his handbook article (1991) provide good overviews of formulations and results for these models.

White (1985b) and (1988a) surveys a wide range of applications of Markov decision processes. He provides short summaries of each application and comments on the extent of implementation. Some areas of application included in the survey are fisheries, forestry, water resources, airline booking, and vehicle replacement. In the 1985 paper, White notes that

“I identified only five pieces of work that have been directly implemented . . .”

However, he goes on to conclude that

“... it appears that in the last few years more efforts are being made to model phenomena using Markov decision processes.”

## PROBLEMS

- 3.1. Derive transition probabilities and expected one-period rewards for the randomized Markov policy of Sec. 3.1.
- 3.2. Consider a generalization of the inventory model of Sec. 3.2 in which unfilled orders may be backlogged indefinitely with a cost of  $b(u)$  if  $u$  units are backlogged for one period. Assume revenue is received at the end of the period in which orders are placed and that backlogging costs are charged only if a unit is backlogged for an entire month, in which case the backlogging cost is incurred at the beginning of that month.
  - a. Identify the state space and derive transition probabilities and expected rewards.
  - b. Calculate  $r_i(s, a)$  and  $p_i(j|s, a)$  for the above model with  $b(u) = 3u$  and all other model parameters identical to Sec. 3.2.2.
  - c. Formulate the model under the assumption that payment for an order is received at the end of the month in which the order is filled.
- 3.3. Consider the network in Fig. 3.3.1, but suppose in addition that it contains an arc connecting node 1 to node 5 with length 10. Give a modified network in which nodes are grouped by stages. Be sure to indicate all arc lengths in the modified network.
- 3.4. Formulate the following stochastic version of the shortest-route problem in Fig. 3.3.1 as a Markov decision problem. In this model, when the decision maker chooses to traverse a particular arc, there is a positive probability of actually



traversing an alternative arc. The probabilities are as follows. If there are two adjacent arcs, such as at node 1, and the decision maker chooses to traverse the arc between nodes 1 and 2, then the probability of traversing that arc is 0.6 and the probability of traversing each of the adjacent arcs is 0.2. When there is only one adjacent arc, then the probability of traversing the chosen arc is 0.7. If there are no adjacent arcs, the chosen arc is traversed with certainty.

- 3.5. Provide a graphical representation for a deterministic version of the stochastic inventory model of Sec. 3.2, in which the demand is known with certainty prior to placing an order. Assume parameter values of Sec. 3.2.2, and that the demand in period 1 is two units and demand in period 2 is one unit.
- 3.6. In most situations where critical path analyses are applied, the length of time to complete a task is not known with certainty. Formulate such a model as a longest-route model and discuss appropriate and practical optimality criteria for such a model.
- 3.7. Provide a graphical representation for the sequential allocation model of Sec. 3.3 in which we restrict each  $x_n$  to be an integer.
- 3.8. At each round of a game of chance, a player may either pay a fee of  $C$  units and play the game or else quit and depart with his current fortune. Upon paying the fee he receives a payoff which is determined by a probability distribution  $F(\cdot)$ . The player's objective is to leave the game with the largest possible fortune.
  - a. Formulate this gambling problem as an optimal stopping problem.
  - b. Explicitly give transition probabilities and rewards when the cost of playing the game is one unit and the player receives two units with probability  $p$  and zero units with probability  $1 - p$  if he plays.
- 3.9. When going out for dinner, I always try to park as close as possible to the restaurant. I do not like to pay to use a parking lot, so I always seek on-street parking. Assume my chosen restaurant is situated on a very long street running east to west which allows parking on one side only. I approach the restaurant from the east starting at a distance of  $Q$  units away. Because traffic is heavy I can only check one potential parking spot at a time.  
 Formulate the parking problem as an optimal stopping problem. Assume the street is divided into one-car-length sections, and the probability that a parking spot at distance  $s$  from the restaurant is unoccupied is  $p_s$ , independent of all others.
- 3.10. Formulate a variant of the secretary problem in which the objective is to maximize the probability that the decision maker chooses
  - a. one of the two best candidates,
  - b. one of the  $k$  best candidates.
- 3.11. Formulate the secretary problem when the employer's objective is to minimize the expected rank of the candidate who receives an offer. Assume the best candidate has rank 1 and the least desirable candidate has rank  $N$ .

- 3.12.** Provide a dynamic equation formulation for a two-product inventory model in which there is a constraint on total warehouse capacity. Derive the appropriate transition probabilities.
- 3.13.** Formulate the inventory model in which there is a random lag between placing and receiving orders; that is, the lead time is a random variable with a known probability distribution.
- 3.14.** Provide a dynamic equation for an inventory model in which a fixed random fraction  $q$  of the stock deteriorates and is unavailable for use in the subsequent period.
- 3.15.** (Love, 1985; Controlling a two city rental car agency) A rental car agency serves two cities with one office in each. The agency has  $M$  cars in total. At the start of each day, the manager must decide how many cars to move from one office to the other to balance stock. Let  $f_i(q)$ ;  $i = 1$  and  $2$ ; denote the probability that the daily demand for cars to be picked up at city  $i$  and returned to city  $i$  equals  $q$ . Let  $g_i(q)$ ,  $i = 1$  and  $2$ ; denote the probability that the daily demand for "one-way" rentals from city  $i$  to the other equals  $q$ . Assume that all rentals are for one day only, that it takes one day to move cars from city to city to balance stock, and, if demand exceeds availability, customers go elsewhere. The economic parameters include a cost of  $K$  per car for moving a car from city to city to balance stock, a revenue of  $R_1$  per car for rentals returned to the rental location, and  $R_2$  per car for one-way rentals.
- Provide dynamic equations for this system and formulate it as a Markov decision process.
  - Suppose, in addition to relocating cars to balance stock, the manager may reserve a certain number of cars for one-way rental only. Formulate this modified model as a Markov decision process.
- 3.16.** (Stengos and Thomas, 1980; The blast furnace problem) Two identical machines are used in a manufacturing process. From time to time, these machines require maintenance which takes three weeks. Maintenance on one machine costs  $c_1$  per week while maintenance on both machines costs  $c_2$  per week. Assume  $c_2 > 2c_1$ . The probability that a machine breaks down if it has been  $i$  periods since its last maintenance is  $p_i$ , with  $p_i$  nondecreasing in  $i$ . Maintenance begins at the start of the week immediately following breakdown, but preventive maintenance may be started at the beginning of any week. The decision maker must choose when to carry out preventive maintenance, if ever, and, if so, how many machines to repair. Assume the two machines fail independently.
- What is the significance of the condition  $c_2 > 2c_1$ ?
  - Formulate this a Markov decision problem.
  - Provide an educated guess about the form of the optimal policy when the decision maker's objective is to minimize expected operating cost.
- 3.17.** (Rosenthal, White, and Young, 1978; A dynamic location model) A repairman who services  $Q$  facilities moves between location  $s$  and location  $j$  in any period

according to the stationary transition probability  $p(j|s)$ . An equipment trailer which carries spare parts and tools may be located at any one of  $M$  sites. If the trailer is at site  $m$  and the repairman is at facility  $j$ , the cost of obtaining material from the trailer is  $c(m, j)$ . The cost of moving the trailer from site  $m$  to site  $j$  is  $d(m, j)$ . The decision maker's objective is to dynamically relocate the trailer so as to minimize expected costs. Assume that the decision maker observes the location of the repairman and trailer, relocates the trailer, and then the repairman moves and services a facility.

Formulate this as a Markov decision process clearly identifying states, actions, transition probabilities, and rewards. What other timing scenarios might be realistic?

- 3.18.** (Stokey and Lucas, 1989; Employment seeking) At the start of each week, a worker receives a wage offer of  $w$  units per week. He may either work at that wage for the entire week or instead seek alternative employment. If he decides to work in the current week, then at the start of the next week, with probability  $p$ , he will have the same wage offer available while, with probability  $1 - p$ , he will be unemployed and unable to seek employment during that week. If he seeks alternative employment, he receives no income in the current week and obtains a wage offer of  $w'$  for the subsequent week according to a transition probability  $p_t(w'|w)$ . Assume his utility when receiving wage  $w$  in week  $t$  is  $\Psi_t(w)$ .

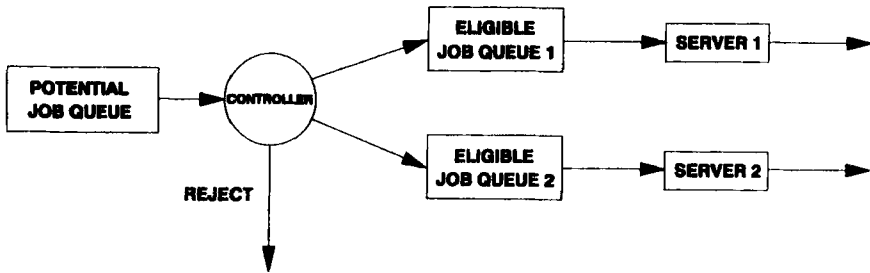
- a. Formulate the worker's decision problem as a Markov decision process in which his objective is to maximize his total expected utility over a finite horizon.
- b. Formulate the worker's decision problem as a bandit problem.

- 3.19.** (Morrison, 1976 and 1986; A hockey coach's dilemma) A hockey team is losing with a few minutes remaining in a game. The coach may either replace his goalie by an extra player or leave his goalie in. By removing the goalie he increases the probability that his team scores the *only* goal in the next  $h$  seconds from  $p_G$  to  $p_{NG}$ ; however, the probability of the opposing team scoring the only goal in that  $h$ -second interval increases from  $q_G$  to  $q_{NG}$ , where  $q_{NG}$  is much larger than  $q_G$ . Of course,  $p_G + q_G < 1$  and  $p_{NG} + q_{NG} < 1$ . Assume that events in successive  $h$ -second intervals are independent.

Formulate the problem facing the coach as a discrete-time Markov decision process in which the objective is to maximize the probability of achieving at least a tie. To do this, let the state represent the point differential between the two teams and let the terminal reward be 1 if the point differential is non-negative and 0 if it is negative. (Note that a continuous-time formulation is more natural for this problem.)

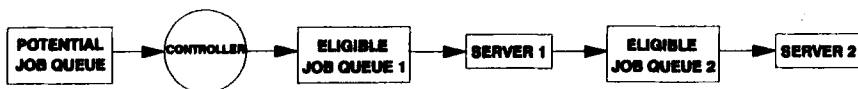
- 3.20.** Provide system equations and a Markov decision process formulation for a modification of the admission control problem of Sec. 3.7.1, in which each job in the "potential job" queue which is not admitted for service returns in the next period, with probability  $p$  independent of the number of new arrivals and the number of jobs in the "potential job queue."

- 3.21.** Provide system equations and a Markov decision process formulation for the following modification of the admission control problem of Sec. 3.7.1, in which there are two parallel servers each accompanied by its own eligible job queue. In this model, the controller may allocate jobs from the potential job queue to either eligible job queue or decide to not serve a job in which case it leaves the system. See Fig. P3.21. Assume that the “possible service distribution” for server  $i$ ,  $i = 1$  and  $2$ , equals  $f_i(n)$ , and that the holding cost equals  $h_i(x)$  per period when there are  $x$  jobs in total at server  $i$  and its eligible queue. Provide a guess regarding the most efficient operating policy for this system.



**Figure P3.21** Admission control to a parallel queueing system.

- 3.22.** Provide system equations and a Markov decision process formulation for the following modification of the admission control problem of Sec. 3.7.1, in which there are two servers in series each accompanied by its own eligible job queue. In this model, the controller may allocate some or all jobs from the potential job queue to the first queue only or decide to refuse to service the job, in which case it leaves the system. Jobs which complete service at the first queue immediately join the eligible job queue preceding the second server. See Fig. P3.22. Assume that the “possible service distribution” for server  $i$ ,  $i = 1$  and  $2$ , equals  $f_i(n)$ , and that the holding cost equals  $h_i(x)$  per period when there are  $x$  jobs in total at server  $i$  and its eligible queue. Provide a guess regarding the form of the most efficient operating policy for this system.



**Figure P3.22** Admission control to a series queueing system.

- 3.23.** Consider a variant of the queueing admission control model in Sec. 3.7.1 in which there are two classes of jobs and a single-server. Suppose  $f(n_1, n_2)$  denotes the probability that  $n_1$  class-1 jobs and  $n_2$  class-2 jobs arrive in a time slot, and  $g(n_1, n_2)$  denotes the probability of  $n_1$  potential class-1 and  $n_2$

potential class-2 service completions in a time slot. Let  $R_i$  denote the revenue from completing a class- $i$  job and  $h_i(u)$  denote the cost of holding  $u$  class- $i$  jobs for one period. At the start of each time slot, the controller determines the number of jobs of each class to admit to service.

- a. Provide dynamic equations for this model.
- b. Formulate it as a Markov decision process.
- c. Speculate on the form of an optimal policy.

**3.24.** A put option gives the holder the right to sell shares of a stock at a specified price at any time up to its expiry date. Formulate the problem of when to exercise a put option as a optimal stopping problem.

**3.25.** (Mangel and Clark, 1988) Suppose that in each period an animal “chooses” in which of  $K$  “patches” of land to forage. Each patch has a different predation risk, different energy reserve requirements, and also yields a random energy return. The energy return probability distribution varies from patch to patch. Suppose that the animal has a minimal energy threshold that will keep it alive, and that it has a maximum energy capacity. Formulate the patch selection problem as a Markov decision problem in which the animal's objective is to maximize its probability of survival for  $T$  periods.

**3.26.** (Mangel and Clark, 1988; The lazy adaptable lion) An adult female lion requires about 6 kg of meat per day and has a gut capacity of about 30 kg; which means that, with a full gut, it can go six days without eating. Zebras have an average edible biomass of 164 kg, large enough to meet the food demands of many lions. If a female selects to hunt in a group, the probability of a successful kill per chase has been observed to increase with group size to a point, and that each chase consumes 0.5 kg of energy. Suppose that the probabilities of a successful hunt are given by  $p(1) = 0.15$ ,  $p(2) = 0.33$ ,  $p(3) = 0.37$ ,  $p(4) = 0.40$ ,  $p(5) = 0.42$ , and  $p(\geq 6) = 0.43$ , where  $p(n)$  represents the kill probability per chase for lions hunting in a group of size  $n$ .

Formulate this as a Markov decision process, in which the state represents the lion's energy reserves, the action is whether to hunt, and, if so, in what size group. Assume one hunt per day and that lion's objective is to maximize its probability of survival over  $T$  days. Assume that if the hunt is successful, lions in the hunting group share the meat equally.

**3.27.** (Feinberg, Kogan, and Smirnov, 1985). Consider the slotted ALOHA model of Sec. 1.5, in which the controller is faced with the problem of dynamically choosing the retransmission probability. Formulate it as a Markov decision process under the assumption that the number of packets generated in a time slot follow a Poisson distribution with parameter  $\mu$ .

**3.28.** Consider the game “But Who's Counting” described in Sec. 1.7. Formulate the problem of maximizing the expected five-digit number as a Markov decision process. Hint: Let the state represent the occupied digits, the numbers in each location and the number observed on the spinner.

## CHAPTER 4

# Finite-Horizon Markov Decision Processes

This chapter concerns finite-horizon, discrete-time Markov decision problems. In it, we introduce and discuss many basic concepts including optimality criteria, inductive computation, The Principle of Optimality, optimality equations, and structured policies. Section 4.5 contains the key algorithm and Section 4.6 illustrates its use through numerical examples and applications. We suggest reviewing Chap. 2 before delving into this chapter. (*We assume throughout that  $N < \infty$ .*)

### 4.1 OPTIMALITY CRITERIA

As a result of choosing and implementing a policy, the decision maker receives rewards in periods  $1, \dots, N$ . Since they are not known prior to implementing the policy, the decision maker must view the reward sequence as random. The decision maker's objective is to choose a policy so that the corresponding random reward sequence is as "large" as possible. This necessitates a method of comparing random reward sequences. We discuss approaches to comparing random sequences below and provide some justification for comparing policies on the basis of their expected total reward.

#### 4.1.1 Some Preliminaries

Let  $\pi = (d_1, d_2, \dots, d_{N-1})$  denote a randomized history-dependent policy. That is, for each  $t = 1, \dots, N-1$ ,  $d_t: H_t \rightarrow \mathcal{P}(A)$ . (Recall that  $H_t = S \times A \times S \times \dots \times A \times S$  is the set of all histories of states and actions prior to decision epoch  $t$ ). As described in Sec. 2.1.6, each  $\pi \in \Pi^{\text{HR}}$  generates a probability distribution  $P^\pi(\cdot)$  on (Borel subsets of)  $H_N$ , the set of histories (sample paths) up to time  $N$ . For each realization of the history  $h_N = (s_1, a_1, s_2, \dots, s_N)$  there corresponds a sequence of rewards  $\{r_1(s_1, a_1), r_2(s_2, a_2), \dots, r_{N-1}(s_{N-1}, a_{N-1}), r_N(s_N)\}$ . Let  $R_t \equiv r_t(X_t, Y_t)$  denote the random reward received in period  $t < N$ ,  $R_N \equiv r_N(X_N)$  denote the terminal reward,  $R = (R_1, R_2, \dots, R_N)$  denote a random sequence of rewards, and  $\mathfrak{R}$  the set of all possible reward sequences. A policy  $\pi$  induces a probability distribution  $P^\pi_{\mathfrak{R}}(\cdot)$

on (Borel subsets of)  $\mathfrak{H}$ :

$$P_{\mathfrak{H}}^{\pi}(\rho_1, \rho_2, \dots, \rho_N) \equiv P^{\pi}[\{(s_1, a_1, \dots, s_N): (r_1(s_1, a_1), \dots, r_N(s_N)) \\ = (\rho_1, \dots, \rho_N)\}].$$

In this definition,  $P^{\pi}[\cdot]$  denotes the probability distribution on the set of all realizations of the Markov decision process under  $\pi$ .

For a deterministic history-dependent  $\pi$ , we may write this reward sequence as

$$\{r_1(X_1, d_1(h_1)), \dots, r_{N-1}(X_{N-1}, d_{N-1}(h_{N-1})), r_N(X_N)\}$$

and, for  $\pi \in \Pi^{\text{MD}}$ , as

$$\{r_1(X_1, d_1(X_1)), \dots, r_{N-1}(X_{N-1}, d_{N-1}(X_{N-1})), r_N(X_N)\}.$$

We compare policies on the basis of the decision maker's preference for different realizations of  $R$  and the probability that each occurs. We now discuss approaches for such comparisons based on stochastic orderings, expected utility, and other criteria.

Let  $W$  be an arbitrary set and let  $>$  denote a *partial ordering* on that set. By a partial ordering we mean a transitive, reflexive, and antisymmetric relationship between elements in the set. That is, for  $u, v$ , and  $w$  in  $W$ ,  $u > v$  and  $v > w$  implies that  $u > w$  (transitivity) and  $w > w$  (reflexivity), and  $v > w$  and  $w > v$  implies  $w = v$  (antisymmetry). Examples of partial orderings include component-wise ordering on  $R^n$  and  $\subset$  on a collection of sets. Two elements  $u$  and  $v$  are *comparable* if either  $u > v$  or  $v > u$ . A partial ordering is a *total ordering* or *linear ordering* on  $W$  if every pair of elements in  $W$  is comparable. Under the usual component-wise ordering the real line is totally ordered while  $n$ -dimensional Euclidean space is not.

Since the models we study are stochastic, we require partial orderings for random vectors or, equivalently, for probability distributions of random vectors. We first define stochastic orderings for random variables. We say that the random variable  $U$  is *stochastically greater than* the random variable  $V$  if

$$P[V > t] \leq P[U > t]$$

for all  $t \in R^1$ . If  $P^1$  and  $P^2$  are two probability distributions on the same real probability space, then  $P^1$  is stochastically greater than  $P^2$  if

$$P^2[(t, \infty)] \leq P^1[(t, \infty)]$$

for all  $t < \infty$ .

Many generalizations to random vectors have been proposed; we adopt the following. We say that the random vector  $U = (U_1, \dots, U_n)$  is *stochastically greater* than the random vector  $V = (V_1, \dots, V_n)$  if

$$E[f(V_1, \dots, V_n)] \leq E[f(U_1, \dots, U_n)] \quad (4.1.1)$$

for all  $f: R^n \rightarrow R^1$  for which the expectation exists and which preserve the partial ordering on  $R^n$  (that is, if  $v_i \leq u_i$  for  $i = 1, \dots, N$ ;  $f(v_1, \dots, v_N) \leq f(u_1, \dots, u_N)$ ).

From a decision-making perspective, the following result which relates the vector and single component stochastic orderings provides a notion of ordering which is more consistent with sequential decision making. It is one of few results on stochastic orderings which takes the order of the vector components into account. We state it without proof.

**Proposition 4.1.1.** Suppose  $U = (U_1, \dots, U_n)$  and  $V = (V_1, \dots, V_n)$  are random vectors such that  $P[V_1 > t] \leq P[U_1 > t]$  for all  $t \in R^1$ , and for  $j = 2, \dots, n$ ,  $v_i \leq u_i$  for  $i = 1, \dots, j-1$  implies

$$P[V_j > t | V_1 = v_1, \dots, V_{j-1} = v_{j-1}] \leq P[U_j > t | U_1 = u_1, \dots, U_{j-1} = u_{j-1}].$$

Then  $U$  is stochastically greater than  $V$ . Further, for any nondecreasing function  $g: R^1 \rightarrow R^1$ ,

$$E[g(V_j) | V_1 = v_1, \dots, V_{j-1} = v_{j-1}] \leq E[g(U_j) | U_1 = u_1, \dots, U_{j-1} = u_{j-1}].$$

The following example illustrates a serious drawback of using stochastic partial orders to compare policies.

**Example 4.1.1.** Let  $N = 2$  and suppose that there are two policies  $\pi$  and  $\nu$  with distributions of rewards given by

$$P^\pi[R = (0, 0)] = P^\pi[R = (0, 1)] = P^\pi[R = (1, 0)] = P^\pi[R = (1, 1)] = \frac{1}{4}$$

and

$$P^\nu[R = (0, 0)] = P^\nu[R = (1, 1)] = \frac{1}{2}.$$

We compare these policies on the basis of  $E^\pi[f(R_1, R_2)]$  and  $E^\nu[f(R_1, R_2)]$ . For  $f(u, v) = u + v$ ,

$$E^\pi[f(R_1, R_2)] = E^\nu[f(R_1, R_2)] = 1.$$

For  $f(u, v) = [u]^+ [v]^+$ ,

$$\frac{1}{4} = E^\pi[f(R_1, R_2)] < E^\nu[f(R_1, R_2)] = \frac{1}{2}.$$

And, for  $f(u, v) = \sup(u, v)$ ,

$$\frac{3}{4} = E^\pi[f(R_1, R_2)] > E^\nu[f(R_1, R_2)] = \frac{1}{2}.$$

Thus, even in this simple model, the policies  $\pi$  and  $\nu$  are not comparable under the vector stochastic partial order defined through (4.1.1).



The difficulty with using a stochastic ordering to compare policies is that for the decision maker to prefer  $\pi$  to  $\nu$ , then

$$E^\pi[f(R_1, \dots, R_N)] \geq E^\nu[f(R_1, \dots, R_N)] \quad (4.1.2)$$

must hold for a large class of functions  $f$ , many of which do not reflect the decision maker's attitude toward risk. By requiring that (4.1.2) hold only for a specified function, utility theory provides an attractive basis for policy comparison. By a *utility*  $\Psi(\cdot)$ , we mean a real-valued function on a set  $W$  which represents the decision maker's preference for elements in  $W$ . If he does not prefer  $\nu$  over  $w$ , then  $\Psi(\nu) \leq \Psi(w)$ . Conversely, if  $\Psi(\nu) \leq \Psi(w)$ , then the decision maker does not prefer  $\nu$  over  $w$ . He is indifferent to  $\nu$  and  $w$  if  $\Psi(\nu) = \Psi(w)$ . Under  $\Psi(\cdot)$ , all elements of  $W$  are comparable.

When the elements of  $W$  represent outcomes of a random process, *expected utility* provides a total ordering on equivalence classes of outcomes. For a discrete random vector  $Y$ , its expected utility is given by

$$E(\Psi(Y)) = \sum_{y \in W} \Psi(y) P\{Y = y\},$$

and for a continuous random vector by

$$E(\Psi(Y)) = \int_W \Psi(y) dF(y),$$

where  $F(\cdot)$  represents the probability distribution of the  $W$ -valued random vector  $Y$ .

For finite-horizon Markov decision processes with discrete state spaces, the expected utility of policy  $\pi$  may be represented by

$$E^\pi[\Psi(R)] = \sum_{(\rho_1, \dots, \rho_N) \in \Re} \Psi(\rho_1, \dots, \rho_N) P_\pi^\pi\{(\rho_1, \dots, \rho_N)\}.$$

Under the expected utility criterion, the decision maker prefers policy  $\pi$  to policy  $\nu$  if

$$E^\pi[\Psi(R_1, \dots, R_N)] > E^\nu[\Psi(R_1, \dots, R_N)].$$

The policies are equivalent if  $E^\pi[\Psi(R_1, \dots, R_N)] = E^\nu[\Psi(R_1, \dots, R_N)]$ .

For arbitrary utility  $\Psi$ , the above computation is difficult to carry out because it requires explicit determination of the probability distribution of the vector  $R$ , or the joint distribution of the histories. However, by exploiting properties of conditional expectations, inductive computation appears possible.

This book will focus on models using *linear additive utility*, that is, if  $(\rho_1, \dots, \rho_N)$  represents a realization of the reward process, then

$$\Psi(\rho_1, \rho_2, \dots, \rho_N) = \sum_{i=1}^N \rho_i.$$

Such utilities represent the preferences of a risk neutral decision maker who is

indifferent to the timing of rewards. We may account for time preferences by including a discount factor in this expression. We discuss this alternative in Sec. 4.1.2. If  $r_t(s, a)$  denotes the utility (instead of the reward) associated with choosing action  $a$  in state  $s$ , then  $\Psi(r_1, r_2, \dots, r_N) = \sum_{i=1}^N r_i$  denotes the decision maker's total utility. A formulation with  $r_t$  as utility may be appropriate when the state of the system represents a measure of wealth such as in the economic examples of Sec. 3.5.3. Multiplicative utilities of the form  $\Psi(\rho_1, \rho_2, \dots, \rho_N) = \prod_{i=1}^N u(\rho_i)$  can also be analyzed using Markov decision process methods.

As an alternative to expected utility criteria, the decision maker might use a criteria which explicitly takes into account both the total expected value and the variability of the sequence of rewards. As a consequence of using such a criteria, he might forego the policy with the largest expected total reward because its variance is too large. Policies with large reward variances are undesirable because, when using them, the chance of receiving a small total reward might be prohibitively high. Using such an approach in Example 4.1.1, a decision maker might prefer policy  $\pi$  with variance of the total reward of  $\frac{1}{2}$  instead of policy  $\nu$  which has variance 1, even though both have total expected reward of 1.

#### 4.1.2 The Expected Total Reward Criterion

Let  $v_N^\pi(s)$  represent the expected total reward over the decision making horizon if policy  $\pi$  is used and the system is in state  $s$  at the first decision epoch. For  $\pi \in \Pi^{\text{HR}}$ , it is defined by

$$v_N^\pi(s) \equiv E_s^\pi \left\{ \sum_{t=1}^{N-1} r_t(X_t, Y_t) + r_N(X_N) \right\}. \quad (4.1.3)$$

For  $\pi \in \Pi^{\text{HD}}$ , the expected total reward can be expressed as

$$v_N^\pi(s) = E_s^\pi \left\{ \sum_{t=1}^{N-1} r_t(X_t, d_t(h_t)) + r_N(X_N) \right\} \quad (4.1.4)$$

since at each decision epoch  $Y_t = d_t(h_t)$ .

Under the assumption that  $|r_t(s, a)| \leq M < \infty$  for  $(s, a) \in S \times A$  and  $t \leq N$ , and that  $S$  and  $A$  are discrete,  $v_N^\pi(s)$  exists and is bounded for each  $\pi \in \Pi^{\text{HR}}$  and each finite  $N$ . In greater generality, we also require measurability assumptions on  $r_t$  and  $\pi$ .

To account for the time value of rewards, we often introduce a discount factor. By a *discount factor*, we mean a scalar  $\lambda, 0 \leq \lambda < 1$  which measures the value at time  $n$  of a one unit reward received at time  $n + 1$ . A one-unit reward received  $t$  periods in the future has present value  $\lambda^t$ . For  $\pi \in \Pi^{\text{HR}}$ , the *expected total discounted reward*

$$v_{N,\lambda}^\pi(s) = E_s^\pi \left\{ \sum_{t=1}^{N-1} \lambda^{t-1} r_t(X_t, Y_t) + \lambda^{N-1} r_N(X_N) \right\}. \quad (4.1.5)$$

Note that the exponent of  $\lambda$  is  $t - 1$  and not  $t$  because  $r_t$  represents the value at decision epoch  $t$  of the reward received throughout period  $t$ , i.e.,  $r_1$  is the value at

the first decision epoch of the reward received in period 1. Substitution of  $\lambda = 1$  into (4.1.5) yields the expected total reward (4.1.3).

Taking the discount factor into account does not effect any theoretical results or algorithms in the finite-horizon case but might effect the decision maker's preference for policies. The discount factor plays a key analytic role in the infinite-horizon models we discuss in Chaps. 6–10.

### 4.1.3 Optimal Policies

Markov decision process theory and algorithms for finite-horizon models primarily concern determining a policy  $\pi^* \in \Pi^{\text{HR}}$  with the largest expected total reward, and characterizing this value. That is, we seek a policy  $\pi^*$  for which

$$v_N^{\pi^*}(s) \geq v_N^\pi(s), \quad s \in S \quad (4.1.6)$$

for all  $\pi \in \Pi^{\text{HR}}$ . We refer to such a policy as an *optimal policy*. In some models such a policy need not exist, so instead we seek an  $\varepsilon$ -*optimal policy*, that is, for an  $\varepsilon > 0$ , a policy  $\pi_\varepsilon^*$  with the property that

$$v_N^{\pi_\varepsilon^*}(s) + \varepsilon > v_N^\pi(s), \quad s \in S \quad (4.1.7)$$

for all  $\pi \in \Pi^{\text{HR}}$ .

We seek to characterize the *value* of the Markov decision problem,  $v_N^*$ , defined by

$$v_N^*(s) \equiv \sup_{\pi \in \Pi^{\text{HR}}} v_N^\pi(s), \quad s \in S, \quad (4.1.8)$$

and when the supremum in (4.1.8) is attained, by

$$v_N^*(s) = \max_{\pi \in \Pi^{\text{HR}}} v_N^\pi(s), \quad s \in S. \quad (4.1.9)$$

The expected total reward of an optimal policy  $\pi^*$  satisfies

$$v_N^{\pi^*}(s) = v_N^*(s), \quad s \in S$$

and the value of an  $\varepsilon$ -optimal policy  $\pi_\varepsilon^*$  satisfies

$$v_N^{\pi_\varepsilon^*}(s) + \varepsilon > v_N^*(s), \quad s \in S.$$

As a consequence of the definition of the supremum, such a policy exists for any  $\varepsilon > 0$ .

We have implicitly assumed above that the decision maker wishes to choose an optimal policy for all possible initial-system states. In practice, all that might be required is an optimal policy for some specified initial state. Alternatively, he might seek a policy prior to knowing the initial state. In this case he seeks a  $\pi \in \Pi^{\text{HR}}$  which maximizes  $\sum_{s \in S} v_N^\pi(s) P_1\{X_1 = s\}$ . Clearly he may find such a policy by maximizing  $v_N^\pi(s)$  for each  $s$  for which  $P_1\{X_1 = s\} > 0$ .

We use the expression *Markov decision problem* (MDP) for a Markov decision process together with an optimality criteria. The problems defined above are discrete-time, finite-horizon Markov decision problems with expected total reward criteria.

## 4.2 FINITE-HORIZON POLICY EVALUATION

Markov decision problem theory and computation is based on using backward induction (dynamic programming) to recursively evaluate expected rewards. In this section, we introduce the fundamental recursion of dynamic programming by providing an efficient method to evaluate the expected total reward of a fixed policy.

Let  $\pi = (d_1, d_2, \dots, d_{N-1})$  be a randomized history-dependent policy. Let  $u_t^\pi: H_t \rightarrow R^1$  denote the total expected reward obtained by using policy  $\pi$  at decision epochs  $t, t+1, \dots, N-1$ . If the history at decision epoch  $t$  is  $h_t \in H_t$ , then define  $u_t^\pi$  for  $t < N$  by

$$u_t^\pi(h_t) = E_{h_t}^\pi \left\{ \sum_{n=t}^{N-1} r_n(X_n, Y_n) + r_N(X_N) \right\}, \quad (4.2.1)$$

where the conditional expectation is as defined in (2.1.15), and let  $u_N^\pi(h_N) = r_N(s)$  when  $h_N = (h_{N-1}, a_{N-1}, s)$ . When  $h_1 = s$ ,  $u_1^\pi(s) = v_N^\pi(s)$ . The difference between  $u_t^\pi$  and  $v_N^\pi$  is that  $v_N^\pi$  includes rewards over the *entire* future, while  $u_t^\pi$  only incorporates rewards from decision epoch  $t$  onward.

We now show how to compute  $v_N^\pi$  by inductively evaluating  $u_t^\pi$ . We call it the *finite-horizon policy evaluation algorithm*. To simplify the notation, assume that a deterministic  $\pi \in \Pi^{\text{HD}}$  has been specified and that  $S$  is discrete. In practice, we will not need to evaluate randomized policies, because subsequent results establish that deterministic policies are optimal under the expected total reward criteria.

### The Finite Horizon-Policy Evaluation Algorithm (for fixed $\pi \in \Pi^{\text{HD}}$ )

1. Set  $t = N$  and  $u_N^\pi(h_N) = r_N(s_N)$  for all  $h_N = (h_{N-1}, a_{N-1}, s_N) \in H_N$ .
2. If  $t = 1$ , stop, otherwise go to step 3.
3. Substitute  $t-1$  for  $t$  and compute  $u_t^\pi(h_t)$  for each  $h_t = (h_{t-1}, a_{t-1}, s_t) \in H_t$  by

$$u_t^\pi(h_t) = r_t(s_t, d_t(h_t)) + \sum_{j \in S} p_t(j|s_t, d_t(h_t)) u_{t+1}^\pi(h_t, d_t(h_t), j), \quad (4.2.2)$$

noting that  $(h_t, d_t(h_t), j) \in H_{t+1}$ .

4. Return to 2.

Some motivation for (4.2.2) follows. The expected value of policy  $\pi$  over periods  $t, t+1, \dots, N$  when the history at epoch  $t$  is  $h_t$  equals the immediate reward received by selecting action  $d_t(h_t)$  plus the expected reward over the remaining periods. The second term contains the product of the probability of being in state  $j$  at

decision epoch  $t + 1$  if action  $d_t(h_t)$  is used, and the expected total reward obtained using policy  $\pi$  over periods  $t + 1, \dots, N$  when the history at epoch  $t + 1$  is  $h_{t+1} = (h_t, d_t(h_t), j)$ . Summing over all possible  $j$  gives the desired expectation expressed in terms of  $u_{t+1}^\pi$ ; that is,

$$u_t^\pi(h_t) = r_t(s_t, d_t(h_t)) + E_{h_t}^\pi\{u_{t+1}^\pi(h_t, d_t(h_t), X_{t+1})\}. \quad (4.2.3)$$

This inductive scheme reduces the problem of computing expected total rewards over  $N$  periods to a sequence of  $N - 1$  similar one-period calculations having immediate rewards  $r_t$  and terminal rewards  $u_{t+1}^\pi$  as in Section 2.2. This reduction is the essence of dynamic programming:

*Multistage problems may be solved by analyzing a sequence of simpler inductively defined single-stage problems.*

By doing this, we avoid the onerous task of explicitly determining the joint probability distribution of histories under  $\pi$ . The procedure for finding optimal policies in the next section generalizes these calculations.

We now formally show that this algorithm does in fact yield  $u^\pi$ . The proof is by induction where the index of induction is  $t$ . Contrary to usual mathematical convention, we apply the induction backwards. This simplifies the subscripts in the recursions.

**Theorem 4.2.1.** Let  $\pi \in \Pi^{\text{HD}}$  and suppose  $u_t^\pi$ ,  $t \leq N$ , has been generated by the policy evaluation algorithm. Then, for all  $t \leq N$ , (4.2.1) holds, and  $v_N^\pi(s) = u_N^\pi(s)$  for all  $s \in S$ .

**Proof.** The result is obviously true when  $t = N$ . Suppose now that (4.2.1) holds for  $t + 1, t + 2, \dots, N$ . Then by using representation (4.2.3) and the induction hypothesis

$$u_t^\pi(h_t) = r_t(s_t, d_t(h_t)) + E_{h_t}^\pi\left\{E_{h_{t+1}}^\pi\left\{\sum_{n=t+1}^{N-1} r_n(X_n, Y_n) + r_N(X_N)\right\}\right\} \quad (4.2.4)$$

$$= r_t(s_t, d_t(h_t)) + E_{h_t}^\pi\left\{\sum_{n=t+1}^{N-1} r_n(X_n, Y_n) + r_N(X_N)\right\}. \quad (4.2.5)$$

Since  $s_t$  and  $h_t$  are known at decision epoch  $t$ ,  $X_t = s_t$ , so that the first term in (4.2.5) can be included inside the expectation yielding the desired result.  $\square$

Observe that in (4.2.3), when taking an expectation conditional on the history up to time  $t + 1$ ,  $X_{t+1}$  is known, but when taking conditional expectation with respect to the history up to time  $t$ ,  $X_{t+1}$  is random. Thus, since the algorithm evaluates  $u_{t+1}^\pi(h_{t+1})$  for all  $h_{t+1}$ , before evaluating  $u_t^\pi$  the expectation over  $X_{t+1}$  can be computed.

Fundamental to justifying the validity of this algorithm is the assumption of linear additive utility. A recursive scheme is also possible for models with multiplicative exponential utility.

In presenting and analyzing this algorithm, we have assumed discrete state space and arbitrary action sets. In theory, this algorithm is also valid for general state and sets provided that integration replaces summation in (4.2.2) and the functions in the expectations are measurable and integrable. From a computational perspective, this degree of generality is unnecessary since numerical calculations are implementable only for problems with finite-state and action sets. On the other hand, results on the structure of optimal policies require such generality.

A key feature of this algorithm is that the induction is backwards. This is required because of the probabilistic nature of the MDP. If the model were deterministic, policies could be evaluated by either forward or backward induction because expectations need not be computed. To evaluate a conditional expectation from decision epoch  $t$  onward, we require the conditional expectation from decision epoch  $t + 1$  onward for each possible history at decision epoch  $t + 1$ . Proceeding in a forward fashion would require enumeration of all histories, and deriving the probability distribution of each under policy  $\pi$ .

We have stated the algorithm for nonrandomized history-dependent policies. To generalize it to randomized policies would require an additional summation in (4.2.2) to account for the probability distribution of the action at decision epoch  $t$  under decision rule  $d_t$  as follows:

$$u_t^\pi(h_t) = \sum_{a \in A_{s_t}} q_{d_t(h_t)}(a) \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}(h_t, a, j) \right\}. \quad (4.2.6)$$

Theorem 4.2.1 may be extended to  $\pi \in \Pi^{\text{HR}}$  as follows.

**Theorem 4.2.2.** Let  $\pi \in \Pi^{\text{HR}}$  and suppose  $u_t^\pi$ ,  $t \leq N$  has been generated by the policy evaluation algorithm with (4.2.6) replacing (4.2.2). Then, for all  $t \leq N$ , (4.2.1) holds and  $v_N^\pi(s) = u_1^\pi(s)$  for all  $s \in S$ .

The policy evaluation algorithm for deterministic Markovian policies replaces (4.2.2) by

$$u_t^\pi(s_t) = r_t(s_t, d_t(s_t)) + \sum_{j \in S} p_t(j|s_t, d_t(s_t)) u_{t+1}^\pi(j). \quad (4.2.7)$$

Because histories need not be retained, this simplifies computation through reduced storage and fewer arithmetic operations. An algorithm based on (4.2.7) evaluates the right-hand side above *only once* for each history that includes  $s_t$  in its realization at time  $t$ . In contrast, the right-hand side of (4.2.2) requires evaluation for each history up to time  $t$  because a different decision rule might be used at time  $t$  for each realization.

When there are  $K$  states and  $L$  actions available in each state, at decision epoch  $t$  there are  $K^{t+1}L^t$  histories. Since each pass through step 3 of the evaluation algorithm requires  $K$  multiplications, evaluation of a nonrandomized history-dependent decision rule requires  $K^2 \sum_{i=0}^{N-1} (KL)^i$  multiplications. For Markov policies, the number of multiplications decreases to  $(N - 1)K^2$ .

### 4.3 OPTIMALITY EQUATIONS AND THE PRINCIPLE OF OPTIMALITY

In this section we introduce optimality equations (sometimes referred to as Bellman equations or functional equations) and investigate their properties. We show that solutions of these equations correspond to optimal value functions and that they also provide a basis for determining optimal policies. We assume either finite or countable  $S$  to avoid technical subtleties.

Let

$$u_t^*(h_t) = \sup_{\pi \in \Pi^{\text{HR}}} u_t^\pi(h_t). \quad (4.3.1)$$

It denotes the supremum over all policies of the expected total reward from decision epoch  $t$  onward when the history up to time  $t$  is  $h_t$ . For  $t > 1$ , we need not consider all policies when taking the above supremum. Since we know  $h_t$ , we only consider portions of policies from decision epoch  $t$  onward; that is, we require only the supremum over  $(d_t, d_{t+1}, \dots, d_{N-1}) \in D_t^{\text{HR}} \times D_{t+1}^{\text{HR}} \times \dots \times D_{N-1}^{\text{HR}}$ . When minimizing costs instead of maximizing rewards, we sometimes refer to  $u_t^*$  as a *cost-to-go* function.

The *optimality equations* are given by

$$u_t(h_t) = \sup_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}(h_t, a, j) \right\} \quad (4.3.2)$$

for  $t = 1, \dots, N-1$  and  $h_t = (h_{t-1}, a_{t-1}, s_t) \in H_t$ . For  $t = N$ , we add the boundary condition

$$u_N(h_N) = r_N(s_N) \quad (4.3.3)$$

for  $h_N = (h_{N-1}, a_{N-1}, s_N) \in H_N$ .

These equations reduce to the policy evaluation equations (4.2.1) when we replace the supremum over all actions in state  $s_t$  by the action corresponding to a specified policy, or equivalently, when  $A_{s_t}$  is a singleton for each  $s \in S$ .

The operation “sup” in (4.3.2) is implemented by evaluating the quantity in brackets for each  $a \in A_{s_t}$  and then choosing the supremum over all of these values. When  $A_{s_t}$  is a continuum, the supremum might be found analytically. If the supremum in (4.3.2) is attained, for example, when each  $A_{s_t}$  is finite, it can be replaced by “max” so (4.3.2) becomes

$$u_t(h_t) = \max_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}(h_t, a, j) \right\}. \quad (4.3.4)$$

A solution to the system of equations (4.3.2) or (4.3.4) and boundary condition (4.3.3) is a sequence of functions  $u_t: H_t \rightarrow R$ ,  $t = 1, \dots, N$ , with the property that  $u_N$  satisfies (4.3.3),  $u_{N-1}$  satisfies the  $(N-1)$ th equation with  $u_N$  substituted into the right-hand side of the  $(N-1)$ th equation, and so forth.

The optimality equations are fundamental tools in Markov decision theory, having the following important and useful properties.

- a. Solutions to the optimality equations are the optimal returns from period  $t$  onward for each  $t$ .
- b. They provide a method for determining whether a policy is optimal. If the expected total reward of policy  $\pi$  from period  $t$  onward satisfies this system of equations for  $t = 1, \dots, N$  it is optimal.
- c. They are the basis for an efficient procedure for computing optimal return functions and policies.
- d. They may be used to determine structural properties of optimal policies and return functions.

Before stating and proving the main result in this chapter, we introduce the following important yet simple lemma.

**Lemma 4.3.1.** Let  $w$  be a real-valued function on an arbitrary discrete set  $W$  and let  $q(\cdot)$  be a probability distribution on  $W$ . Then

$$\sup_{u \in W} w(u) \geq \sum_{u \in W} q(u)w(u)$$

*Proof.* Let  $w^* = \sup_{u \in W} w(u)$ . Then,

$$w^* = \sum_{u \in W} q(u)w^* \geq \sum_{u \in W} q(u)w(u).$$

□

Note that the lemma remains valid with  $W$  a Borel subset of a measurable space,  $w(u)$  an integrable function on  $W$ , and the summation replaced by integration.

The following theorem summarizes the optimality properties of solutions of the optimality equation. Its inductive proof illustrates several dynamic programming principles and consists of two parts. First we establish that solutions provide upper bounds on  $u_t^*$  and then we establish existence of a policy  $\pi'$  for which  $u_t^{\pi'}$  is arbitrarily close to  $u_t$ .

**Theorem 4.3.2.** Suppose  $u_t$  is a solution of (4.3.2) for  $t = 1, \dots, N - 1$ , and  $u_N$  satisfies (4.3.3). Then

- a.  $u_t(h_t) = u_t^*(h_t)$  for all  $h_t \in H_t$ ,  $t = 1, \dots, N$ , and
- b.  $u_1(s_1) = v_N^*(s_1)$  for all  $s_1 \in S$ .

*Proof.* The proof is in two parts. First we establish by induction that  $u_n(h_n) \geq u_n^*(h_n)$  for all  $h_n \in H_n$  and  $n = 1, 2, \dots, N$ .

Since no decision is made in period  $N$ ,  $u_N(h_N) = r_N(s_N) = u_N^{\pi}(h_N)$  for all  $h_N \in H_N$  and  $\pi \in \Pi^{\text{HR}}$ . Therefore  $u_N(h_N) = u_N^*(h_N)$  for all  $h_N \in H_N$ . Now assume



that  $u_t(h_t) \geq u_t^*(h_t)$  for all  $h_t \in H_t$  for  $t = n + 1, \dots, N$ . Let  $\pi' = (d'_1, d'_2, \dots, d'_{N-1})$  be an arbitrary policy in  $\Pi^{\text{HR}}$ . For  $t = n$ , the optimality equation is

$$u_n(h_n) = \sup_{a \in A_{s_n}} \left\{ r_n(s_n, a) + \sum_{j \in S} p_n(j|s_n, a) u_{n+1}(h_n, a, j) \right\}.$$

By the induction hypothesis

$$u_n(h_n) \geq \sup_{a \in A_{s_n}} \left\{ r_n(s_n, a) + \sum_{j \in S} p_n(j|s_n, a) u_{n+1}^*(h_n, a, j) \right\} \quad (4.3.5)$$

$$\geq \sup_{a \in A_{s_n}} \left\{ r_n(s_n, a) + \sum_{j \in S} p_n(j|s_n, a) u_{n+1}^{\pi'}(h_n, a, j) \right\} \quad (4.3.6)$$

$$\geq \sum_{a \in A_{s_n}} q_{d'_n(h_n)}(a) \left\{ r_n(s_n, a) + \sum_{j \in S} p_n(j|s_n, a) u_{n+1}^{\pi'}(h_n, a, j) \right\} \quad (4.3.7)$$

$$= u_n^{\pi'}(h_n).$$

The inequality in (4.3.5) follows from the induction hypothesis and the non-negativity of  $p_n$ , and that in (4.3.6) from the definition of  $u_{n+1}^*$ . That in (4.3.7) follows from Lemma 4.3.1 with  $W = A_{s_n}$  and  $w$  equal to the expression in brackets. The last equality follows from (4.2.6) and Theorem 4.2.2. Since  $\pi'$  is arbitrary,

$$u_n(h_n) \geq u_n^{\pi}(h_n) \quad \text{for all } \pi \in \Pi^{\text{HR}}.$$

Thus  $u_n(h_n) \geq u_n^*(h_n)$  and the induction hypothesis holds.

Now we establish that for any  $\varepsilon > 0$ , there exists a  $\pi' \in \Pi^{\text{HD}}$  for which

$$u_n^{\pi'}(h_n) + (N - n)\varepsilon \geq u_n(h_n) \quad (4.3.8)$$

for all  $h_n \in H_n$  and  $n = 1, 2, \dots, N$ . To do this, construct a policy  $\pi' = (d_1, d_2, \dots, d_{N-1})$  by choosing  $d_n(h_n)$  to satisfy

$$r_n(s_n, d_n(h_n)) + \sum_{j \in S} p_n(j|s_n, d_n(h_n)) u_{n+1}(s_n, d_n(h_n), j) + \varepsilon \geq u_n(h_n). \quad (4.3.9)$$

We establish (4.3.8) by induction. Since  $u_N^{\pi'}(h_N) = u_N(h_N)$ , the induction hypothesis holds for  $t = N$ . Assume that  $u_t^{\pi'}(h_t) + (N - t)\varepsilon \geq u_t(h_t)$  for  $t = n + 1, \dots, N$ .

Then it follows from Theorem 4.2.1 and (4.3.9) that

$$\begin{aligned}
 u_n^{\pi'}(h_n) &= r_n(s_n, d_n(h_n)) + \sum_{j \in S} p_n(j|s_n, d_n(h_n)) u_{n+1}^{\pi'}(s_n, d_n(h_n), j) \\
 &\geq r_n(s_n, d_n(h_n)) + \sum_{j \in S} p_n(j|s_n, d_n(h_n)) u_{n+1}(s_n, d_n(h_n), j) \\
 &\quad - (N - n - 1)\varepsilon \\
 &\geq u_n(h_n) - (N - n)\varepsilon.
 \end{aligned}$$

Thus the induction hypothesis is satisfied and (4.3.8) holds for  $n = 1, 2, \dots, N$ .

Therefore for any  $\varepsilon > 0$ , there exists a  $\pi' \in \Pi^{\text{HR}}$  for which

$$u_n^*(h_n) + (N - n)\varepsilon \geq u_n^{\pi'}(h_n) + (N - n)\varepsilon \geq u_n(h_n) \geq u_n^*(h_n)$$

so that (a) follows. Part (b) follows from the definitions of the quantities.  $\square$

Part (a) of Theorem 4.3.2 means that solutions of the optimality equation are the optimal value functions from period  $t$  onward, and result (b) means that the solution to the equation with  $n = 1$  is the value function for the MDP, that is, it is the optimal value from decision epoch 1 onward.

The following result shows how to use the optimality equations to find optimal policies, and to verify that a policy is optimal. Theorem 4.3.3 uses optimality equations (4.3.4) in which maxima are attained. Theorem 4.3.4 considers the case of suprema.

**Theorem 4.3.3.** Suppose  $u_t^*$ ,  $t = 1, \dots, N$  are solutions of the optimality equations (4.3.4) subject to boundary condition (4.3.3), and that policy  $\pi^* = (d_1^*, d_2^*, \dots, d_{N-1}^*) \in \Pi^{\text{HD}}$  satisfies

$$\begin{aligned}
 &r_t(s_t, d_t^*(h_t)) + \sum_{j \in S} p_t(j|s_t, d_t^*(h_t)) u_{t+1}^*(h_t, d_t^*(h_t), j) \\
 &= \max_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(h_t, a, j) \right\} \quad (4.3.10)
 \end{aligned}$$

for  $t = 1, \dots, N - 1$ .

Then

a. For each  $t = 1, 2, \dots, N$ ,

$$u_t^{\pi^*}(h_t) = u_t^*(h_t), \quad h_t \in H_t. \quad (4.3.11)$$

b.  $\pi^*$  is an optimal policy, and

$$v_N^{\pi^*}(s) = v_N^*(s), \quad s \in S. \quad (4.3.12)$$

**Proof.** We establish part (a); part (b) follows from Theorem 4.2.1 and Theorem 4.3.2b. The proof is by induction. Clearly

$$u_N^*(h_n) = u_N^*(h_n), \quad h_n \in H_n.$$

Assume the result holds for  $t = n + 1, \dots, N$ . Then, for  $h_n = (h_{n-1}, d_{n-1}^*(h_{n-1}), s_n)$ ,

$$\begin{aligned} u_n^*(h_n) &= \max_{a \in A_{s_n}} \left\{ r_n(s_n, a) + \sum_{j \in S} p_n(j|s_n, a) u_{n+1}^*(h_n, a, j) \right\} \\ &= r_n(s_n, d_n^*(h_n)) + \sum_{j \in S} p_n(j|s_n, d_n^*(h_n)) u_{n+1}^*(h_n, d_n^*(h_n), j) \\ &= u_n^*(h_n). \end{aligned}$$

The second equality is a consequence of (4.3.10), and the induction hypothesis and the last equality follows from Theorem 4.2.1. Thus the induction hypothesis is satisfied and the result follows.  $\square$

We frequently write equation (4.3.10) as

$$d_t^*(h_t) \in \arg \max_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S_t} p_t(j|s_t, a) u_{t+1}^*(h_t, a, j) \right\},$$

where “arg max” has been defined in Sec. 2.2.

The theorem implies that an optimal policy is found by first solving the optimality equations, and then for each history choosing a decision rule which selects any action which attains the maximum on the right-hand side of (4.3.10) for  $t = 1, 2, \dots, N$ . When using this equation in computation, for each history the right-hand side is evaluated for all  $a \in A_{s_t}$  and the set of maximizing actions is recorded. When there is more than one maximizing action in this set, there is more than one optimal policy.

Note that we have restricted attention to history-dependent deterministic policies in Theorem 4.3.3. This is because if there existed a history-dependent randomized policy which satisfied the obvious generalization of (4.3.10), as a result of Lemma 4.3.1, we could find a deterministic policy which satisfied (4.3.10). We expand on this point in the next section.

This theorem provides a formal statement of “The Principle of Optimality,” a fundamental result of dynamic programming. An early verbal statement appeared in Bellman (1957, p. 83).

“An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.”

Denardo (1982, p. 15) provides a related statement.

“There exists a policy that is optimal for every state (at every stage).”

Any deterministic policy  $\pi^*$  satisfying (4.3.10) has these properties and such a policy must *exist* because maxima are attained. Alternatively, Theorem 4.3.3 provides *sufficient* conditions for verifying the optimality of a policy.

Note that “The Principle of Optimality” may not be valid for other optimality criteria. In Sec. 4.6.2, we provide an example in which it does not hold, albeit under a nonstandard optimality criteria.

In case the supremum in (4.3.2) is not attained, the decision maker must be content with  $\varepsilon$ -optimal policies. To account for this we modify Theorem 4.3.3 as follows. Arguments in the second part of the proof of Theorem 4.3.2 can be used to establish it.

**Theorem 4.3.4.** Let  $\varepsilon > 0$  be arbitrary and suppose  $u_t^*$ ,  $t = 1, \dots, N$  are solutions of the optimality equations (4.3.2) and (4.3.3). Let  $\pi^\varepsilon = (d_1^\varepsilon, d_2^\varepsilon, \dots, d_{N-1}^\varepsilon) \in \Pi^{\text{HD}}$  satisfy

$$\begin{aligned} & r_t(s_t, d_t^\varepsilon(h_t)) + \sum_{j \in S} p_t(j|s_t, d_t^\varepsilon(h_t)) u_{t+1}^*(h_t, d_t^\varepsilon(h_t), j) + \frac{\varepsilon}{N-1} \\ & \geq \sup_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(h_t, a, j) \right\} \end{aligned} \quad (4.3.13)$$

for  $t = 1, 2, \dots, N-1$ . Then,

a. For each  $t = 1, 2, \dots, N-1$ ,

$$u_t^{\pi^\varepsilon}(h_t) + (N-t) \frac{\varepsilon}{N-1} \geq u_t^*(h_t), \quad h_t \in H_t. \quad (4.3.14)$$

b.  $\pi^\varepsilon$  is an  $\varepsilon$ -optimal policy, that is

$$v_N^{\pi^\varepsilon}(s) + \varepsilon \geq v_N^*(s), \quad s \in S. \quad (4.3.15)$$

## 4.4 OPTIMALITY OF DETERMINISTIC MARKOV POLICIES

This section provides conditions under which there exists an optimal policy which is deterministic and Markovian, and illustrates how backward induction can be used to determine the structure of an optimal policy.

From the perspective of application, we find it comforting that by restricting attention to nonrandomized Markov policies, which are simple to implement and evaluate, we may achieve as large an expected total reward as if we used randomized history-dependent policies. We show that when the immediate rewards and transition probabilities depend on the past only through the current state of the system (as assumed throughout this book), the optimal value functions depend on the history only through the current state of the system. This enables us to impose assumptions on the action sets, rewards, and transition probabilities which ensure existence of optimal policies which depend only on the system state.

Inspection of the proof of Theorem 4.3.2 reveals that it constructs an  $\varepsilon$ -optimal deterministic history-dependent policy. Theorem 4.3.3 and 4.3.4 identify optimal and  $\varepsilon$ -optimal policies. We summarize these results as follows.

**Theorem 4.4.1.**

- a. For any  $\varepsilon > 0$ , there exists an  $\varepsilon$ -optimal policy which is deterministic history dependent. Any policy in  $\Pi^{\text{HD}}$  which satisfies (4.3.13) is  $\varepsilon$ -optimal.
- b. Let  $u_t^*$  be a solution of (4.3.2) and (4.3.3) and suppose that for each  $t$  and  $s_t \in S$ , there exists an  $a' \in A_{s_t}$  for which

$$\begin{aligned} r_t(s_t, a') + \sum_{j \in S} p_t(j|s_t, a') u_{t+1}^*(h_t, a', j) \\ = \sup_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(h_t, a, j) \right\} \end{aligned} \quad (4.4.1)$$

for all  $h_t = (s_{t-1}, a_{t-1}, s_t) \in H_t$ . Then there exists a deterministic history-dependent policy which is optimal.

**Proof.** Part (a) follows from the second part of the proof of Theorem 4.3.2. The policy  $\pi^\varepsilon$  in Theorem 4.3.4 is optimal and deterministic. When there exists an  $a' \in A_{s_t}$  for which (4.4.1) holds, the policy  $\pi^* \in \Pi^{\text{HD}}$  of Theorem 4.3.3 is optimal.  $\square$

We next show by induction that there exists an optimal policy which is Markovian and deterministic.

**Theorem 4.4.2.** Let  $u_t^*$ ,  $t = 1, \dots, N$  be solutions of (4.3.2) and (4.3.3). Then

- a. For each  $t = 1, \dots, N$ ,  $u_t^*(h_t)$  depends on  $h_t$  only through  $s_t$ .
- b. For any  $\varepsilon > 0$ , there exists an  $\varepsilon$ -optimal policy which is deterministic and Markov.
- c. If there exists an  $a' \in A_{s_t}$  such that (4.4.1) holds for each  $s_t \in S$  and  $t = 1, 2, \dots, N-1$ , there exists an optimal policy which is deterministic and Markov.

**Proof.** We show that (a) holds by induction. Since  $u_N^*(h_N) = u_N^*(h_{N-1}, a_{N-1}, s) = r_N(s)$  for all  $h_{N-1} \in H_{N-1}$  and  $a_{N-1} \in A_{s_{N-1}}$ ,  $u_N^*(h_N) = u_N^*(s_N)$ . Assume now that (a) is valid for  $n = t+1, \dots, N$ . Then

$$u_t^*(h_t) = \sup_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(h_t, a, j) \right\},$$

which by the induction hypothesis gives

$$u_t^*(h_t) = \sup_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(j) \right\}. \quad (4.4.2)$$

Since the quantity in brackets depend on  $h_t$  only through  $s_t$ , (a) holds for all  $t$ .

Choose  $\varepsilon > 0$ , and let  $\pi^\varepsilon = (d_1^\varepsilon, d_2^\varepsilon, \dots, d_{N-1}^\varepsilon)$  be any policy in  $\Pi^{\text{MD}}$  satisfying

$$\begin{aligned} & r_t(s_t, d_t^\varepsilon(s_t)) + \sum_{j \in S} p_t(j|s_t, d_t^\varepsilon(s_t)) u_{t+1}^*(j) + \frac{\varepsilon}{N-1} \\ & \geq \sup_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(j) \right\}. \end{aligned} \quad (4.4.3)$$

Then, by part (a),  $\pi^\varepsilon$  satisfies the hypotheses of Theorem 4.3.4b so it is  $\varepsilon$ -optimal.

Part (c) follows by noting that under the hypotheses of Theorem 4.4.1b, there exists a  $\pi^* = (d_1^*, d_2^*, \dots, d_{N-1}^*) \in \Pi^{\text{MD}}$ , which satisfies

$$\begin{aligned} & r_t(s_t, d_t^*(s_t)) + \sum_{j \in S} p_t(j|s_t, d_t^*(s_t)) u_{t+1}^*(j) \\ & = \max_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(j) \right\}. \end{aligned} \quad (4.4.4)$$

Therefore by part (a) and Theorem 4.3.3b,  $\pi^*$  is optimal.  $\square$

Thus we have established that

$$v_N^*(s) = \sup_{\pi \in \Pi^{\text{HR}}} v_N^\pi(s) = \sup_{\pi \in \Pi^{\text{MD}}} v_N^\pi(s), \quad s \in S.$$

We now provide conditions under which the supremum in (4.4.1) is attained, so that we may easily determine when there exists a deterministic Markovian policy which is optimal. Following Appendix B, we say that a set  $X$  is *compact* if it is a compact subset of a complete separable metric space. In many applications we consider only compact subsets of  $R^n$ . The proof of part (c) below is quite technical and relies on properties of semicontinuous functions described in Appendix B. Note that more subtle arguments are required for general  $S$ .

**Proposition 4.4.3.** Assume  $S$  is finite or countable, and that

- $A_s$  is finite for each  $s \in S$ , or
- $A_s$  is compact,  $r_t(s, a)$  is continuous in  $a$  for each  $s \in S$ , there exists an  $M < \infty$  for which  $|r_t(s, a)| \leq M$  for all  $a \in A_s$ ,  $s \in S$ , and  $p_t(j|s, a)$  is continuous in  $a$  for each  $j \in S$  and  $s \in S$  and  $t = 1, 2, \dots, N$ , or
- $A_s$  is a compact,  $r_t(s, a)$  is upper semicontinuous (u.s.c.) in  $a$  for each  $s \in S$ , there exists an  $M < \infty$  for which  $|r_t(s, a)| \leq M$  for all  $a \in A_s$ ,  $s \in S$ , and for each  $j \in S$  and  $s \in S$ ,  $p_t(j|s, a)$  is lower semi-continuous (l.s.c.) in  $a$  and  $t = 1, 2, \dots, N$ .

Then there exists a deterministic Markovian policy which is optimal.

**Proof.** We show that there exists an  $a'$  which satisfies (4.4.1) under hypothesis (a), (b), or (c), in which case the result follows from Theorem 4.4.2. Note that as a consequence of Theorem 4.4.2c, we require that, for each  $s \in S$ , there exists an

$a' \in A_s$ , for which

$$\begin{aligned} r_t(s, a') + \sum_{j \in S} p_t(j|s_t, a') u_{t+1}^*(j) \\ = \sup_{a \in A_s} \left\{ r_t(s, a) + \sum_{j \in S} p_t(j|s, a) u_{t+1}^*(j) \right\}. \end{aligned} \quad (4.4.5)$$

Clearly such an  $a'$  exists when  $A_s$  is finite so the result follows under a.

Suppose that the hypotheses in part (c) hold. By assumption  $|r_t(s, a)| \leq M$  for all  $s \in S$  and  $a \in A_s$ ,  $|u_t^*(s)| \leq NM$  for all  $s \in S$  and  $t = 1, 2, \dots, N$ . Therefore, for each  $t$ ,  $u_t^*(s) - NM \leq 0$ . Now apply Proposition B.3, with  $s$  fixed and  $X$  identified with  $S_Y$ ,  $A_s$  identified with  $q(x, y)$ ,  $p_t(j|s, a)$  identified with  $f(w, x)$  and  $u_t^*(s) - NM$  identified with  $f(x)$ , to obtain that

$$\sum_{j \in S} p_t(j|s, a) [u_{t+1}^*(j) - NM]$$

is u.s.c., from which we conclude that  $\sum_{j \in S} p_t(j|s, a) u_{t+1}^*(j)$  is u.s.c. By Proposition B.1.a,  $r_t(s, a) + \sum_{j \in S} p_t(j|s, a) u_{t+1}^*(j)$  is u.s.c. in  $a$  for each  $s \in S$ . Therefore by Theorem B.2, the supremum over  $a$  in (4.4.5) is attained, from which the result follows.

Conclusion (b) follows from (c) since continuous functions are both upper and lower semicontinuous.  $\square$

To illustrate this result, consider the following example which we analyze in further detail in Sec. 6.4.

**Example 4.4.1.** Let  $N = 3$ ;  $S = \{s_1, s_2\}$ ;  $A_{s_1} = [0, 2]$ , and  $A_{s_2} = \{a_{2,1}\}$ ;  $r_t(s_1, a) = -a^2$ ,  $r_t(s_2, a_{2,1}) = -\frac{1}{2}$ ,  $p_t(s_1|s_1, a) = \frac{1}{2}a$ ,  $p_t(s_2|s_1, a) = 1 - \frac{1}{2}a$ , and  $p_t(s_2|s_2, a_{2,1}) = 1$  for  $t = 1, 2$ ,  $r_3(s_1) = -1$ , and  $r_3(s_2) = -\frac{1}{2}$ . (See Fig. 4.4.1)

Since  $A_{s_1}$  is compact and  $r_t(s_1, \cdot)$  and  $p_t(j|s_1, \cdot)$  are continuous functions on  $A_{s_1}$ , there exists a deterministic Markov policy which is optimal.

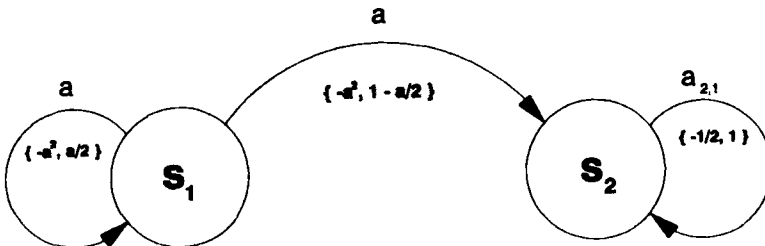


Figure 4.4.1 Graphical representation of Example 4.4.1.

## 4.5 BACKWARD INDUCTION

Backward induction provides an efficient method for solving finite-horizon discrete-time MDPs. For stochastic problems, enumeration and evaluation of *all* policies is the only alternative, but forward induction and reaching methods provide alternative solution methods for deterministic systems. The terms “backward induction” and “dynamic programming” are synonymous, although the expression “dynamic programming” often refers to all results and methods for sequential decision processes. This section presents the backward induction algorithm and shows how to use it to find optimal policies and value functions. The algorithm generalizes the policy evaluation algorithm of Sec. 4.2.

We present the algorithm for a model in which maxima are obtained in (4.3.2), so that we are assured of obtaining an optimal (instead of an  $\varepsilon$ -optimal) Markovian deterministic policy. The algorithm solves optimality equations (4.3.4) subject to boundary condition (4.3.3). Generalization to models based on (4.3.2) is left as an exercise.

### The Backward Induction Algorithm

1. Set  $t = N$  and

$$u_N^*(s_N) = r_N(s_N) \quad \text{for all } s_N \in S,$$

2. Substitute  $t - 1$  for  $t$  and compute  $u_t^*(s_t)$  for each  $s_t \in S$  by

$$u_t^*(s_t) = \max_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(j) \right\}. \quad (4.5.1)$$

Set

$$A_{s_t, t}^* = \arg \max_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(j) \right\}. \quad (4.5.2)$$

3. If  $t = 1$ , stop. Otherwise return to step 2.

As a consequence of Theorem 4.4.2, which among other results shows that  $u_t^*$  depends on  $h_t = (s_{t-1}, a_{t-1}, s_t)$  only through  $s_t$ , we need not evaluate  $u_t^*$  for all  $h_t \in H_t$ . This significantly reduces computational effort. Combining results from Theorems 4.3.2 and 4.4.2 yields the following properties of the iterates of the backward induction algorithm.

**Theorem 4.5.1.** Suppose  $u_t^*$ ,  $t = 1, \dots, N$  and  $A_{s_t, t}^*$ ,  $t = 1, \dots, N - 1$  satisfy (4.5.1) and (4.5.2); then,

- a. for  $t = 1, \dots, N$  and  $h_t = (h_{t-1}, a_{t-1}, s_t)$

$$u_t^*(s_t) = \sup_{\pi \in \Pi^{\text{HR}}} u_t^\pi(h_t), \quad s_t \in S.$$



- b. Let  $d_t^*(s_t) \in A_{s_t,t}^*$  for all  $s_t \in S$ ,  $t = 1, \dots, N-1$ , and let  $\pi^* = (d_1^*, \dots, d_{N-1}^*)$ . Then  $\pi^* \in \Pi^{\text{MD}}$  is optimal and satisfies

$$v_N^{\pi^*}(s) = \sup_{\pi \in \Pi^{\text{HR}}} v_N^{\pi}(s), \quad s \in S$$

and

$$u_t^{\pi^*}(s_t) = u_t^*(s_t), \quad s_t \in S$$

for  $t = 1, \dots, N$ .

This theorem represents a formal statement of the following properties of the backward induction algorithm.

- For  $t = 1, 2, \dots, N-1$ , it finds sets  $A_{s_t,t}^*$  which contain all actions in  $A_{s_t}$  which attain the maximum in (4.5.1).
- It evaluates any policy which selects an action in  $A_{s_t,t}^*$  for each  $s_t \in S$  for all  $t = 1, 2, \dots, N-1$ .
- It computes the expected total reward for the entire decision-making horizon, and from each period to the end of the horizon for any optimal policy.

Let  $D_t^* \equiv \times_{s_t \in S} A_{s_t,t}^*$ . Then any  $\pi^* \in \Pi^* \equiv D_1^* \times \dots \times D_{N-1}^*$  is an optimal policy. If more than one such  $\pi^*$  exists, each yields the same expected total reward. This occurs if for some  $s_t$ ,  $A_{s_t,t}^*$  contains more than one action. To obtain a particular optimal policy, it is only necessary to retain a single action from  $A_{s_t,t}^*$  for each  $t \leq N-1$  and  $s_t \in S$ .

Although this chapter emphasizes models with finite  $S$  and  $A$ , the algorithm and results of Theorem 4.5.1 are valid in greater generality. It applies to models with countable, compact, or Polish state and action spaces. In nondiscrete models, regularity conditions are required to ensure that  $u_t$  is measurable, integrals exist, and maxima are attained. Often we discretize the state and action spaces prior to computation; however, backward induction may be used to find optimal policies when the maxima and maximizing actions can be determined analytically. More importantly, a considerable portion of stochastic optimization literature uses backward induction to characterize the form of optimal policies under structural assumptions on rewards and transition probabilities. When such results can be obtained, specialized algorithms may be developed to determine the best policy of that type. We expand on this point in Sec. 4.7.

When there are  $K$  states with  $L$  actions in each, the backward induction algorithm requires  $(N-1)LK^2$  multiplications to evaluate and determine an optimal policy. Since there are  $(L^K)^{(N-1)}$  deterministic Markovian policies, and direct evaluation of each requires  $(N-1)K^2$  multiplications, this represents considerable reduction in computation.

A further advantage of using backward induction is that, at pass  $t$  through the algorithm, only  $r_t$ ,  $p_t$  and  $u_{t+1}^*$  need be in high-speed memory. The data from iterations  $t+1$ ,  $t+2$ ,  $\dots$ ,  $N$  are not required since they are summarized through  $u_{t+1}^*$ , and the data from decision epochs  $1, 2, \dots, t-1$  are not needed until subsequent iterations. Of course,  $A_{s_t,t}^*$  must be stored for all  $t$  to recover all optimal policies.

## 4.6 EXAMPLES

In this section, we use the backward induction algorithm to find optimal policies for several finite-horizon Markov decision problems. In the first two problems, we use enumeration to perform maximization. The deterministic model of Sec. 4.6.3 illustrates the use of analytic methods. In Sec. 4.6.4, we solve the secretary problem. Recall that  $u_t^*$ ,  $t = 1, 2, \dots, N$  denote solutions of the optimality equations.

### 4.6.1 The Stochastic Inventory Model

We find the optimal policy for the model of Sec. 3.2. Because the model parameters are stationary, we drop the time index on the transition probability and the reward. Numerical values for these quantities appear in Sec. 3.2.2. Define  $u_t^*(s, a)$  by

$$u_t^*(s, a) = r(s, a) + \sum_{j \in S} p(j|s, a) u_{t+1}^*(j).$$

We implement the backward induction algorithm as follows.

1. Set  $t = 4$  and  $u_4^*(s) = r_4(s) = 0$ ,  $s = 0, 1, 2, 3$ .
2. Since  $t \neq 1$ , continue. Set  $t = 3$  and

$$\begin{aligned} u_3^*(s) &= \max_{a \in A_s} \left\{ r(s, a) + \sum_{j \in S} p(j|s, a) u_4^*(j) \right\}, \quad s = 0, 1, 2, 3 \\ &= \max_{a \in A_s} \{ r(s, a) \}. \end{aligned}$$

Inspecting the values of  $r(s, a)$  shows that in each state the maximizing action is 0, that is, do not order. We obtain

| $s$ | $u_3^*(s)$ | $A_{s,3}^*$ |
|-----|------------|-------------|
| 0   | 0          | 0           |
| 1   | 5          | 0           |
| 2   | 6          | 0           |
| 3   | 5          | 0           |

3. Since  $t \neq 1$ , continue. Set  $t = 2$  and

$$u_2^*(s) = \max_{a \in A_s} \{ u_2^*(s, a) \},$$

where, for example,

$$\begin{aligned} u_2^*(0, 2) &= r(0, 2) + p(0|0, 2)u_3^*(0) + p(1|0, 2)u_3^*(1) + p(2|0, 2)u_3^*(2) + p(3|0, 2)u_3^*(3) \\ &= -2 + \left(\frac{1}{4}\right) \times 0 + \left(\frac{1}{2}\right) \times 5 + \left(\frac{1}{4}\right) \times 6 + 0 \times 5 = 2 \end{aligned}$$

The quantities  $u_2^*(s, a)$ ,  $u_2^*(s)$  and  $A_{s,2}^*$  are summarized in the following table with  $\times$ 's denoting infeasible actions.

| $s$ | $u_2^*(s, a)$  |               |               |               | $u_2^*(s)$     | $A_{s,2}^*$ |
|-----|----------------|---------------|---------------|---------------|----------------|-------------|
|     | $a = 0$        | $a = 1$       | $a = 2$       | $a = 3$       |                |             |
| 0   | 0              | $\frac{1}{4}$ | 2             | $\frac{1}{2}$ | 2              | 2           |
| 1   | $\frac{25}{4}$ | 4             | $\frac{5}{2}$ | $\times$      | $\frac{25}{4}$ | 0           |
| 2   | 10             | $\frac{9}{2}$ | $\times$      | $\times$      | 10             | 0           |
| 3   | $\frac{21}{2}$ | $\times$      | $\times$      | $\times$      | $\frac{21}{2}$ | 0           |

4. Since  $t \neq 1$ , continue. Set  $t = 1$  and

$$u_1^*(s) = \max_{a \in A_s} \{u_1^*(s, a)\}.$$

The quantities  $u_1^*(s, a)$ ,  $u_1^*(s)$ , and  $A_{s,1}^*$  are summarized in the following table.

| $s$ | $u_1^*(s, a)$    |                  |                 |                 | $u_1^*(s)$       | $A_{s,1}^*$ |
|-----|------------------|------------------|-----------------|-----------------|------------------|-------------|
|     | $a = 0$          | $a = 1$          | $a = 2$         | $a = 3$         |                  |             |
| 0   | 2                | $\frac{33}{16}$  | $\frac{66}{16}$ | $\frac{67}{16}$ | $\frac{67}{16}$  | 3           |
| 1   | $\frac{129}{16}$ | $\frac{98}{16}$  | $\frac{99}{16}$ | $\times$        | $\frac{129}{16}$ | 0           |
| 2   | $\frac{194}{16}$ | $\frac{131}{16}$ | $\times$        | $\times$        | $\frac{194}{16}$ | 0           |
| 3   | $\frac{227}{16}$ | $\times$         | $\times$        | $\times$        | $\frac{227}{16}$ | 0           |

5. Since  $t = 1$ , stop.

This algorithm yields the optimal expected total reward function  $v_4^*(s)$  and the optimal policy  $\pi^* = (d_1^*(s), d_2^*(s), d_3^*(s))$ , which is tabulated below. Note in this example that the optimal policy is *unique*.

| $s$ | $d_1^*(s)$ | $d_2^*(s)$ | $d_3^*(s)$ | $v_4^*(s)$       |
|-----|------------|------------|------------|------------------|
| 0   | 3          | 2          | 0          | $\frac{67}{16}$  |
| 1   | 0          | 0          | 0          | $\frac{129}{16}$ |
| 2   | 0          | 0          | 0          | $\frac{194}{16}$ |
| 3   | 0          | 0          | 0          | $\frac{227}{16}$ |

The quantity  $v_4^*(s)$  gives the expected total reward obtained using the optimal policy when the inventory at the start of the month is  $s$  units. The optimal policy has a particularly simple form; if at the start of month 1 the inventory is 0 units, order three units, otherwise do not order; if at the start of month 2 the inventory is two units, order two units, otherwise do not order, and do not order in month 3 for any inventory level. This policy is a nonstationary  $(\sigma, \Sigma)$  policy as defined in Sec. 3.2. The reordering point  $\sigma_t = 0$  at each decision epoch and the target stock  $\Sigma_t$  equals 3 at

decision epoch 1; 2 at decision epoch 2; and 0 at decision epoch 3. For example,

$$d_1^*(s) = \begin{cases} 0 & s > 0 \\ 3 & s = 0. \end{cases}$$

Policies of this form are optimal in inventory models in which the ordering cost has the form

$$O(u) = \begin{cases} 0 & u = 0 \\ K + cu & u > 0, \end{cases}$$

where  $K > 0$  and  $c > 0$ , the shortage and holding cost  $h(u)$  is convex, and backlogging of unfilled orders is permitted. A proof that  $(\sigma, \Sigma)$  policies are optimal under these assumptions may be based on backward induction; however, the technical details are quite complicated. We refer the interested reader to references in the bibliographic summary for more on results of this type.

#### 4.6.2 Routing Problems

In this section, we apply backward induction to find the longest route in the model in Fig. 3.3.1. Here states correspond to nodes and action sets correspond to the collection of arcs originating at each node. Below, the action  $(i, j)$  corresponds to choosing the arc between nodes  $i$  and  $j$ .

1. Set  $t = 4$  and  $u_4^*(8) = 0$ .
2. Since  $t \neq 1$ , continue. Set  $t = 3$  and

$$\begin{aligned} u_3^*(5) &= 1 + u_4^*(8) = 1, & u_3^*(6) &= 2 + u_4^*(8) = 2, \\ u_3^*(7) &= 6 + u_4^*(8) = 6. \end{aligned}$$

Because there is only one action in each state,

$$A_{5,3}^* = (5, 8), \quad A_{6,3}^* = (6, 8), \quad A_{7,3}^* = (7, 8).$$

3. Since  $t \neq 1$ , continue. Set  $t = 2$  and

$$\begin{aligned} u_2^*(2) &= \max\{4 + u_3^*(5), 5 + u_3^*(6)\} = \max\{5, 7\} = 7, \\ u_2^*(3) &= \max\{5 + u_3^*(5), 6 + u_3^*(6), 1 + u_3^*(7)\} = \max\{6, 8, 7\} = 8, \\ u_2^*(4) &= 2 + u_3^*(7) = 8. \end{aligned}$$

The optimal action sets are

$$A_{2,2}^* = (2, 6), \quad A_{3,2}^* = (3, 6), \quad A_{4,2}^* = (4, 7).$$

4. Since  $t \neq 1$ , continue. Set  $t = 1$  and

$$u_1^*(1) = \max\{2 + u_2^*(2), 4 + u_2^*(3), 3 + u_2^*(4)\} = \max\{9, 12, 11\} = 12.$$

The optimal action set is

$$A_{1,1}^* = (1, 3).$$

5. Since  $t = 1$ , stop.

Since the optimal action set at each state is a singleton, there is a unique optimal policy  $\pi^* = (d_1^*, d_2^*, d_3^*)$  where

$$\begin{aligned} d_1^*(1) &= (1, 3), \\ d_2^*(2) &= (2, 6), \quad d_2^*(3) = (3, 6), \quad d_2^*(4) = (4, 7), \\ d_3^*(5) &= (5, 8), \quad d_3^*(6) = (6, 8), \quad d_3^*(7) = (7, 8). \end{aligned}$$

This policy selects which arc to traverse from each node in order to travel the longest route from that node to node 8. Starting at node 1, the optimal path is  $1 \rightarrow 3 \rightarrow 6 \rightarrow 8$ . While starting at node 2, the optimal path is  $2 \rightarrow 6 \rightarrow 8$ .

From the critical path analysis perspective, if the distances represent weeks, this result means that the project requires 12 weeks to complete and the critical path consists of the tasks represented by arcs (1, 3), (3, 6), and (6, 8). If any of these are delayed, the entire project will be delayed.

This example illustrates both statements of the Principle of Optimality in Sec. 4.3. Starting at node 1 and using the optimal policy moves the system from node 3 to node 6 to node 8. However, if the system started at node 3, the optimal policy would be to go to node 6 and then node 8. This is identical to the optimal policy for the whole problem as stated in the Principle of Optimality. Because the policy  $\pi^*$  above gives the optimal path from each node to node 8, the second statement of the Principle of Optimality is also satisfied.

We now provide an example in which the Principle of Optimality does not hold. Consider the network in Figure 4.6.1. Define the “value” of a path from a given node

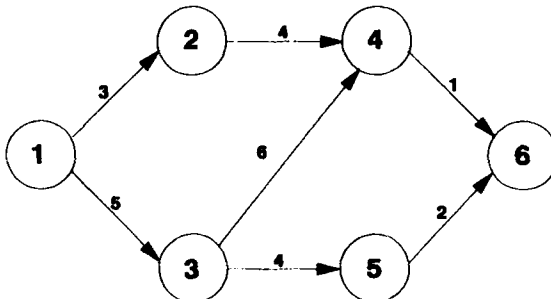


Figure 4.6.1 Network example for Sec. 4.6.2.

to node 6, to be the length of the *second-longest arc* on that path when there are two or more arcs between that node and node 6; if there is a single arc, then its length is the arc length. Our objective is to find the path from node 1 to node 6 with greatest value. Clearly this is the routing  $1 \rightarrow 3 \rightarrow 4 \rightarrow 6$ . Its value is 5 corresponding to arc (1, 3). Applying the optimal decision in node 1 moves the system to node 3. If the Principle of Optimality were valid, the optimal path from that node to node 6 would be  $3 \rightarrow 4 \rightarrow 6$ . Clearly it is not. The optimal path is  $3 \rightarrow 5 \rightarrow 6$  with value 2 corresponding to arc (5, 6) while the value of  $3 \rightarrow 4 \rightarrow 6$  is 1 corresponding to (4, 6). Under this optimality criterion, backward induction does not yield an optimal policy. This is because to find an optimal policy using backward induction requires a separable policy evaluation function such as expected total reward (see also Prob. 4.2).

### 4.6.3 The Sequential Allocation Model

In this section we adapt the backward induction algorithm to find an optimal solution to the sequential allocation model of Sec. 3.3.3 and use it to solve the problem with  $f(x_1, \dots, x_N) = -x_1^2 - x_2^2 - \dots - x_N^2$ . Since maximization of a negative function is equivalent to minimization of a positive function, this problem has the following nonlinear (quadratic) programming representation:

$$\text{Minimize } x_1^2 + x_2^2 + \dots + x_N^2$$

subject to

$$x_1 + x_2 + \dots + x_N = M \quad (4.6.1)$$

and  $x_t \geq 0$ ,  $t = 1, \dots, N$ . We solve this problem using backward induction with "min" replacing "max" throughout.

We first describe the algorithm for a separable sequential allocation problem. Let  $u_t(s)$  be the minimal cost if  $s$  units are available for use in periods  $t$  through  $N$ . Clearly  $u_N(s) = g_N(s)$ . For  $t = 1, \dots, N - 1$ ,

$$u_t(s) = \text{minimum}_{(x_t, \dots, x_N) \in B_t(s)} g_t(x_t) + \dots + g_N(x_N),$$

where  $B_t(s) = \{x_n: x_n \geq 0, n = t, \dots, N \text{ and } x_t + \dots + x_N = s\}$ . To solve the problem, we seek  $u_1(M)$  and the corresponding allocations. These are found by first using the backward induction algorithm to find  $u_t(s)$  for each  $t$ . Since the problem is deterministic, we require actions only at states that can be achieved by implementing the optimal policy. We find these by applying the optimal decision rule in period 1 at the specified initial system state, determining the resulting state at decision epoch 2, applying the optimal decision epoch 2 decision rule at this state to determine the state at decision epoch 3, etc.

The optimality equations for this problem are

$$u_t(s) = \min_{a \in [0, s]} \{g_t(a) + u_{t+1}(s - a)\}. \quad (4.6.2)$$

$0 \leq s \leq M$  and  $t = 1, \dots, N - 1$  and  $u_N(s) = g_N(s)$ . We derive them by noting that the optimal value over periods  $t$  through  $N$  equals the minimal value that can be found by allocating  $a$  units in period  $t$  and then optimally allocating the remaining  $s - a$  over periods  $t + 1$  to  $N$ .

We solve problem (4.6.1) by applying backward induction for a few iterations to suggest the form of the optimal policy and the corresponding cost-to-go functions and then using mathematical induction to prove that the optimal policy has this form. Notation follows Sec. 3.3.3 with  $g_t(s) = s^2$  for  $t = 1, \dots, N$ . Implementation of the algorithm proceeds as follows.

1. Set  $t = N$  and  $u_N^*(s) = g_N(s) = s^2$ , for  $s \in [0, M]$ .
2. Since  $t \neq 1$ , set  $t = N - 1$  and

$$u_{N-1}^*(s) = \min_{a \in [0, s]} \{a^2 + u_N^*(s - a)\} = \min_{a \in [0, s]} \{a^2 + (s - a)^2\}.$$

Differentiating the expression in brackets with respect to  $a$ , setting the derivative equal to zero and solving for  $a$  shows that

$$\frac{s}{2} = \arg \min_{a \in [0, s]} \{a^2 + (s - a)^2\},$$

so that  $A_{s, N-1}^* = s/2$  and  $u_{N-1}^*(s) = s^2/2$ .

3. Since  $t \neq 1$ , set  $t = N - 2$ , and

$$u_{N-2}^*(s) = \min_{a \in [0, s]} \left\{a^2 + \frac{1}{2}(s - a)^2\right\}.$$

We find that  $A_{s, N-2}^* = s/3$  and  $u_{N-2}^*(s) = s^2/3$ .

The pattern is now obvious. We hypothesize that  $A_{s, N-n+1}^* = s/n$  and that  $u_{N-n+1}^*(s) = s^2/n$  and apply induction. As a consequence of the induction hypothesis, the optimality equation becomes

$$u_{N-n}^*(s) = \min_{a \in [0, s]} \left\{a^2 + \frac{1}{n}(s - a)^2\right\},$$

so that  $A_{s, N-n}^* = s/(n+1)$  and  $u_{N-n}^*(s) = s^2/(n+1)$ . Thus  $A_{s, N-n}^* = s/(n+1)$  and  $u_{N-n}^*(s) = s^2/(n+1)$  for all  $n$ .

The optimal value of the objective function is  $u_1^*(M) = M^2/N$  and the optimal decision in period 1 is the unique element of  $A_{M, 1}^*$ ,  $M/N$ . When action  $s/N$  is used in period 1, the state of the system at decision epoch 2 is  $M - (M/N) = [(N-1)/N]M$ , so the optimal decision in period 2 is the unique element of  $A_{[(N-1)/N]M, 2}^*$  which equals  $[(N-1)/N]M/(N-1)$  or  $M/N$ . Applying a formal induction argument shows that the optimal allocation in each period is  $M/N$  so that the optimal solution to the problem is  $x_t = M/N$  for  $t = 1, \dots, N$ .

The same procedure applied to (3.3.4) subject to (3.3.2) and  $q_i \geq 0$ ,  $i = 1, \dots, N$  yields the following maximum likelihood estimates for the multinomial distribution:

$$\hat{q}_i = \frac{y_i}{y_1 + \dots + y_N}, \quad i = 1, \dots, N. \quad (4.6.3)$$

Demonstrating this is left as an exercise.

#### 4.6.4 The Secretary Problem

We now solve the secretary problem of Sec. 3.4.3. Recall that  $N$  candidates are available. Let  $u_t^*(1)$  denote the maximum probability of choosing the best candidate, when the current candidate has the highest relative rank among the first  $t$  interviewed, and  $u_t^*(0)$  denote the maximum probability of choosing the best overall candidate if the current candidate does not have highest relative rank among the first  $t$  interviewed. Recall that  $\Delta$  denotes the stopped state.

Then  $u_t^*$  satisfies the following recursions:

$$u_N^*(1) = h(1) = 1, \quad u_N^*(0) = h(0) = 0, \quad u_N^*(\Delta) = 0,$$

and, for  $t < N$ ,

$$\begin{aligned} u_t^*(1) &= \max\{g_t(1) + u_{t+1}^*(\Delta), -f_t(0) + p_t(1|1)u_{t+1}^*(1) + p_t(0|1)u_{t+1}^*(0)\} \\ &= \max\left\{\frac{t}{N}, \frac{1}{t+1}u_{t+1}^*(1) + \frac{t}{t+1}u_{t+1}^*(0)\right\}, \end{aligned} \quad (4.6.4)$$

$$\begin{aligned} u_t^*(0) &= \max\{g_t(0) + u_{t+1}^*(\Delta), -f_t(0) + p_t(1|0)u_{t+1}^*(1) + p_t(0|0)u_{t+1}^*(0)\} \\ &= \max\left\{0, \frac{1}{t+1}u_{t+1}^*(1) + \frac{t}{t+1}u_{t+1}^*(0)\right\}, \end{aligned} \quad (4.6.5)$$

and

$$u_t^*(\Delta) = u_{t+1}^*(\Delta) = 0.$$

Noting that  $u_t^* \geq 0$  allows simplification of (4.6.5) and (4.6.4) to:

$$u_t^*(0) = \frac{1}{t+1}u_{t+1}^*(1) + \frac{t}{t+1}u_{t+1}^*(0), \quad (4.6.6)$$

$$u_t^*(1) = \max\left\{\frac{t}{N}, u_t^*(0)\right\}. \quad (4.6.7)$$

Solution of (4.6.6) and (4.6.7) yields an optimal policy for this problem as follows. In state 1 when  $t/N > u_t^*(0)$  the optimal action is to stop, when  $t/N < u_t^*(0)$  the optimal action is to continue and when  $t/N = u_t^*(0)$  either action is optimal. In state 0 the optimal action is to continue.

Applying induction formally or appealing to intuition suggests that the optimal policy has the form "Observe the first  $\tau$  candidates; after that, select the first candidate who is better than all the previous ones." This translates to the policy  $\pi^* = (d_1^*, d_2^*, \dots, d_{N-1}^*)$ , with  $d_t^*(0) = C$ ,  $d_t^*(1) = C$  if  $t \leq \tau$  and  $d_t^*(1) = Q$  if  $t > \tau$ .

We now show that an optimal policy has this form by demonstrating that if  $u_\tau^*(1) > \tau/N$  or  $u_\tau^*(1) = \tau/N = u_\tau^*(0)$  for some  $\tau$ , in which case it is optimal to continue, then  $u_t^*(1) > t/N$  for  $t < \tau$  so that it is again optimal to continue. Assume



either  $u_\tau^*(1) > \tau/N$  so that from (4.6.7),  $u_\tau^*(1) = u_\tau^*(0)$ , or  $u_\tau^*(1) = \tau/N = u_\tau^*(0)$  so that

$$u_{\tau-1}^*(0) = \frac{1}{\tau} u_\tau^*(1) + \frac{\tau-1}{\tau} u_\tau^*(0) = u_\tau^*(0) \geq \frac{\tau}{N}.$$

Thus

$$u_{\tau-1}^*(1) = \max\left\{\frac{\tau-1}{N}, u_{\tau-1}^*(0)\right\} \geq \frac{\tau}{N} > \frac{\tau-1}{N}.$$

Repeating this argument with  $\tau-1$  replacing  $\tau$  and so forth shows that the result holds for all  $t < \tau$ . This implies that an optimal policy cannot have the form  $d_t^*(1) = C$  for  $t \leq t'$ ,  $d_t^*(1) = Q$  for  $t' < t \leq t''$ , and  $d_t^*(1) = C$  for  $t > t''$ .

By definition  $\tau < N$ . We show that, if  $N > 2$ ,  $\tau \geq 1$ . Supposing it is not, then, for all  $t$ ,  $u_t^*(1) = t/N$ , so that, by (4.6.6),

$$u_t^*(0) = \frac{1}{t+1} \frac{t+1}{N} + \frac{t}{t+1} u_{t+1}^*(0) = \frac{1}{N} + \frac{t}{t+1} u_{t+1}^*(0). \quad (4.6.8)$$

Noting that  $u_N^*(0) = 0$ , and solving Eq. (4.6.8) by backward induction yields

$$u_t^*(0) = \frac{t}{N} \left[ \frac{1}{t} + \frac{1}{t+1} + \cdots + \frac{1}{N-1} \right], \quad 1 \leq t < N, \quad (4.6.9)$$

for  $N > 2$ . This implies that  $u_1^*(0) > 1/N = u_1^*(1) \geq u_1^*(0)$ ; the second inequality is a consequence of (4.6.7). This is a contradiction, so we may conclude that  $\tau \geq 1$  when  $N > 2$ .

Thus, when  $N > 2$ ,

$$u_1^*(0) = u_1^*(1) = \cdots = u_\tau^*(0) = u_\tau^*(1), \quad (4.6.10)$$

$u_t^*(1) = t/N$  for  $t > \tau$ , and

$$u_t^*(0) = \frac{t}{N} \left[ \frac{1}{t} + \frac{1}{t+1} + \cdots + \frac{1}{N-1} \right], \quad t > \tau.$$

Thus it is optimal to continue to observe candidates as long as  $u_t^*(0) > t/N$ , so that

$$\tau = \max_{t \geq 1} \left\{ \left[ \frac{1}{t} + \frac{1}{t+1} + \cdots + \frac{1}{N-1} \right] > 1 \right\}. \quad (4.6.11)$$

When  $N \leq 2$ , choose  $\tau = 0$ , but of course, any policy is optimal in this case.

For example, suppose  $N = 4$ , since  $1 + \frac{1}{2} + \frac{1}{3} > 1$  and  $\frac{1}{2} + \frac{1}{3} < 1$ ,  $\tau = 1$ . This means that, with four candidates, the decision maker should observe the first candidate and then choose the next candidate whose relative rank is 1. Figure 4.6.2 shows how  $\tau/N$  and  $u_1^*(1)$ , the probability of choosing the best candidate, vary with  $N$ . It suggests that, as  $N$  becomes large,  $\tau/N$  and  $u_1^*(1)$  approach limits. Let  $\tau(N)$  denote the optimal value of  $\tau$  when there are  $N$  candidates to consider. Further calculations

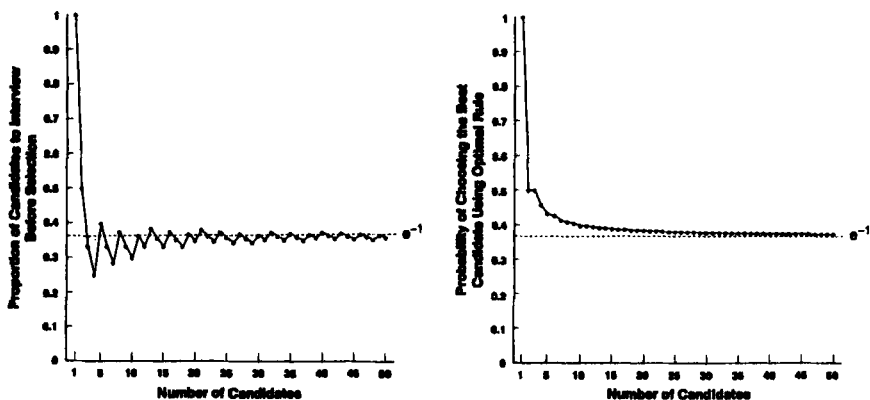


Figure 4.6.2 Graphical representation of the solution of the secretary problem.

reveal that  $\tau(1000) = 368$ ,  $\tau(10,000) = 3679$ , and  $\tau(100,000) = 36788$ . From (4.6.10), it follows that, for  $N$  sufficiently large,

$$1 \approx \left[ \frac{1}{\tau(N)} + \frac{1}{\tau(N) + 1} + \cdots + \frac{1}{N-1} \right] \approx \int_{\tau(N)}^N \frac{1}{x} dx, \quad (4.6.12)$$

so that, for large  $N$ ,

$$\log \left( \frac{N}{\tau(N)} \right) \approx 1.$$

This implies that

$$\lim_{N \rightarrow \infty} \frac{\tau(N)}{N} = e^{-1},$$

and, as a consequence of (4.6.10), (4.6.11) and the first equality in (4.6.12),

$$u_1^*(0) = u_1^*(1) = u_{\tau(N)}^*(0) = \frac{\tau(N)}{N} \rightarrow e^{-1}$$

Thus with a large number of candidates the decision maker should observe  $Ne^{-1}$  or 36.8% of them and subsequently choose the first candidate with highest relative rank. The probability of choosing the best candidate using this policy is  $1/e$  or 0.36788, which agrees to five places with the value when  $N = 100,000$ .

## 4.7 OPTIMALITY OF MONOTONE POLICIES

In Sec. 4.4, we provided conditions under which there exists a deterministic Markov optimal policy. In this section we investigate more detailed results concerning the structure of an optimal policy. We provide conditions which ensure that optimal

policies are monotone (nondecreasing or nonincreasing) in the system state, interpret them in the context of a product price determination model, and show that they are satisfied in an equipment replacement model. For such a concept to be meaningful, we require that the state have a physical interpretation and some natural ordering. Throughout this book the expression “monotone policy” refers to a monotone deterministic Markov policy.

### 4.7.1 Structured Policies

One of the principal uses of Markov decision process methods is to establish the existence of optimal policies with special structure. The importance of results regarding the optimality of structured policies lies in

- a. their appeal to decision makers,
- b. their ease in implementation, and
- c. their enabling efficient computation.

The last point is particularly noteworthy because, when the optimal policy has simple form, specialized algorithms can be developed to search only among policies that have the same form as the optimal policy. This avoids the need for the less efficient general backward induction algorithm.

Examples of policies with simple structure are  $(\sigma, \Sigma)$  policies in inventory models, and control limit or critical number policies in queueing control models. A *control limit policy* is a deterministic Markov policy composed of decision rules of the form

$$d_t(s) = \begin{cases} a_1 & s < s^* \\ a_2 & s \geq s^*, \end{cases}$$

where  $a_1$  and  $a_2$  are distinct actions and  $s^*$  is a control limit. We interpret such a policy as follows; when the state of the system is less than  $s^*$ , it is optimal to use action  $a_1$ , and when the system state is  $s^*$  or greater, it is optimal to use action  $a_2$ . In a queueing control model,  $s$  might represent the number of customers in the system,  $a_1$  the minimum service rate, and  $a_2$  the maximum service rate; while, in a replacement problem,  $s$  might represent the age or condition of the component,  $a_1$  the action not to replace, and  $a_2$  the action to replace. If we establish that such policies are optimal, the problem of finding an optimal policy reduces to that of determining  $s^*$ .

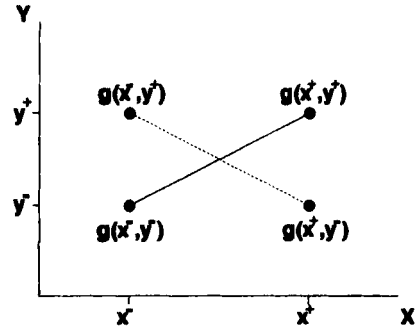
### 4.7.2 Superadditive Functions

We begin with some technical material. Let  $X$  and  $Y$  be partially ordered sets and  $g(x, y)$  a real-valued function on  $X \times Y$ . We say that  $g$  is *superadditive* if for  $x^+ \geq x^-$  in  $X$  and  $y^+ \geq y^-$  in  $Y$ ,

$$g(x^+, y^+) + g(x^-, y^-) \geq g(x^+, y^-) + g(x^-, y^+). \quad (4.7.1)$$

If the reverse inequality above holds,  $g(x, y)$  is said to be *subadditive*. Thus  $g(x, y)$  is

**Figure 4.7.1** In a superadditive function, the sum of the quantities joined by the solid line exceeds that of the sum of the quantities joined by the dashed line.



subadditive if  $-g(x, y)$  is superadditive. We sometimes refer to functions satisfying (4.7.1) as *supermodular* and expression (4.7.1) as the *quadrangle inequality* (Fig. 4.7.1).

Equivalently, we say that  $g(x, y)$  is superadditive if it has *monotone increasing differences*; that is,

$$g(x^+, y^+) - g(x^+, y^-) \geq g(x^-, y^+) - g(x^-, y^-). \quad (4.7.2)$$

Examples of superadditive functions include the following.

1.  $g(x, y) = h(x) + e(y)$ , where  $h$  and  $e$  are arbitrary functions on  $X$  and  $Y$  respectively.
2.  $g(x, y) = h(x + y)$ , where  $X = Y = R^1$  and  $h$  is convex increasing.
3.  $g(x, y) = xy$ , where  $X = Y = R^1$ .

In particular,  $(x + y)^2$  and  $-(x - y)^2$  are superadditive while  $(x - y)^2$  and  $-(x + y)^2$  are subadditive. A consequence of (4.7.2) is that, when  $X = Y = R^1$ , and  $g(x, y)$  is twice differentiable, then  $g$  is superadditive whenever

$$\frac{\partial^2 g(x, y)}{\partial x \partial y} \geq 0.$$

The following lemma provides the key property of superadditive functions from our perspective.

**Lemma 4.7.1.** Suppose  $g$  is a superadditive function on  $X \times Y$  and for each  $x \in X$ ,  $\max_{y \in Y} g(x, y)$  exists. Then

$$f(x) = \max_{y \in Y} \left\{ y' \in \arg \max_{y \in Y} g(x, y) \right\} \quad (4.7.3)$$

is monotone nondecreasing in  $x$ .

**Proof.** Let  $x^+ \geq x^-$  and choose  $y \leq f(x^-)$ . Then, by the definition of  $f$ ,

$$g(x^-, f(x^-)) - g(x^-, y) \geq 0,$$

and by (4.7.1),

$$g(x^-, y) + g(x^+, f(x^-)) \geq g(x^-, f(x^-)) + g(x^+, y).$$

Rewriting the second inequality as

$$g(x^+, f(x^-)) \geq g(x^+, y) + [g(x^-, f(x^-)) - g(x^-, y)]$$

and combining it with the first inequality above yields

$$g(x^+, f(x^-)) \geq g(x^+, y)$$

for all  $y \leq f(x^-)$ . Consequently,  $f(x^+) \geq f(x^-)$ .  $\square$

When  $\arg \max_{y \in Y} g(x, y)$  contains a unique element, (4.7.3) may be written as  $f(x) = \arg \max_{y \in Y} g(x, y)$ . Problems at the end of the chapter address analogous conditions for minimization problems.

### 4.7.3 Optimality of Monotone Policies

We now provide conditions on the states, actions, rewards, and transition probabilities under which monotone policies are optimal. Assume that  $S$  represents the set of non-negative integers and  $A_s = A'$  for all  $s \in S$ . Define

$$q_t(k|s, a) = \sum_{j=k}^{\infty} p_t(j|s, a)$$

for  $t = 1, \dots, N-1$ . It represents the probability that the state at decision epoch  $t+1$  exceeds  $k-1$  when choosing action  $a$  in state  $s$  at decision epoch  $t$ .

We demonstrate optimality of monotone policies by

1. Inductively showing that the optimal value functions from  $t$  onward,  $u_t^*(s)$ , are nondecreasing or nonincreasing in  $S$ , and
2. then showing that

$$w_t(s, a) = r_t(s, a) + \sum_{j=0}^{\infty} p_t(j|s, a) u_t^*(j) \quad (4.7.4)$$

is superadditive or subadditive.

We begin with the following important technical lemma.

**Lemma 4.7.2.** Let  $\{x_j\}, \{x'_j\}$  be real-valued non-negative sequences satisfying

$$\sum_{j=k}^{\infty} x_j \geq \sum_{j=k}^{\infty} x'_j \quad (4.7.5)$$

for all  $k$ , with equality holding in (4.7.5) for  $k = 0$ .

Suppose  $v_{j+1} \geq v_j$  for  $j = 0, 1, \dots$ , then

$$\sum_{j=0}^{\infty} v_j x_j \geq \sum_{j=0}^{\infty} v_j x'_j, \quad (4.7.6)$$

where limits in (4.7.6) exist but may be infinite.

**Proof.** Let  $k$  be arbitrary and  $v_{-1} = 0$ . Then

$$\begin{aligned} \sum_{j=0}^{\infty} v_j x_j &= \sum_{j=0}^{\infty} x_j \sum_{i=0}^j (v_i - v_{i-1}) = \sum_{j=0}^{\infty} (v_j - v_{j-1}) \sum_{i=j}^{\infty} x_i \\ &= \sum_{j=1}^{\infty} (v_j - v_{j-1}) \sum_{i=j}^{\infty} x_i + v_0 \sum_{i=0}^{\infty} x_i \geq \sum_{j=1}^{\infty} (v_j - v_{j-1}) \sum_{i=j}^{\infty} x'_i + v_0 \sum_{i=0}^{\infty} x'_i \\ &= \sum_{j=0}^{\infty} v_j x'_j. \end{aligned} \quad (4.7.7)$$

The last equality follows by reversing the steps in (4.7.7). As a consequence of representation (4.7.7), both limits in (4.7.6) exist.  $\square$

Note that Lemma 4.7.2 allows us to derive an important property of stochastically ordered random variables. Suppose  $Y$  and  $Y'$  denote random variables with  $x_j = P\{Y = j\}$  and  $x'_j = P\{Y' = j\}$ . When (4.7.5) holds,  $Y$  is stochastically greater than  $Y'$  (Sec. 4.1). As a consequence of (4.7.6), it follows that, for any nondecreasing function  $f(j)$ ,  $E\{f(Y)\} \geq E\{f(Y')\}$ .

We now provide conditions under which  $u_t^*$  is monotone. We assume that  $A'$ ,  $r_t$ , and  $p_t$  are such that

$$\max_{a \in A'} \left\{ r_t(s, a) + \sum_{j=0}^{\infty} p_t(j|s, a) u(j) \right\} \quad (4.7.8)$$

is attained for all monotone  $u$ . This occurs under the conditions in Proposition 4.4.3. Note Proposition 4.7.3 is also valid when “sup” replaces “max” in (4.7.8).

**Proposition 4.7.3.** Suppose the maximum in (4.7.8) is attained and that

1.  $r_t(s, a)$  is nondecreasing (nonincreasing) in  $s$  for all  $a \in A'$  and  $t = 1, \dots, N-1$ ,
2.  $q_t(k|s, a)$  is nondecreasing in  $s$  for all  $k \in S$ ,  $a \in A'$ , and  $t = 1, \dots, N-1$ , and
3.  $r_N(s)$  is nondecreasing (nonincreasing) in  $s$ .

Then  $u_t^*(s)$  is nondecreasing (nonincreasing) in  $s$  for  $t = 1, \dots, N$ .

**Proof.** We prove the result in the nondecreasing case by using backward induction to show that  $u_t^*(s)$  is nondecreasing for all  $t$ . Since  $u_N^*(s) = r_N(s)$ , the result holds for  $t = N$  by 3.

Assume now that  $u_n^*(s)$  is nondecreasing for  $n = t + 1, \dots, N$ . By assumption, there exists an  $a_s^* \in A'$  which attains the maximum in

$$u_t^*(s) = \max_{a \in A'} \left\{ r_t(s, a) + \sum_{j=0}^{\infty} p_t(j|s, a) u_{t+1}^*(j) \right\},$$

so that

$$u_t^*(s) = r_t(s, a_s^*) + \sum_{j=0}^{\infty} p_t(j|s, a_s^*) u_{t+1}^*(j).$$

Let  $s' \geq s$ . By (1) and (2), the induction hypothesis, and Lemma 4.7.2 applied with  $x_j' = p_t(j|s, a_s^*)$ ,  $x_j = p_t(j|s', a_s^*)$  and  $v_j = u_{t+1}^*(j)$ ,

$$\begin{aligned} u_t^*(s) &\leq r_t(s', a_s^*) + \sum_{j=0}^{\infty} p_t(j|s', a_s^*) u_{t+1}^*(j) \\ &\leq \max_{a \in A'} \left\{ r_t(s', a) + \sum_{j=0}^{\infty} p_t(j|s', a) u_{t+1}^*(j) \right\} = u_t^*(s'). \end{aligned}$$

Thus  $u_t^*(s)$  is nondecreasing, the induction hypothesis is satisfied, and the result follows.  $\square$

The following theorem provides conditions under which there exist monotone optimal policies. Note there might exist other optimal policies which are not monotone.

**Theorem 4.7.4.** Suppose for  $t = 1, \dots, N - 1$  that

1.  $r_t(s, a)$  is nondecreasing in  $s$  for all  $a \in A'$ ,
2.  $q_t(k|s, a)$  is nondecreasing in  $s$  for all  $k \in S$  and  $a \in A'$ ,
3.  $r_t(s, a)$  is a superadditive (subadditive) function on  $S \times A'$ ,
4.  $q_t(k|s, a)$  is a superadditive (subadditive) function on  $S \times A'$  for all  $k \in S$ , and
5.  $r_N(s)$  is nondecreasing in  $s$ .

Then there exist optimal decision rules  $d_t^*(s)$  which are nondecreasing (nonincreasing) in  $s$  for  $t = 1, \dots, N - 1$ .

**Proof.** We prove the result in the superadditive case. We show that  $w_t(s, a)$  defined in (4.7.4) is superadditive whenever  $q$  and  $r$  are. By condition (4), and the definition of superadditivity, for  $s^- \leq s^+$  and all  $k \in S$ ,

$$\sum_{j=k}^{\infty} [p_t(j|s^-, a^-) + p_t(j|s^+, a^+)] \geq \sum_{j=k}^{\infty} [p_t(j|s^-, a^+) + p_t(j|s^+, a^-)].$$

By Proposition 4.7.3,  $u_t^*(s)$  is nondecreasing in  $s$  for all  $t$ , so applying Lemma 4.7.2 yields

$$\sum_{j=0}^{\infty} [p_t(j|s^-, a^-) + p_t(j|s^+, a^+)] u_t(j) \geq \sum_{j=0}^{\infty} [p_t(j|s^-, a^+) + p_t(j|s^+, a^-)] u_t(j).$$

Thus for each  $t$ ,  $\sum_{j=0}^{\infty} p_t(j|s, a) u_t(j)$  is superadditive.

From condition (3),  $r_t(s, a)$  is superadditive and, since the sum of superadditive functions is superadditive,  $w_t(s, a)$  is superadditive. The result follows from Lemma 4.7.1.  $\square$

Theorem 4.7.5 below provides alternative conditions which lead to monotone optimal policies. It follows from the last three lines of the proof of Theorem 4.7.4. Note that, although condition (4) includes the function  $u$  which is not part of the basic problem data, the condition may be verified directly in many applications.

**Theorem 4.7.5.** Suppose for  $t = 1, \dots, N-1$  that

1.  $r_t(s, a)$  is nonincreasing in  $s$  for all  $a \in A'$ ,
2.  $q_t(k|s, a)$  is nondecreasing in  $s$  for all  $k \in S$  and  $a \in A'$ ,
3.  $r_t(s, a)$  is a superadditive function on  $S \times A'$ ,
4.  $\sum_{j=0}^{\infty} p_t(j|s, a) u(j)$  is a superadditive function on  $S \times A'$  for nonincreasing  $u$ , and
5.  $r_N(s)$  is nonincreasing in  $s$ .

Then there exist optimal decision rules  $d_t^*(s)$  which are nondecreasing in  $s$  for  $t = 1, \dots, N$ .

For applications to queueing and inventory models, the assumption that  $A_s = A'$  for all  $s \in S$  is often overly restrictive. Slight modifications of the theorem hypotheses above allow extension to  $A_s$  which satisfy

1.  $A_s \subset A'$  for  $s \in S$ ,
2.  $A_s \subset A_{s'}$  for  $s' \geq s$ , and
3. for each  $s, a \in A_s$  and  $a' \leq a$  implies  $a' \in A_s$ .

#### 4.7.4 A Price Determination Model

We apply Theorem 4.7.4 to a product price determination model to interpret conditions under which the optimal price level is monotone nondecreasing in sales. A statement of the problem follows.

A marketing manager wishes to determine optimal price levels based only on current sales. Let the system state  $s \in S = \{0, 1, \dots\}$  represent the monthly sales of a particular product. At the start of each month, the manager notes sales in the previous month. Based on this information, he selects the product price  $a$ , for the current month from the set  $A' \approx \{a: a_L \leq a \leq a_U\}$ , where  $a_L$  and  $a_U$  represent the minimum and maximum price levels. Price effects both current revenue and sales.



The quantity  $r_t(s, a)$  represents expected revenue in month  $t$  if the previous month's sales were  $s$  and the price in month  $t$  was set at  $a$ . The probability that sales in month  $t$  are  $j$  units when the price is  $a$  and sales in month  $t - 1$  were  $s$  equals  $p_t(j|s, a)$ . The product lifetime is  $N - 1$  months so that  $r_N(s) = 0$  for all  $s$ . The manager's objective is to choose an adaptive price setting rule that maximizes expected total revenue over the product lifetime.

We discuss the hypotheses of Theorem 4.7.4 in the context of this model. Assume that  $r_t(s, a)$  and  $p_t(j|s, a)$  are continuous in  $a$  for fixed  $s$  and that  $\mathcal{A}'$  is compact so that the maximum in (4.7.8) is attained.

1. That  $r_t(s, a)$  is nondecreasing in  $s$  for fixed  $a$  means that, for a fixed price  $a$ , the expected revenue in the current month will be greater when previous months sales are greater.
2. That  $q_t(k|s, a)$  is nondecreasing in  $s$  for fixed  $a$  and  $k$  means that the probability that sales exceed  $k$  in the current month is greater when the previous month's sales are greater. In other words, sales one month ahead are stochastically increasing with respect to current sales.
3. From (4.7.2), the superadditivity of  $r_t(s, a)$  implies

$$r_t(s^+, a^+) - r_t(s^+, a^-) \geq r_t(s^-, a^+) - r_t(s^-, a^-)$$

for  $s^+ \geq s^-$  and  $a^+ \geq a^-$ . This condition is satisfied whenever the incremental effect on revenue of decreasing price is greater when the previous month's sales is greater. If  $r_t(s, a)$  is differentiable in  $a$ , superadditivity is equivalent to the partial derivative of  $r_t(s, a)$  with respect to  $a$  being nondecreasing in  $s$ .

4. Superadditivity of  $q_t(k|s, a)$  implies that

$$q_t(k|s^+, a^+) - q_t(k|s^+, a^-) \geq q_t(k|s^-, a^+) - q_t(k|s^-, a^-)$$

This condition is satisfied when the incremental effect of a price decrease on the probability that sales exceed a fixed level is greater if current sales are greater.

When all of these conditions are satisfied, the optimal pricing level is a nondecreasing function of sales. Thus the higher the current sales, the higher the optimal price in the subsequent period.

### 4.7.5 An Equipment Replacement Model

In this section, we apply Theorem 4.7.5 to the following equipment replacement model to establish that a control limit policy is optimal.

The condition of a piece of equipment used in a manufacturing process deteriorates over time. Let the state  $s \in S = \{0, 1, \dots\}$ , represent the condition of the equipment at each decision epoch. The higher the value of  $s$ , the worse the condition of the equipment. At each decision epoch, the decision maker can choose one of two actions from the set  $\mathcal{A}' = \{0, 1\}$ . Action 0 corresponds to operating the equipment as is for an additional period, while action 1 corresponds to scrapping the equipment and replacing it immediately with a new and identical piece of equipment. We assume that, in each period, the equipment deteriorates by  $i$  states with probability  $p(i)$

independent of the state at the beginning of the period. The transition probabilities for this model satisfy

$$p_i(j|s, 0) = \begin{cases} 0, & j < s \\ p(j - s), & j \geq s, \end{cases}$$

and  $p_i(j|s, 1) = p(j)$ ,  $j \geq 0$ .

The reward consists of three parts: a fixed income of  $R$  units per period; a state-dependent operating cost  $h(s)$ , with  $h(s)$  nondecreasing in  $s$ ; and a replacement cost of  $K$  units. Consequently,

$$r_i(s, a) = \begin{cases} R - h(s), & a = 0 \\ R - K - h(0), & a = 1. \end{cases}$$

The scrap value of the equipment at the end of the planning horizon is  $r_N(s)$  which we assume to be nonincreasing in  $s$ .

We show that this model satisfies the hypotheses of Theorem 4.7.5.

1. By the assumption on  $h$ ,  $r_i(s, a)$  is non-increasing in  $s$  for  $a = 0, 1$ .
2. For  $a = 1$ ,  $q_i(k|s, a)$  is independent of  $s$  so it is nondecreasing. For  $a = 0$ , define

$$\Delta q_i(k, s) \equiv q_i(k|s + 1, 0) - q_i(k|s, 0)$$

so that

$$\Delta q_i(k, s) = \begin{cases} \sum_{j=k}^{\infty} [p(j - s - 1) - p(j - s)] = p(k - s - 1), & k > s \\ 0, & k \leq s. \end{cases}$$

Thus  $\Delta q_i(k, s) \geq 0$ , and it follows that  $q_i$  is nondecreasing.

We apply the following easily proved lemma to verify conditions (3) and (4) of Theorem 4.7.5.

**Lemma 4.7.6.** Let  $g(s, a)$  be a real-valued function on  $S \times A$ , with  $A = \{0, 1\}$  and  $S = \{0, 1, \dots\}$ . If  $g(s, a)$  satisfies

$$[g(s + 1, 1) - g(s + 1, 0)] - [g(s, 1) - g(s, 0)] \geq 0 \quad (4.7.9)$$

for all  $s$ , it is superadditive.

3. Since

$$[r_i(s + 1, 1) - r_i(s + 1, 0)] - [r_i(s, 1) - r_i(s, 0)] = h(s + 1) - h(s) \geq 0,$$

where the last inequality follows from the assumed monotonicity of  $h$ , Lemma 4.7.6 establishes that  $r_i(s, a)$  is superadditive.

4. We show that  $\sum_{j=0}^{\infty} p_t(j|s, a)u(j)$  is superadditive for nonincreasing  $u$ . The left-hand side of (4.7.9) applied to this quantity equals

$$\begin{aligned} & \sum_{j=0}^{\infty} p(j)u(j) - \sum_{j=s+1}^{\infty} p(j-s-1)u(j) - \sum_{j=0}^{\infty} p(j)u(j) + \sum_{j=s}^{\infty} p(j-s)u(j) \\ &= \sum_{j=0}^{\infty} p(j)[u(j+s) - u(j+s+1)] \geq 0. \end{aligned}$$

The inequality follows because  $u$  is nonincreasing and  $p$  is non-negative. Thus, by Lemma 4.7.6,  $\sum_{j=0}^{\infty} p_t(j|s, a)u(j)$  is superadditive.

Consequently, this model satisfies the hypotheses of Theorem 4.7.5, so that for each  $t$  there is an optimal decision rule which is nondecreasing. Since there are only two actions, a control limit policy is optimal. That is, for each  $t$  there exists an  $s_t^*$  with the property that if  $s$  exceeds  $s_t^*$ , the optimal decision is to replace the equipment; and if  $s \leq s_t^*$  then it is optimal to operate the equipment as is for another period.

#### 4.7.6 Monotone Backward Induction

We now develop an efficient backward induction algorithm for finding optimal monotone decision rules. We assume that for each  $t$  there is a monotone optimal decision rule, that  $S = \{0, 1, \dots, M\}$  with  $M$  finite and  $A_s = A'$  for all  $s \in S$ .

##### The Monotone Backward Induction Algorithm

1. Set  $t = N$  and

$$u_N^*(s) = r_N(s) \quad \text{for all } s \in S.$$

2. Substitute  $t - 1$  for  $t$ , set  $s = 1$  and  $A'_1 = A'$ .

2a. Set

$$u_t^*(s) = \max_{a \in A'_s} \left\{ r_t(s, a) + \sum_{j \in S} p_t(j|s, a)u_{t+1}^*(j) \right\}.$$

2b. Set

$$A_{s,t}^* = \arg \max_{a \in A'_s} \left\{ r_t(s, a) + \sum_{j \in S} p_t(j|s, a)u_{t+1}^*(j) \right\}.$$

- 2c. If  $s = M$ , go to 3, otherwise set

$$A'_{s+1} = \{a \in A' : a \geq \max[a' \in A_{s,t}^*]\}$$

- 2d. Substitute  $s + 1$  for  $s$ , and return to 2a.

3. If  $t = 1$ , stop, otherwise return to (2).

Any decision rule which chooses actions from  $A_{s,t}^*$  in state  $s$  at decision epoch  $t$  is monotone and optimal.

This algorithm differs from the backward induction algorithm of Sec. 4.5 in that maximization is carried out over the sets  $A'_s$  which become smaller with increasing  $s$ . In the worse case,  $A'_s = A'$  for all  $s$  and computational effort equals that of backward induction; however, when an optimal decision rule is strictly increasing, the sets  $A'_s$  will decrease in size with increasing  $s$  and hence reduce the number of actions which need to be evaluated in step (2). If at some  $s'$ ,  $A'_{s'}$  contains a single element, say  $a^*$ , then no further maximization is necessary since that action will be optimal at all  $s \geq s'$ . In this case we need not carry out any further maximizations at iteration  $t$ , and instead set

$$u_t^*(s) = r_t(s, a^*) + \sum_{j \in S} p_t(j|s, a^*) u_{t+1}^*(j)$$

for all  $s \geq s'$ . For this reason, such an algorithm is particularly appealing in a two action model such as those in Secs. 3.4 and 4.7.5. Note also that in countable state models, this algorithm provides the only way of explicitly finding an optimal control limit policy.

## BIBLIOGRAPHIC REMARKS

Heyman and Sobel (1984) provide background on utility theory relevant to sequential decision making. Our discussion on stochastic partial orderings in Sec. 4.1 follows Chap. 17 in Marshall and Olkin (1979). Example 4.1 is adopted from Example A.2 therein. Howard and Matheson, (1972), Jaquette (1976), and Rothblum (1984) study models with multiplicative utilities. Kreps (1977) adopts a different approach which defines utilities on histories instead of reward sequences. His model may be transformed into one with rewards by expanding the state space. White (1988b) and Sobel (1988) provide surveys of Markov decision processes with nonstandard optimality criteria such as expected utility, preference orderings, mean and variance comparisons, and constraints. These topics are currently receiving considerable attention in Markov decision process theory research.

Section 4.2 introduces inductive computation. The algorithm therein is a special case of the general dynamic programming algorithm of Sec. 4.5 in which each  $A_s$  contains a single action.

Bellman's book (1957) presented the optimality equations and the Principle of Optimality together with references to his earlier papers (dating back to 1952) which introduced and illustrated many of the key ideas of dynamic programming. Karlin (1955), recognizing the need for a formal proof of the Principle of Optimality, introduced a formal mathematical structure for the analysis of dynamic programming models and demonstrated its validity within this context. Hinderer (1970) discussed the relationship between the Principle of Optimality and the optimality equations in considerable generality, distinguishing and identifying many mathematical subtleties. The results in Sec. 4.4 are in the spirit of Chap. 2 of Derman (1970).

Proofs of the optimality of  $(\sigma, \Sigma)$  or  $(s, S)$  policies appear in Heyman and Sobel (1984) and Bertsekas (1987). As mentioned earlier, the first proof of this result was provided by Scarf (1960). Notable extensions include works of Veinott (1966), Porteus (1971), Schal (1976), and Zheng (1990). Kalin (1980) extends these concepts to multidimensional problems. Johnson (1968), Federgruen and Zipkin (1984), Tijms (1986) and Zheng and Federgruen (1991) provide algorithms for computing  $\sigma$  and  $\Sigma$ .

Our solution of the secretary problem in Sec. 4.6.4 follows Bather (1980). The result seems to have some empirical validity. We note the following article which appeared in *The Globe and Mail* on September 12, 1990, page A22.

#### “The Last Shall be First

The last person interviewed for a job gets it 55.8 per cent of the time according to Runzheimer Canada, Inc. Early applicants are hired only 17.6 per cent of the time; the management-consulting firm suggests that job-seekers who find they are among the first to be grilled ‘tactfully ask to be rescheduled for a later date.’ Mondays are also poor days to be interviewed and any day just before quitting time is also bad.”

The elegant paper of Serfozo (1976) and the development in Albright and Winston (1979) provide the basis for Sec. 4.7. Our proof of Lemma 4.7.2 is adapted from Marshall and Olkin (1979), which contains considerable details on superadditive functions. They attribute the concept of superadditivity to Fan (1967). Derman (1970) gives an alternative proof of Lemma 4.7.2. Topkis (1978) provides a general theory of submodular functions on lattices which extends our results. His work appears to be responsible for introducing these ideas in the operations research literature. Further discussions of monotonicity appear in Ross (1983) and Heyman and Sobel (1984). Stidham (1985) discusses the use of superadditivity in queuing control models.

## PROBLEMS

- 4.1. Provide a policy evaluation algorithm analogous to that in Sec. 4.2, which computes the expected total reward for a randomized history-dependent policy. Use it to evaluate the randomized history-dependent problem defined in Sec. 3.1.
- 4.2. (Howard and Matheson, 1972) Consider a Markov decision problem in which the decision maker wishes to maximize the expected utility of the random reward over a finite horizon where the utility is multiplicative and has the form

$$\Psi(r_1, \dots, r_N) = \psi(r_1)\psi(r_2) \cdots \psi(r_N).$$

- a. Provide a backward algorithm for solving this problem when the reward is a function of the current state, the current action, and the subsequent state. How would the algorithm simplify when the reward depends only on the current state and action?
- b. Solve a two-period ( $N = 3$ ) version of the example in Sec. 3.1 with  $\psi(x) = e^{\gamma x}$ ,  $r_3(s) = 0$ , and  $\gamma = 0.2$ .

- c. Show by example that there exist random variables  $X$  and  $Y$  for which  $E[X] > E[Y]$  but  $E[e^{rX}] < E[e^{rY}]$ . Conclude that different policies may be optimal under the expected total reward and the expected utility criteria.

- 4.3. Prove Theorem 4.3.4.
- 4.4. For the separable sequential allocation problem of Sec. 4.6.3 show that if  $g_i$  is concave and has a maximum on  $[0, M]$  then the optimal allocation pattern is to consume  $M/(N - 1)$  units each period.
- 4.5. For the separable sequential allocation problem, show that, if  $g_i$  is convex, then an optimal consumption pattern is to consume  $M$  units in one period and 0 units in all other periods.
- 4.6. Solve Problem 3.4 using backward induction. Explicitly state the optimal policy.
- 4.7. Show that the maximum likelihood estimates, for parameters of a multinomial probability distribution are given by (4.6.3).
- 4.8. For the secretary problem, show by induction that, for all  $t < \tau$ ,  $u_t^*(1) > t/n$ .
- 4.9. Solve the secretary problem, in which the objective is to maximize the probability of choosing one of the two best candidates, i.e., a candidate with rank 1 or 2.
- 4.10. Solve the secretary problem when the objective is to maximize the expected rank of the candidate selected.
- 4.11. Verify directly that each of the functions in Sec. 4.7.2 is superadditive.
- 4.12. Show that a twice differentiable function  $g(x, y)$  on  $R^1 \times R^1$  is superadditive (subadditive) whenever its second mixed partial derivatives are non-negative (nonpositive).
- 4.13. Prove the following generalization of Lemma 4.7.6. Let  $g(s, a)$  be a function on  $S \times A$ , where  $S = A = \{0, 1, \dots\}$ , and suppose
- $$g(s + 1, a + 1) + g(s, a) \geq g(s, a + 1) + g(s + 1, a)$$
- for all  $a \in A$  and  $s \in S$ . Then  $g$  is superadditive.
- 4.14. Show that the sum of superadditive functions is superadditive. Is the product of superadditive functions superadditive?
- 4.15. Generalize Theorem 4.7.4 to a model in which  $S$  is the non-negative real line.
- 4.16. Prove a result analogous to Lemma 4.7.1 for  $g$  subadditive. That is, show that, whenever  $g$  is subadditive,  $f(x)$  defined by (4.7.3) is nonincreasing.

**4.17.** Show that, if  $g(x, y)$  is superadditive (subadditive) on  $X \times Y$ , then

$$f(x) = \min \left\{ y' : y' = \arg \min_{y \in Y} g(x, y) \right\}$$

is nonincreasing, (nondecreasing) in  $x$ .

**4.18.** In a deterministic problem show that the graph corresponding to an optimal policy is always a maximal spanning tree.

**4.19.** Prove Proposition 4.7.3 in the case where all quantities are nonincreasing.

**4.20.** Solve a two-period ( $N = 3$ ) version of the replacement model in Sec. 4.7.5 using the monotone backward induction algorithm. Take  $p(j)$  to be geometric with  $\pi = 0.4$ ,  $R = 0$ ,  $K = 5$ ,  $h(s) = 2s$ , and  $r_3(s) = \max\{5 - s, 0\}$ . Recall that a geometric random variable has a probability mass function  $p(j) = (1 - \pi)\pi^j$  for  $j = 0, 1, 2, \dots$ , and equals 0 otherwise. Compare the computational effort to that for the general backward induction algorithm.

**4.21.** (A simple bandit model) Suppose there are two projects available for selection in each of three periods. Project 1 yields a reward of one unit and always occupies state  $s$  and the other, project 2, occupies either state  $t$  or state  $u$ . When project 2 is selected, and it occupies state  $u$ , it yields a reward of 2 and moves to state  $t$  at the next decision epoch with probability 0.5. When selected in state  $t$ , it yields a reward of 0 and moves to state  $u$  at the next decision epoch with probability 1. Assume a terminal reward of 0, and that project 2 does not change state when it is not selected.

Using backward induction determine a strategy that maximizes the expected total reward.

**4.22.** Consider the hockey model described in Problem 3.19. Suppose your team is losing by one goal with two minutes remaining in the game, and decisions to “pull the goalie” may be made every 20 seconds. Assume that  $p_G = q_G = 0.02$ .

a. Assume that  $p_{NG} = 0.05$  and  $q_{NG} = 0.10$ ; find the strategy that maximizes the probability of winning the game; of obtaining at least a tie. Find the indicated probabilities.

b. Repeat part (a) when  $q_{NG} = 0.20$ .

c. Determine the optimal strategy as a function of the model parameters.

**4.23.** Consider the employment seeking model in Problem 3.18.

a. Impose assumptions on  $p_i(w|w')$  and  $\Psi_i(w)$  so that the optimal policy is monotone in the wage  $w$ .

b. Explicitly solve the problem with  $N = 3$ , when  $\Psi_i(w) = w$ , and  $p_i(w|w')$  is uniform on  $[0, W]$ .

**4.24.** Prove the following modification of Theorem 4.7.4 in which we drop the assumption that  $A_s = A'$  for all  $s \in S$ .

Suppose that

- i.  $A_s \subset A'$  for  $s \in S$ ,
- ii.  $A_s \subset A_{s'}$  for  $s' \geq s$ , and
- iii. for each  $s, a \in A_s$  and  $a' \in a$  implies  $a' \in A_s$ ,  
and, for  $t = 1, \dots, N-1$ ,
  - 1.  $r_t(s, a)$  is nondecreasing in  $s$  for all  $a \in A_s$ ,
  - 2.  $q_t(k|s, a)$  is nondecreasing in  $s$  for all  $k \in S$  and  $a \in A_s$ ,
  - 3.  $r_t(s, a)$  is a superadditive (subadditive) function,
  - 4.  $q_t(k|s, a)$  is a superadditive (subadditive) function for all  $k \in S$ , and
  - 5.  $r_N(s)$  is nondecreasing in  $s$ .

Then there exist optimal decision rules  $d_t^*(s)$  which are nondecreasing (nonincreasing) in  $s$  for  $t = 1, \dots, N$ .

**4.25.** Use the following approach to establish that  $u_t^*$ ,  $t = 1, 2, \dots, N$ , satisfies the optimality equations (4.3.2) subject to (4.3.3).

a. Show inductively that

$$u_t^*(h_t) \geq \sup_{a \in A_s} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(h_t, a, j) \right\}$$

by observing that, since

$$u_{t+1}^*(h_{t+1}) = \sup_{\pi \in \Pi^{\text{HR}}} u_{t+1}^\pi(h_{t+1})$$

for every  $\varepsilon > 0$ , there exists a  $\pi' \in \Pi^{\text{HR}}$  such that  $u_{t+1}^{\pi'}(h_{t+1}) \geq u_{t+1}^*(h_{t+1}) - \varepsilon$ . Show that this implies that, for all  $a \in A_s$ ,

$$u_t^*(h_t) \geq r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^{\pi'}(h_t, a, j) - \varepsilon,$$

from which the result follows.

b. Show inductively that

$$u_t^*(h_t) \leq \sup_{a \in A_s} \left\{ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^*(h_t, a, j) \right\}$$

by using the result that, for every  $\varepsilon > 0$ , there exists a  $\pi' = (d_1, d_2, \dots, d_{N-1}) \in \Pi^{\text{HR}}$ , for which  $u_t^{\pi'}(h_t) \geq u_t^*(h_t) - \varepsilon$ , to establish that

$$u_t^*(h_t) \leq \sum_{a \in A_s} q_{d_t(h_t)}(a) \left[ r_t(s_t, a) + \sum_{j \in S} p_t(j|s_t, a) u_{t+1}^{\pi'}(h_t, a, j) \right] + \varepsilon.$$

c. Combine (a) and (b) to conclude the result.



- 4.26.** Numerically solve the following version of the service rate control model of Sec. 3.7.2. Objectives are to determine if there is any obvious structure for the optimal policy and to investigate its sensitivity to model parameters.

Assume that there is a finite system capacity of eight units; that is, if arriving jobs cause the system content to exceed eight units, excess jobs do not enter the system and are lost. Let  $R = 5$ ,  $h(s) = 2s$ ,  $B = \{0, 1, 2\}$ ,  $K = 3$ ,  $d(0) = 0$ ,  $d(1) = 2$ , and  $d(2) = 5$ . Take  $r_N(s, b) = 0$  for all  $s$  and  $b$  and  $N = 10$ . Assume that jobs arrive following a Poisson distribution with a rate of 1.5 jobs per period so that

$$g(s) = \frac{e^{-1.5}(1.5)^s}{s!}, \quad s = 0, 1, 2, \dots,$$

and that the service distribution  $f_b(s)$  satisfies  $f_0(0) = 0.8$ ,  $f_0(1) = 0.2$ ;  $f_1(0) = 0.5$ ,  $f_1(1) = 0.5$ ; and  $f_2(0) = 0.2$ ,  $f_2(1) = 0.8$ .

Of course this problem requires developing a computer program to carry out calculations.

- 4.27.** Find an optimal policy for the game “But Who’s Counting” described in Sec. 1.7, using the description of the state space suggested in Problem 3.28. Show that the optimal policy depends only on the position of the unoccupied digits; that is, it does not depend on the numbers previously placed in the occupied digits.
- 4.28.** In the model in Problem 3.25, suppose there are three patches in which the animal can forage. In patch 1, the risk of predation is 0, the probability of finding food is 0.0, and its energy value is 0. In patch 2, the risk of predation is 0.004 in each period, the probability of finding food is 0.4, and the energy gain is 3 if food is found. In patch 3, the predation risk is 0.020, the probability of finding food is 0.6, and its energy gain is 5. Foraging in any patch uses one unit of energy reserves. Energy reserves below 4 indicate death and the animal’s maximum energy capacity is ten units. Solve this problem for 20 foraging periods to find a patch selection strategy which maximizes the animal’s probability of survival over this period.
- 4.29.** Consider the model in Problem 3.26 regarding hunting group size selection in lions.
- Find a strategy (determining whether or not to hunt, and selection of hunting group size) that maximizes the probability of survival of the lion over 30 hunting days, assuming one hunt per day seeking a prey of zebras.
  - Investigate the sensitivity of this policy to the number of hunts per day and the edible biomass of available prey.
  - Suppose a lion’s stomach content is known at the beginning of the 30-day period. How could the model results be compared to field observations?
- 4.30.** Suppose the current price of some stock is \$30 per share, and its daily price increases by \$0.10 with probability 0.6, remains the same with probability 0.1 and decreases by \$0.10 with probability 0.3. Find the value of a call option to

purchase 100 shares of this stock at \$31 per share any time in the next 30 days by finding an optimal policy for exercising this option. Assume a transaction cost of \$50. Investigate the effect of the time value of money and the stock price change distribution on the optimal policy and option value.

- 4.31.** Suppose the current price of some stock is \$30 per share and its daily price increases by \$0.10 with probability 0.4, remains the same with probability 0.1, and decreases by \$0.10 with probability 0.5. Find the value of a put option to sell 100 shares of this stock at \$29 per share any time in the next 30 days by finding an optimal policy for exercising this option. Assume a transaction cost of \$50. Investigate the effect of the time value of money and the stock price change distribution on the optimal policy and option value.
- 4.32.** Find an optimal policy for a variant of the inventory model of Secs. 3.2.2 and 4.6.2 which allows backlogging of unfilled orders, and in which there is a penalty cost for backlogged orders of 0.5 per unit per period.
- 4.33.** (Canadian Tire inventory control, Section 1.2) Daily demand for paint brushes at a particular store follows the demand distribution:

|             |     |      |     |      |      |
|-------------|-----|------|-----|------|------|
| demand      | 0   | 1    | 2   | 3    | 4    |
| probability | 0.7 | 0.15 | 0.1 | 0.04 | 0.01 |

The stock level is reviewed in the evening every four days and when warranted an order is placed at the central warehouse to augment stock. Orders arrive two days later (a two day lead time) and are available to meet demand on the morning of the third day following the review. For example, an order placed Sunday evening can be used to meet demand on Wednesday. Demand not satisfied from stock on hand is never filled. Management imposes a penalty to account for this.

- Find a reordering policy that minimizes expected total ordering, holding and shortage costs under the assumption that the fixed cost for placing an order is \$0.20, the daily per unit holding cost is \$0.01 and the per unit penalty cost for unfilled orders is \$0.50. Assume a planning horizon of 84 days and assign a value of \$0.70 for each unit on hand at the end of this period.
- Investigate the sensitivity of the optimal policy to the penalty cost.
- Investigate the sensitivity of the expected total cost to the review period. In particular what is the effect of reviewing the inventory level every 7 days instead. (Hint: Write a system equation which describes the system state every review period as a function of the system state at the previous review period and the daily demand at each day between reviews. Note further that the two day demand can be found by computing the convolution of the above demand distribution with itself. Of course, solving this problem involves writing a computer program.)